

THE DRASIL FRAMEWORK

SUCCINCTLY VERBOSE: THE DRASIL FRAMEWORK

BY

DANIEL M. SZYMCZAK, M.A.Sc., B.Eng

A THESIS

SUBMITTED TO THE DEPARTMENT OF COMPUTING & SOFTWARE

AND THE SCHOOL OF GRADUATE STUDIES

OF MCMASTER UNIVERSITY

IN PARTIAL FULFILMENT OF THE REQUIREMENTS

FOR THE DEGREE OF

DOCTOR OF PHILOSOPHY

© Copyright by Daniel M. Szymczak, TBD 2022

All Rights Reserved

Doctor of Philosophy (2022)
(department of computing & software)

McMaster University
Hamilton, Ontario, Canada

TITLE: Succinctly Verbose: The Drasil Framework

AUTHOR: Daniel M. Szymczak
M.A.Sc.

SUPERVISOR: Jacques Carette and Spencer Smith

NUMBER OF PAGES: xvii, 152

Lay Abstract

A lay abstract of not more 150 words must be included explaining the key goals and contributions of the thesis in lay terms that is accessible to the general public.

Abstract

Abstract here (no more than 300 words)

Your Dedication
Optional second line

Acknowledgements

Acknowledgements go here.

Contents

Lay Abstract	iii
Abstract	iv
Acknowledgements	vi
List of Figures	xii
List of Tables	xiii
Notation, Definitions, and Abbreviations	xiv
Declaration of Academic Achievement	xvi
1 Introduction	1
1.1 Problem Statement and Motivation	4
1.2 The mundane and why it matters	5
1.3 Scope	6
1.4 Roadmap	8
1.5 Contributions	9

2	Background	11
2.1	Software Artifacts	12
2.2	Software Reuse and Software Families	18
2.3	Literate Approaches to Software Development	24
2.4	Generative Programming	28
3	A look under the hood: Our process	31
3.1	A minimal motivating example: HGHC	32
3.2	A (very) brief introduction to our case study systems	34
3.3	Breaking down softifacts	36
3.4	Identifying Repetitive Redundancy	50
3.5	Organizing knowledge - a fluid approach	58
3.6	Summary - The seeds of Drasil	62
4	Drasil	64
4.1	Drasil in a nutshell	65
4.2	Our Ingredients: Organizing and Capturing Knowledge	67
4.3	Recipes: Codifying Structure	81
4.4	Cooking it all up: Generation/Rendering	96
4.5	Seasoning to Taste: Iteration and Refinement	107
4.6	Summary	109
5	Results	112
5.1	Value in the mundane	112
5.2	Observations: Case Studies and Reimplementations	115
5.3	Usability and Adoption	118

5.4	Summary and Takeaways	119
6	Discussion	120
6.1	Interpreting the Results	120
6.2	Human Factors and Usability	124
6.3	Limitations and Directions for Validation	125
6.4	Lessons Learned	126
6.5	Summary and Takeaways	128
7	Conclusion	130
7.1	Summary of Main Contributions	131
7.2	Limitations and Open Questions	132
7.3	Future Work	133
7.4	Broader Impact and Personal Reflection	138
7.5	Closing Remarks	139
A	HGHC SRS	141

List of Figures

2.1	The Waterfall Model of Software Development	13
3.1	The Table of Contents from the expanded Smith et al. template . . .	38
3.2	The reference sections of GlassBR	40
3.3	Table of Contents for GlassBR Module Guide	43
3.4	Calc Module from the GlassBR Module Guide	44
3.5	Python source code directory structure for GlassBR	46
3.6	Source code of the Calc.py module for GlassBR	48
3.7	Table of Symbols (truncated) Section from GamePhysics	51
3.8	Data Definition for Dimensionless Load ($\hat{q}$) from GlassBR SRS	52
3.9	Theoretical Model of conservation of thermal energy found in both the SWHS and NoPCM SRS	55
3.10	Instance Model difference between SWHS and NoPCM	57
4.1	NamedChunk Definition	69
4.2	Some example instances of ConceptChunk using the dcc smart con- structor	70
4.3	Projecting knowledge into a chunk's definition	72
4.4	The force ConceptChunk	73
4.5	The force and acceleration UnitalChunks	73

4.6	The fundamental SI Units as Captured in Drasil	74
4.7	Examples of chunks for units derived from other SI Unit chunks . . .	74
4.8	Defining real number multiplication in Expr	76
4.9	Encoding Newton’s second law of motion	77
4.10	Newton’s second law of motion as a <i>Chunk</i>	77
4.11	A subset of Drasil’s Chunk Classification system showing some of the encoded property relationships	80
4.12	A subset of Drasil’s chunks showing how chunks may belong to multiple classifications based on the underlying knowledge they project.	81
4.13	Recipe Language for SRS	83
4.14	ReqmntSec Decomposition and Definitions	84
4.15	SRS Recipe for GlassBR	85
4.16	System Information for GlassBR	86
4.17	The iMods definition	88
4.18	The Recipe for GlassBR’s Executable Code	91
4.19	The Recipe Language for Executable Code (CodeSpec)	91
4.20	The readTableMod module definition	92
4.21	The Choices object definition	93
4.22	genTeX function for LaTeX rendering	98
4.23	generateCode and genPackage for code rendering	99
4.24	prntDoc’ function for document output	100
4.25	prntMake function for document output	100
4.26	createCodeFiles for code output	101
4.27	Definition of $\hat{q}$ as a chunk in the GlassBR knowledge base	102

4.28	Raw generated $\text{\LaTeX}$ of the $\hat{q}$ Data Definition	103
4.29	$\hat{q}$ as defined in the generated python code	106

List of Tables

1.1	The same information as natural language, equation, and code	2
2.1	Mapping of Rational Design Process to Waterfall Model and Common Software Artifacts	14
2.2	A summary of the Audience for each of the most common softifacts and what problem that softifact is solving	17
3.1	Summary of the case study systems	35
4.1	System information object breakdown - every property is represented by chunks encoding given information	87
4.2	CodeSpec object breakdown	92
4.3	Choices object breakdown	94
5.1	Summary statistics for Drasil case study SRS reimplementations . . .	115
5.2	Key differences between SWHS and NoPCM Drasil implementations .	118

Notation, Definitions, and Abbreviations

[TODO: Update this —DS]

Notation

$A \leq B$ A is less than or equal to B

Definitions

Softifact A portmanteau of ‘software’ and ‘artifact’. The term refers to any of the artifacts (documentation, code, test cases, build instructions, etc.) created during a software project’s development.

Abbreviations

QA Quality Assurance

SI Système International d’Unités

SRS

Software Requirements Specification

Declaration of Academic Achievement

This thesis was completed by Daniel M. Szymczak under the supervision of Jacques Carette and Spencer Smith. The principal original contributions of the author are the analysis, design, implementation, and evaluation presented here for the Drasil framework as a knowledge-centric approach to generating software artifacts from a centralized representation of domain knowledge. In particular, the author carried out the systematic analysis of redundancy and inconsistency across scientific software artifacts, developed the thesis account of Drasil’s chunk- and recipe-based architecture, implemented substantial portions of the framework and its supporting representations, and reimplemented the case studies used in this thesis within Drasil in order to evaluate the approach.

This thesis also draws on prior and collaborative work. The broader Drasil project is part of an ongoing research effort, and the author’s work should not be taken to mean that all ideas, infrastructure, or repository contents originated solely with the author. In addition, most of the case studies discussed in this thesis were originally developed by other teams at McMaster University; the author’s contribution in this thesis is their analysis and re-expression within Drasil, rather than authorship of

all original source artifacts for those systems. Although Drasil uses GOOL as an intermediate layer for code generation, GOOL itself was not developed by the author and is not claimed as a contribution of this thesis. Where prior templates, existing artifacts, supervisory guidance, or contributions from other project members informed the work, they are acknowledged as part of the broader research context so as to avoid misattribution.

Chapter 1

Introduction

There is near universal agreement on the value of documentation [1, 2, 7, 11, 12, 13, 15, 16, 18, 22, 23, 24, 25, 26, 27, 28, 29, 30, 39, 40, 41, 45, 47, 48, 49, 50, 52, 53, 55, 56, 57, 58, 60, 61, 63]. Despite this, it is not often prioritized on software projects. Many projects focus heavily on source code, treating it as the primary artifact, while documentation is often neglected despite its necessary role. Documentation serves as a critical bridge between stakeholders, enabling effective communication, knowledge transfer, and long-term maintainability. It helps clarify requirements, supports onboarding of new team members, facilitates regulatory compliance, and provides a reference for future maintenance and evolution. Well-maintained documentation reduces the risk of misunderstandings, errors, and knowledge loss, ultimately contributing to higher quality and more reliable software systems.

To simplify our discussion of both source code and documentation, this thesis introduces the term *softifact* to refer to any artifact produced during software development that conveys knowledge, including source code, requirements, design documents, test cases, user manuals, and any other relevant materials. Code and other

software artifacts (softifacts) by their nature must say the same thing, albeit to different audiences (i.e. developers, maintainers, regulators, or end users); if they did not, they would be describing different systems. This principle applies regardless of the context of the softifact. Take this thesis for example: it is written for an audience familiar with classical software engineering, including those who have experience with requirements documents, design documents, and the challenges of maintaining large, long-lived software systems. The examples and case studies are drawn from scientific computing, but the pain points and solutions discussed are relevant to wider software engineering contexts as a whole.

Speaking broadly about softifacts, let us consider a few different documents. A software requirements document is a human-readable abstraction of *what* the software is supposed to do. Whereas a design document is a human-readable version of *how* the software is supposed to fulfill its requirements. The source code itself is a computer-readable list of instructions combining *what* must be done and, in many languages, *how* that is to be accomplished. Importantly, this multiplicity of views is not limited to separate softifacts; even within a single artifact the same information may appear in multiple forms. For example, a requirements document may include natural language descriptions, mathematical equations, and other relationships expressed in a multitude of ways (tables, graphs, diagrams, etc.).

Table 1.1: The same information as natural language, equation, and code

Description	Equation	Code (Python)
P_b is the Probability of Breakage	$P_b = 1 - e^{-B}$	<pre> 1 def func_P_b(B): 2 return 1.0 - math.exp(-B) </pre>

Table 1.1 shows how the same information (the definition of P_b) can be represented in several different views: as a natural language description, equation, and code definition. We aim to take advantage of the inherent redundancy across these views to distill a single source of information, thus removing the need to manually duplicate information across softifacts. This example illustrates a broader challenge: when the same knowledge must be expressed in multiple forms for different audiences or purposes, the effort to keep these representations consistent becomes significant.

Manually writing and maintaining a full range of softifacts (i.e. multiple documents for different audiences plus the source code) is redundant and tedious. Factor in deadlines, changing requirements, and other common issues faced during development and you have a perfect storm for inter-artifact synchronization issues.

How can we avoid having our artifacts fall out of sync with each other? Some would argue “just write code!” And that is exactly what a number of other approaches have tried. Documentation generators like Doxygen, Javadoc, Pandoc, and more take a code-centric view of the problem. Typically, they work by having natural-language descriptions and/or explanations written as specially delimited comments in the code which are later automatically compiled into a human-readable document.

While these approaches definitely have their place and can come in quite handy, they do not solve the underlying redundancy problem. The developers are still forced to manually write descriptions of systems in both code and comments. They also do not generate all softifacts - commonly they are used to generate only API documentation targeted towards developers or user manuals.

If we can truly understand the differences and commonalities between softifacts, we can develop tools to capitalize on them and streamline the process of creating and

maintaining them. To accomplish this, we first need to break down common softifacts to develop an understanding of their contents, intended audience, and how they differ whether on intra- or inter-project scales.

We propose a new framework, Drasil¹, alongside a knowledge-centric view of software, to help take advantage of inherent redundancy while reducing manual duplication and synchronization problems. Our approach looks at what underlies the problems we solve using software and capturing that “common” or “core” knowledge. We then use that knowledge to generate our softifacts, thus gaining the benefits inherent to the generation process: lack of manual duplication, one source to maintain, and “free” traceability of information.

1.1 Problem Statement and Motivation

Despite the recognized value of documentation and softifacts, the process of creating and maintaining them is fraught with redundancy, tedium, and potential risks of inconsistency. Change is inevitable in software projects: requirements evolve, bugs are fixed, and features are added. As software evolves, artifacts (requirements documents, design documents, source code, etc.) often fall out of sync, leading to errors, increased maintenance costs, and reduced trust in documentation. This thesis addresses the challenge: *How can we systematically reduce unnecessary redundancy and improve consistency across softifacts, especially in domains where correctness and traceability are critical?*

The consequences of these challenges are particularly acute in scientific computing

¹The name Drasil comes from a shortening of the name Yggdrasil from Norse mythology, as we believe Drasil can span the many worlds (domains) of software engineering.

and other domains where correctness, traceability, and regulatory compliance are essential. In such contexts, even minor inconsistencies can undermine confidence in the software and its documentation, potentially leading to costly errors or rework.

This thesis therefore investigates a knowledge-centric approach that captures the core information underlying a system and uses it to generate the relevant softifacts. The goal is to reduce manual duplication while improving consistency, maintainability, and traceability across the software lifecycle.

1.2 The mundane and why it matters

What we are doing is not big and flashy; instead, it is focused on the tedium of day-to-day software engineering work. Much of this thesis is motivated by the small but recurring annoyances that consume time and attention: updating multiple artifacts after a single change, tracking down mismatches between documents and code, or fixing errors caused by duplicated information.

These little pain points, including having to manually update softifacts after modifying code, especially in regulated industries, accumulate over time and can have a significant impact on productivity and morale. While each instance may seem minor, together they represent a substantial barrier to efficient and reliable software development, particularly in domains where correctness and traceability are essential.

The contrast here is important. By “mundane” we mean the recurring work of keeping softifacts aligned, propagating changes consistently, maintaining interfaces, and preserving traceability. By contrast, the “non-mundane” work often lies in the domain-specific numerical methods themselves: designing models, selecting solvers, or implementing and optimizing specialized algorithms. Drasil does not need to generate

every line of that code. In many cases, it is more appropriate to rely on existing numerical libraries and use Drasil to generate the surrounding softifacts and the code that interfaces with those libraries.

The value of this approach is most apparent in the small details: the hours saved by not having to update documentation by hand, the confidence that comes from knowing all softifacts are synchronized, and the ability to trace the origin of any piece of information. These improvements may not be as immediately dramatic as solving a major technical challenge, but they accumulate to create a more reliable, maintainable, and satisfying development experience. In this way, Drasil demonstrates that addressing the mundane is not only worthwhile, but essential for advancing the practice of software engineering.

1.3 Scope

We are well aware of the ambitious nature of attempting to solve the problem of manual duplication and unnecessary redundancy across all possible software systems. Frankly, it would be highly impractical to attempt to solve such a broad spectrum of problems. Each software domain poses its own challenges, alongside specific benefits and drawbacks.

Accordingly, this thesis focuses on the generation of source code, requirements, and build instructions. Although Drasil is extensible and can be adapted to generate additional artifact types, those artifacts are outside the scope of the present work.

Our work on Drasil is most relevant to software that is well-understood [6] and undergoes frequent change (maintenance). Good candidates for development using

Drasil are long-lived (10+ years) software projects with artifacts of interest to multiple stakeholders. With that in mind, we have decided to focus on Scientific Computing (SC) software. Specifically, we are looking at software that follows the pattern $input \rightarrow process \rightarrow output$. This focus allows us to make the initial implementation of Drasil feasible and tractable, while still capturing a broad class of scientific computing problems.

SC software has a strong fundamental underpinning of well-understood concepts. It also has the benefit of seldomly changing, and when it does, existing models are not necessarily invalidated. For example, rigid-body problems in physics are well-understood and the underlying modeling equations are unlikely to change. However, should they change, the current models will likely remain as good approximations under a specific set of assumptions. For instance, who hasn't heard "assume each body is a sphere" during a physics lecture?

SC software could also benefit from buy-in to good software development practices as many SC software developers put the emphasis on science and not development [24]. Rather than following rigid, process-heavy approaches deemed unfavourable [7], developers of SC software choose to use knowledge acquisition driven [23], agile [1, 7, 11, 51], or amethododical [22] processes instead.

Finally, while recent advances in Machine Learning (ML), Large Language Models (LLMs), and other Artificial Intelligence (AI) based tooling have shown promise in automating aspects of software engineering, Drasil intentionally does not employ these approaches. This is because our focus is on achieving determinism and reproducibility: properties that are essential in scientific computing. The artifacts generated by Drasil must be consistent and repeatable, which is not guaranteed by current AI-based

methods.

1.4 Roadmap

The remainder of this thesis is organized as follows:

- **Chapter 2: Background** reviews relevant literature on softifacts, redundancy, reuse, literate programming, and generative programming.
- **Chapter 3: Our Process** details the methods used to analyze softifacts, identify redundancy, and motivate our knowledge-centric approach.
- **Chapter 4: The Drasil Framework** presents the design, architecture, and implementation of Drasil, including its realization as a collection of lightweight domain-specific languages embedded in Haskell for knowledge capture and artifact generation.
- **Chapter 5: Results** presents observations and analysis from reimplementing existing case studies using Drasil, highlighting practical benefits.
- **Chapter 6: Discussion** interprets the results, explores the implications of the knowledge-centric approach, and discusses limitations and challenges encountered.
- **Chapter 7: Conclusion** summarizes the findings, contributions, and implications of this work and outlines potential extensions, open questions, and directions for further research.

1.5 Contributions

The main contributions of this thesis are:

- **Systematic analysis of redundancy in softifacts and its operationalization as knowledge-centric generation.** Through detailed breakdowns of softifacts across multiple scientific computing case studies, this work identifies patterns of unnecessary duplication and inconsistency and formalizes a process for capturing, organizing, and projecting domain knowledge to enable automated artifact generation from a single source of truth.
- **Design and implementation of the Drasil framework.** Drasil is presented as a novel, extensible framework for knowledge-centric software artifact generation. Its architecture includes robust mechanisms for knowledge capture (chunks), projection (recipes), and rendering (multi-format output), supporting traceability and maintainability by construction.
- **Empirical evidence that single-source generation enforces consistency, traceability, and reproducibility by construction.** Through reimplementations of several scientific computing systems using Drasil, the thesis shows that local changes to domain knowledge propagate automatically across generated artifacts, that generated content retains explicit provenance, and that artifact generation is deterministic and reproducible.
- **Increased error visibility and clarification of tacit knowledge.** By forcing explicit encoding of domain knowledge, Drasil surfaces hidden assumptions, semantic inconsistencies, and other defects, enabling single-point correction and consistent documentation across all generated artifacts.

- **Separation of knowledge from presentation through the recipe DSL.**

Drasil’s recipe-based generation approach allows the structure of generated artifacts to be modified without altering the underlying knowledge base, improving maintainability and adaptability.

- **Framework for rapid generation and maintenance of software families.**

The single-source approach enables efficient creation and evolution of software family variants through selective extension of the knowledge base and adaptation of recipes, minimizing duplication and ensuring consistency across related products.

- **Lessons learned and open questions for knowledge-centric development.** The thesis distills practical lessons from iterative development, onboarding, and refinement, highlighting the importance of centralized knowledge, improved authoring tools, and further validation for extending Drasil to new domains.

Drasil allows developers to focus on the system being built as a whole rather than the individual artifacts. We can still produce a “rational design” [40] without the explicit tedium of manually keeping everything in sync.

Chapter 2

Background

This chapter surveys foundational concepts and approaches that underpin the thesis. Each section is selected to clarify the context, motivations, and limitations of the work. We structure the review as follows:

- First we introduce literature around software artifacts, emphasizing their role in communication, maintenance, and verification within software engineering.
- Next we examine software reuse, software families, and reproducible research, highlighting strategies for leveraging prior work and domain knowledge.
- Then we discuss literate approaches to software development to motivate the integration of documentation and code.
- Finally, we review Generative programming as a means of automating artifact creation from high-level specifications.

We have selected specific topic areas following a criterion for inclusion regarding

the direct relevance to our central problem: improving the production and maintenance of software artifacts through automation and knowledge reuse. Topics are selected if they inform the design, implementation, or evaluation of the proposed approach. Areas such as Model-Driven Engineering (MDE), or Knowledge-Based Software Engineering (KBSE)¹ while significant in the broader literature, are excluded unless they directly intersect with the thesis’s scope. This ensures the chapter remains focused and purposeful, rather than exhaustive.

2.1 Software Artifacts

Software artifacts (or softifacts) come in a wide variety of forms and have existed since the first programs were created. In the broadest sense, we can think of softifacts as anything and everything produced during the creation of a piece of software that serves some purpose. Any document detailing what the software should do, how it was designed, how it was implemented, how to test it, and so on would be considered a softifact, as would the source code whether as a text file, stack of punched cards, magnetic tapes, or other media. Softifacts also include items such as build instructions, Continuous Integration/Continuous Delivery (CI/CD) pipeline definitions, automated test cases, READMEs, etc.

Softifacts beyond just the source code are important to us for a number of reasons. They serve as a means of communication between different stakeholders involved in the software development process, or even different generations of developers involved in the production of a piece of software. They provide a common understanding

¹While KBSE shares some conceptual overlap, its primary concerns (such as expert systems, rule-based reasoning, and formal methods) do not directly inform the design, implementation, or evaluation of Drasil.

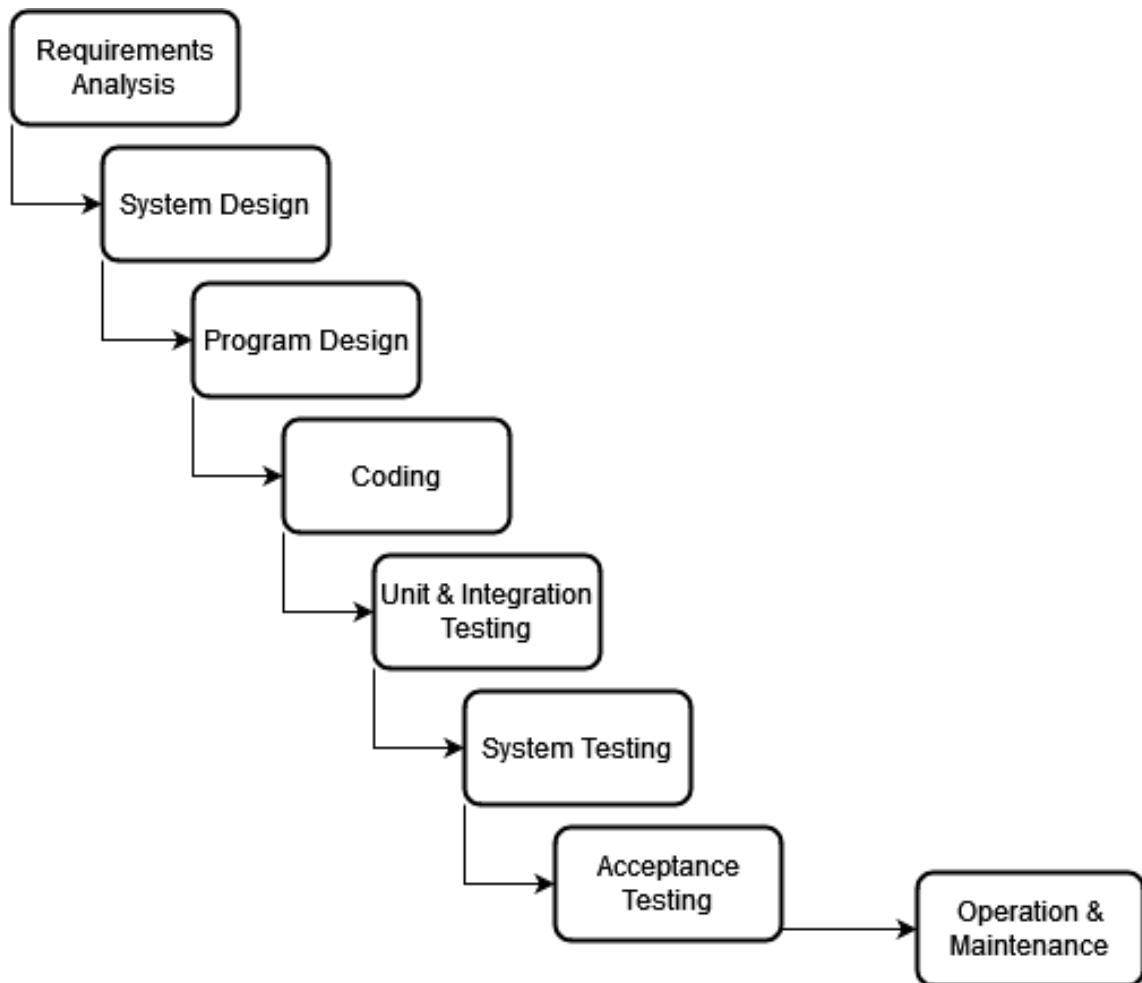


Figure 2.1: The Waterfall Model of Software Development

of what the software is supposed to do, how it will be built, and how it will be tested. Softifacts also provide a record of the decisions that were made during the development process, which is important for future maintenance/modification of the software or for compliance reasons. Combined with the former, this gives us a means to verify and validate the software against its original requirements and design. The production, maintenance, and use of softifacts is largely dependent on the specific design process being used within a team.

Table 2.1: Mapping of Rational Design Process to Waterfall Model and Common Software Artifacts

Rational Design Phase	Waterfall Phase(s)	Common Artifacts
Establish/Document Requirements	Requirements Analysis	System Requirements Specification; Verification and Validation Plan
Design and Document Module Structure	System Design	Design Document
Design and Document Module Interfaces	Program Design	Module Interface Specification
Design and Document Module Internal Structures	Program Design	Module Guide
Write Programs	Coding	Source Code; Build Instructions
Maintain	Testing (Unit, Integration, System, Acceptance) Operation & Maintenance	Test Cases; Verification and Validation Report

Software design can follow a number of different design processes, each with their own collection of softifacts. A common, traditional approach is the Waterfall model (Figure 2.1) of software development [43], which often does not work in practice [?]. [\[Need this source. —DS\]](#) However, Parnas and Clements [40] detailed what they dubbed a “rational” design process; an idealized version of software development that includes what needs to be documented in corresponding softifacts. The rational design process is not meant to be a linear process like the Waterfall model, but instead an iterative process using section stubs for information that is not yet available or not fully clear during the time of writing. Those stubs are then filled in over the

development process, and existing documentation is updated so it appears to have been created in a linear fashion for ease of review later in the software lifecycle. The rational design process involves the steps listed in the first column of Table 2.1.

Parnas provided a list of the most important documents [39] required for the rational design process, which Smith [59] expanded upon by including complimentary artifacts such as source code, verification and validation plans, test cases, and build instructions. While there have been many proposed artifacts, the following collection covers those most relevant to this thesis:

Name: Software Requirements Specification
Description: Contains the functional and nonfunctional requirements detailing what the desired software system should do.
Name: System Design Document
Description: Explains how the system should be broken down and documents implementation-level decisions that have been made for the design of the system.
Name: Module Guide
Description: In-depth explanation of the modules outlined in the System Design Document.
Name: Module Interface Specification
Description: Interface specification for each of the modules outlined in the System Design Document/Module Guide.
Name: Program Source Code
Description: The source code of the implemented software system.

Name: Verification and Validation Plan
Description: Uses the system requirements to document acceptance criteria for each requirement that can be validated.
Name: Test cases
Description: Implementation of the Verification and Validation Plan in source code (where applicable) or as a step-by-step guide for testers.
Name: Verification and Validation Report
Description: Report outlining the results after undertaking all of the testing initiatives outlined in the Verification and Validation plan and test cases.

This collection is not exhaustive of all the types of softifacts, as there are many other design processes that use different types of softifacts. For example, looking at an agile approach using Scrum/Kanban, the softifacts tend to be distributed in different ways [21]. Requirements are documented in tickets under so-called “epics”, “stories”, and “tasks” as opposed to a singular requirements artifact, and the acceptance criteria listed on those tickets make up the validation and verification plan.

Regardless of the process, most attempt to document very similar information to that of the rational design process. Using the waterfall model as an example, we can see (Table 2.1) the rational design process and its artifacts map onto the model in a very straightforward manner.

As mentioned earlier, softifacts are important to development in a number of ways, such as easing the burden of maintenance and training. We outline the artifacts we are most interested in below with a brief description of their purpose.

Parnas [39] does an excellent job of defining the target audience for each of the most common softifacts and we extend that alongside our work. A summary can be found in Table 2.2.

Table 2.2: A summary of the Audience for each of the most common softifacts and what problem that softifact is solving

Softifact	Who (Audience)	What (Problem/Solution)
SRS	Software Architects QA analysts	Defines exactly what specification the software system must adhere to.
Module Guide	All developers QA analysts	Outlines implementation decisions around the separation of functionality into modules and give an overview of each module.
Module Interface Specifications	Developers who implement the module Developers who use the module QA analysts	Details the exact interfaces of the modules from the Module Guide.
Source Code / Executable	All developers QA analysts End users	Implements the machine instructions designed to address the overall problem for which the software system has been specified
Verification and Validation Plan	Developers who implement the module QA analysts	Describes how the software should be verified using tests that can be validated. Includes module-specific and system-wide plans.

2.2 Software Reuse and Software Families

In this section we look at ways in which others have attempted to avoid “reinventing the wheel” by providing means to reuse software, in part or whole, and reuse the analysis and design efforts of those that came before. We focus on software families (software product lines), component-based software engineering (CBSE), frameworks for reuse such as Draco, and the concept of reproducible research. These techniques were selected because they represent significant strategies for leveraging prior work, domain expertise, and automation in software development. Software families and CBSE illustrate structural and compositional reuse, Draco demonstrates early domain-specific reuse and code generation, while reproducible research highlights the importance of transparency and repeatability in computational work. Together, these approaches inform our emphasis on automating artifact generation and promoting knowledge reuse across the software lifecycle.

2.2.1 Software/Program Families

Software/program families, or software product lines, refer to a group of related systems sharing a common set of features, functionality, and design [46]. These systems are typically designed to serve a specific domain or market, and are often developed using a common set of core technologies and design principles.

A central concept in software/program families is the distinction between *commonalities* and *variabilities* [5, 46]. *Commonalities* refer to the features, components, or design decisions shared by all members of the family, forming the stable core that defines the family’s identity. In contrast, *variabilities* capture the aspects that can

differ among family members, enabling customization or adaptation to specific requirements or contexts. These variabilities are typically managed through explicit *parameters of variation*, which may include configuration options, feature selections, or architectural choices. Identifying and systematically managing these parameters is essential for effective reuse and automation in software product lines, as it allows developers to generate tailored systems while maintaining consistency and reducing duplication.

One well-known example of a software family is the Microsoft Office suite. Each program in the suite is used for a specific application (word processing, spreadsheet management, presentation tools, etc), yet they all have similar design principles and user interface features. A user who understands how to use one of the software family members will have an intuitive sense of how to use the others thanks to the common design features.

Another, much larger scale, software family is that of the GNU/Linux operating system (OS) and its various distributions[3]. There are many variations on the OS depending on the user's needs. For an everyday computer desktop experience, there are general purpose distributions (Ubuntu, Linux Mint, Fedora, Red Hat, etc.). For server/data center applications, there are specialized server distributions (Ubuntu Server, Fedora Server, Debian, etc.). For systems running on embedded hardware there are lightweight, specialized, embedded distributions built for specific architecture (Armbian, RaspbianOS, RedSleeve, etc.). There are even specialized distributions that are meant to be run without persistent storage for specific applications like penetration/network testing (Kali Linux, Backbox, Parrot Security OS, BlackArch, etc.). However, if you are familiar with one particular flavour of Linux, you'll likely be

comfortable moving between several of the distributions built upon the same cores. You may even be familiar moving to other *NIX based systems like MacOS/Unix.

[Add a blurb here about spinning up a kernel through parameterized options and how that can create a fully customized family member. —DS]

Software/program families can provide a range of benefits to both developers and end-users. For developers, software families can increase productivity by providing a reusable set of core technologies and design principles [46]. This can help reduce the time and effort required to develop new systems, and improve the quality and consistency of the resulting software. For end-users, program families provide a range of usability and functional benefits. Common features and UI elements improve the user experience by making it easier for users to learn and use multiple systems [5]. By also providing a range of related applications, such as those provided by the Microsoft office suite, software families help meet users' needs across a wider range of domains.

2.2.2 Software Reuse

Reusing software is an ideal means of reducing costs. If we can avoid spending time developing new software and instead use some existing application (or a part therein), we save time and money. There have been many proposals on ways to encourage software reuse, each with their own merits and drawbacks.

Component based software engineering (CBSE) is one such example. CBSE is an approach to software development that emphasizes using pre-built software components as the building blocks of larger systems. These reusable components can be developed independently and tested in isolation before being integrated into the larger system software.

A key benefit of CBSE is it can help to reduce the cost and time required for software development, since developers do not need to implement everything from scratch. Additionally, CBSE can improve the quality and reliability of software systems, since components have typically been thoroughly tested and previously used in other contexts.

One CBSE framework is the Common Object Request Broker Architecture (CORBA), which provides a standardized mechanism for components to communicate with each other across a network. CORBA defines a set of interfaces and protocols that allow components written in different programming languages to interact with each other in a distributed environment [36].

Another CBSE framework is the JavaBeans Component Architecture. It is a standard for creating reusable software components in Java. JavaBeans are self-contained software modules with a well-defined interface that can be easily integrated into a variety of development environments and combined to form larger applications [38].

The largest challenge of CBSE is ensuring components are compatible with others and can be integrated into a larger system without conflicts or errors. In an effort to address this challenge, numerous approaches to component compatibility testing have been proposed [64]. This limitation motivates our approach to explore reuse beyond mere composition, focusing instead on capturing and reusing domain knowledge to generate consistent, compatible artifacts.

Others have attempted to provide frameworks for reuse as well. For example, Neighbors [33, 34, 35] proposed a method for engineering reusable software systems known as “Draco”. Draco was among the earliest frameworks for software reuse, particularly in the context of domain-specific software generation, though its direct

influence was more limited compared to later approaches. The core idea behind Draco is to construct software systems by composing reusable, domain-specific components called “domain knowledge modules”. Unlike CBSE, which focuses on assembling systems from independently developed, interchangeable components, Draco emphasizes the reuse of domain-specific knowledge and automated code generation through configurable transformations.

In Draco, a software engineer first defines a domain model, capturing the essential abstractions and operations relevant to a particular problem area. The system then provides a set of reusable transformations and code generation templates tailored to that domain. By selecting and configuring these modules, developers can automatically generate significant portions of a software system, reducing manual coding effort. While Draco itself is not widely used, its approach anticipated later developments in software product lines and generative programming by emphasizing the explicit capture and reuse of domain knowledge [9, 34]. We draw inspiration from Draco’s focus on domain knowledge reuse and automated artifact generation in our own approach.

2.2.3 Reproducible Research

Being able to reproduce results, is fundamental to the idea of good science. When it comes to software projects, there are often many undocumented assumptions or modifications (including hacks) involved in the finished product. This can make replication impossible without the help of the original author, and in some cases reveal errors in the original author’s work [17].

Reproducible research has been used to mean embedding executable code in research papers (example: using Jupyter Notebooks [20]) to allow readers to reproduce

the results described [50]. [In contrast to stuff covered in the problem statement Drasil is doing X —DS] We have designed Drasil to go beyond this narrow definition of reproducible research by emphasizing the generation of multiple, consistent artifacts from a single knowledge base. This approach not only facilitates the reproduction of computational results but also ensures that all related documentation, code, and tests remain synchronized and up-to-date, thereby enhancing the overall reproducibility and reliability of scientific software.

Combining research reports with relevant code, data, etc. is not necessarily easy, especially when dealing with the publication versions of an author’s work. As such, the idea of *compendia* were introduced [14] to provide a means of encapsulating the full scope of the work. Compendia allow readers to see computational details, as well as re-run computations performed by the author. Gentleman and Lang proposed that compendia should be used for peer review and distribution of scientific work [14].

Currently, several tools have been developed for reproducible research including, but not limited to, Sweave [27], SASweave [29], Statweave [28], Scribble [12], and Orgmode [50]. The most popular of those being Sweave [50]. The aforementioned tools maintain a focus on code and certain computational details. Sweave, specifically, allows for embedding code into a document that is run as the document is being typeset so that up to date results are always included. However, Sweave (along with many other tools), still maintains a focus on producing a single, linear document. Drasil, on the other hand, focuses on producing multiple, interrelated artifacts from a single knowledge base, providing a breadth of output options and ensuring consistency and reproducibility across all generated outputs.

2.3 Literate Approaches to Software Development

There have been several approaches attempting to combine development of program code with documentation. Literate Programming and literate software are two such approaches that have influenced the direction of this thesis. Each of these approaches is outlined in the following sections.

2.3.1 Literate Programming

Literate Programming (LP) is a method for writing software introduced by Knuth that focuses on explaining to a human what we want a computer to do rather than simply writing a set of instructions for the computer on how to perform the task [25].

Developing literate programs involves breaking algorithms down into *chunks* [18] or *sections* [25] which are small and easily understandable. The chunks are ordered to follow a “psychological order” [45] if you will, that promotes understanding. They do not have to be written in the same order that a computer would read them. It should also be noted that in a literate program, the code and documentation are kept together in one source. To extract runnable code, a process known as *tangle* must be performed on the source. A similar process known as *weave* is used to extract and typeset the documentation.

There are many advantages to LP beyond understandability. As a program is developed and updated, the documentation surrounding the source code is more likely to be updated simultaneously. It has been experimentally found that using LP ends up with more consistent documentation and code [52]. There are many downsides to having inconsistent documentation while developing or maintaining code [26, 63],

while the benefits of consistent documentation are numerous [16, 26] including improved maintainability and ease of future modifications, reduced risk of introducing errors during updates, enhanced onboarding and training for new team members, better communication among stakeholders, and increased confidence in overall software quality. With the advantages and disadvantages of good documentation in mind we see that more effective, maintainable code can be produced if properly using LP [45].

Regardless of the benefits of LP, it has not been very popular with developers [52]. However, there are several successful examples of LP’s use in SC. Two such literate programs that come to mind are VNODE-LP [32] and “Physically Based Rendering: From Theory to Implementation” [44] a literate program and textbook on the subject matter. Shum and Cook [52] discuss the main issues behind LP’s lack of popularity which can be summed up as dependency on a particular output language or text processor, and the lack of flexibility on what should be presented or suppressed in the output.

There are several other factors which contribute to LP’s lack of popularity and slow adoption thus far. While LP allows a developer to write their code and its documentation simultaneously, that documentation is comprised of a single artifact which may not cover the same material as standard artifacts software engineers expect (see Section 2.1 for more details). LP also does not simplify the development process: documentation and code are written as usual, and there is the additional effort of re-ordering the chunks. It is also challenging to use LP when requirements and design choices change, as those changes may affect many different chunks. That is not to say LP does not have benefits, the LP development process allows developers to follow a more natural flow in development by writing chunks in whichever order they wish,

keep the documentation and code updated simultaneously (in theory) because of their co-location, and automatically incorporate code chunks into the documentation to reduce some information duplication.

There have been many attempts to increase LP’s popularity by focusing on changing the output language or removing the text processor dependency. Several new tools such as CWeb (for the C language), DOC++ (for C++), noweb (programming language independent), and others were developed. Other tools such as javadoc (for Java) and Doxygen (for multiple languages) were also influenced by LP, but differ in that they are merely document extraction tools. They do not contain the chunking features which allow for re-ordering algorithms.

With new tools came new features including, but not limited to, phantom abstracting [52], a “What You See Is What You Get” (WYSIWYG) editor [13], and even movement away from the “one source” idea [53].

While LP is still not mainstream [47], these more robust tools helped drive the understanding behind what exactly LP tools must do. In certain domains LP is becoming more standardized, for example: Agda, Haskell, and R support LP to some extent, even though it is not yet common practice. R has good tool support, with the most popular being Sweave [27], however it is designed to dynamically create up-to-date reports or manuals by running embedded code as opposed to being used as part of the software development process.

Drasil draws inspiration from LP in its emphasis on integrating documentation and code, and in its goal of improving the understandability and maintainability of scientific software. Drasil differs from LP in several important ways. First, while LP focuses on producing a single, human-readable artifact that interleaves code and

documentation, Drasil is designed to generate multiple, distinct artifacts from a single, unified knowledge base. This approach addresses the needs of software engineering processes that require a variety of artifacts for different stakeholders and purposes, going beyond the single-document focus of traditional LP.

Second, Drasil explicitly models domain knowledge, enabling automated consistency across all generated artifacts. In contrast, LP relies on manual synchronization between code and documentation. Drasil’s knowledge-centric approach facilitates reuse, traceability, and automated updates, making it better suited for environments where change is frequent and artifact consistency is critical. Drasil can be seen as a generalization and extension of LP, retaining its core strengths while addressing its limitations.

2.3.2 Literate Software

A combination of LP and Box Structure [31] was proposed as a new method called “Literate Software Development” (LSD) [2]. Box structure can be summarized as the idea of different views that are abstractions that communicate the same information in different levels of detail, for different purposes. Box structures consist of black box, state machine, and clear box structures. The black box gives an external (user) view of the system and consists of stimuli and responses; the state machine makes the state data of the system visible (it defines the data stored between stimuli); and the clear box gives an internal (designer’s) view describing how data are processed, typically referring to smaller black boxes [31]. These three structures can be nested as many times as necessary to describe a system.

LSD was developed with the intent to overcome the disadvantages of both LP

and box structure. It was intended to overcome LP’s inability to specify interfaces between modules, the inability to decompose boxes and implement the design created by box structures, as well as the lack of tools to support box structure [10].

The framework developed for LSD, “WebBox”, expanded LP and box structures in a variety of ways. It included new chunk types, the ability to refine chunks, the ability to specify interfaces and communication between boxes, and the ability to decompose boxes at any level. However, literate software (and LSD) remains primarily code-focused with very little support for creating other software artifacts, in much the same way as LP. Drasil aims to extend the principles of LSD by generating a comprehensive suite of software artifacts from a single knowledge base. This broader scope addresses the needs of modern software engineering practices, which require diverse artifacts for various stakeholders, thereby enhancing the utility and applicability of the literate approach and making it better suited for environments where artifact diversity and traceability are essential.

2.4 Generative Programming

Generative programming is an approach to software development that focuses on automating the process of generating code from high-level specifications [9, 62]. By writing a program specification and feeding it to the generator, one does not have to manually implement the desired program.

One of the primary benefits of generative programming is that it can help to increase productivity and reduce the time and effort required to develop software [9]. By automating the generation of code, developers can focus on high-level design and specification, rather than low-level implementation details.

Generative programming has the added benefit of helping to improve the quality of software by reducing the risk of errors and inconsistencies [9, 62]. Since the code is generated automatically from high-level specifications, there is less room for human error, and the generated code is typically more consistent and predictable.

There are also some potential drawbacks to generative programming. For instance, the generated code may not always be optimal or efficient [9, 62]. As the code is generated automatically, it may not take into account all of the nuance or complexity of the underlying system potentially leading to suboptimal performance or other issues.

Generative programming also requires a significant upfront investment in time and effort to develop the generators and other tools needed to automate the process of code generation [9, 62]. This means it is often not worth the effort to use generative programming for one-off projects.

There are a large number of generative programming tools available today. Some, like template metaprogramming (TMP) tools, are built into a number of programming languages (C++, Rust, Scala, Java, Go, Python, etc.) and offer varying levels of support for generative programming. Others are widely used across a multitude of disparate domains. For example, **ANTLR** (Another Tool for Language Recognition) automatically generates parsers and lexers from grammar specifications, enabling rapid development of compilers and interpreters [42]. **YACC** (Yet Another Compiler Compiler) and its derivatives serve a similar purpose for C-based toolchains [19]. In web development, frameworks like **Yeoman** scaffold entire web applications by generating boilerplate code, configuration files, and project structure [4]. **Swagger Codegen** and the **OpenAPI Generator** produce client libraries, server stubs, and

API documentation from OpenAPI specifications, streamlining the development of RESTful services [37, 54]. In embedded systems, tools such as **Simulink Coder** generate C/C++ code from graphical models, facilitating deployment on hardware targets [8]. These examples illustrate the breadth of generative programming, from language processing to web and embedded systems, and demonstrate how code generators can accelerate development and improve consistency across software projects.

However, current generators still tend to focus primarily on the code, but the same techniques can be used to generate other software artifacts as well. Drasil aims to utilize code generation to produce a comprehensive suite of software artifacts, from a single knowledge base. This holistic approach ensures consistency across all artifacts and addresses the diverse needs of modern software engineering practices.

[Not using LLMs or AI models for generation —DS]

[cover meta-programming here as well -j broad strokes. We rely on GOOL so in many ways it doesn't matter how much I understand about code gen. According to Jacques -j can pass the buck. —DS]

[Add section for model driven engineering explaining that we did not look into it to avoid biasing our approach (practical example-driven requirements for our model) —DS]

Chapter 3

A look under the hood:

Our process

This chapter examines the process by which we developed the Drasil framework itself. Rather than describing the general software development process that Drasil is intended to support or automate, we turn our attention inward to examine the steps, decisions, and rationale that guided its creation. Our goal is to provide transparency into the reasoning that shaped Drasil, highlighting the challenges encountered and the strategies employed to address them. By doing so, we aim to give the reader a clear understanding of how the framework’s design emerged from a careful analysis of redundancy and knowledge organization across software artifacts.

As one of the central motivations for developing Drasil was the observation that softifacts contain significant, unnecessary redundancy, we begin by motivating our approach through concrete examples of redundancy in softifacts. We then describe how we selected and analyzed a set of representative case studies, breaking down their artifacts to identify recurring patterns and organizational structures. This analysis

leads to a general methodology for knowledge capture and projection, which ultimately shaped the core principles of Drasil. Throughout, we emphasize the interplay between practical experience and theoretical insight, illustrating how each informed our process and contributed to the framework’s evolution.

3.1 A minimal motivating example: HGHC

Before delving into the full complexity of our case studies and artifact templates, it is helpful to illustrate the core problem of knowledge redundancy with a minimal, toy example. For this, we use the “HGHC” system, a simple heat transfer case study included in the Drasil repository.

The HGHC example’s SRS specifies what problem the software is trying to solve. In this case, it should model heat transfer coefficients in nuclear fuel rods, specifically h_g (gap conductance) and h_c (effective heat transfer coefficient between clad and coolant). Even in this minimal system, the same core knowledge appears in multiple places within the SRS.

To illustrate, consider the following: The SRS defines variables such as h_g , h_c , τ_c (clad thickness), h_p (initial gap film conductance), h_b (initial coolant film conductance), and k_c (clad conductivity). The defining equations for h_g and h_c are also given as:

$$h_g = \frac{2k_ch_p}{2k_c + \tau_ch_p}$$

$$h_c = \frac{2k_ch_b}{2k_c + \tau_ch_b}$$

We can already see repeated knowledge (k_c and τ_c) in these two equations. The redundancy goes even deeper in the full SRS (which can be found in Appendix A).

In the given SRS we can see repeated descriptions and units for h_g and h_c in the table of symbols and the data definitions. We also observe something more troubling: missing symbols. None of k_c , h_p , h_b , or τ_c have units associated with them nor are they defined in the table of symbols, despite their presence in the equations and data definitions for h_g and h_c .

As this is a toy example, it may seem contrived to have missing information. However we ran into exactly these sorts of issues with our real-world case studies. In this example, while we can easily identify and rectify omissions, it highlights a key issue: even in a simple system, knowledge is repeated across different sections of a single softifact, and critical definitions can be overlooked. Once we add more complexity, with multiple softifacts and larger systems, these issues compound.

Imagine now that we wish to validate the units of both h_g and h_c . The knowledge required for this validation is currently missing. Suppose we add the appropriate symbol definitions and units to the SRS; if we then discover an inconsistency in the units due to an error in the defining equations, we would need to update multiple sections of the SRS to correct it, increasing the risk of further mistakes. If additional softifacts, particularly code, have already been created based on these definitions, the challenge of maintaining consistency becomes even greater. Corrections would now be required across multiple input languages (for example, $\text{\LaTeX}$ and Python), significantly increasing the verification burden. Ultimately, these softifacts are communicating much of the same knowledge (i.e. the definitions of h_g and h_c), merely formatted for different audiences. In the case of the SRS and the source code, the audiences are human stakeholders and computers, respectively.

This small example foreshadows the larger patterns of redundancy and risks of

inconsistency that we observe in more complex systems. The rest of this chapter generalizes from such examples, showing how we systematically identified, categorized, and addressed these issues in the development of Drasil.

3.2 A (very) brief introduction to our case study systems

To ground our analysis in reality and move beyond abstract principles, this thesis draws on a set of detailed case studies. These case studies serve not only to illustrate the prevalence and impact of redundant knowledge in softifacts, but also to provide a concrete basis for systematically identifying, comparing, and generalizing patterns of redundancy and knowledge organization. By examining systems developed using common artifact templates, we are able to motivate the need for the Drasil framework and inform its design.

To simplify the process of identifying redundancies and patterns, we have chosen several case studies developed using common artifact templates, specifically those used by Smith et al. [\[source? —DS\]](#) Also, as previously mentioned in Section 1.3, our case study software systems follow the ‘*input*’ $\rightarrow$ ‘*process*’ $\rightarrow$ ‘*output*’ pattern. These systems cover a variety of use cases, to help avoid over-specializing into one particular system type.

The majority of the aforementioned case studies were developed at McMaster University to address real engineering problems outside of our research group. Table 3.1 provides a high-level reference to each case study. For the specifics of each system, all relevant case study artifacts can be found in the GitHub repository.

Table 3.1: Summary of the case study systems

Case study	Problem
GlassBR	Predict whether a glass slab can withstand a blast under specified conditions.
SWHS	Investigate how phase change material (PCM) affects heating and energy storage in a solar water heating tank.
NoPCM	Investigate heating and energy storage in a solar water heating tank without phase change material.
SSP	Evaluate slope stability by identifying the critical slip surface and determining the associated factor of safety and interslice forces.
Projectile	Predict the landing position of a projectile.
GamePhysics	Provide a physics library that simulates objects under varied physical conditions efficiently enough for soft real-time use in games.

The NoPCM case study was created as a software family member for the SWHS case study. It was manually written, removing all references to PCM and thus re-modeling the system.

The Projectile case study, was the first example of a system created solely in Drasil, i.e. we did not have a manually created version to compare and contrast with through development. As such, it will not be referenced often since it did not inform Drasil’s design or development until much further in our process. The Projectile case study was created post-facto to provide a simple, understandable example for a general audience as it requires, at most, a high-school level understanding of physics.

After carefully selecting our case studies, we went about a practical approach to find and remove redundancies. The first step was to break down each artifact type and understand exactly what they are trying to convey.

3.3 Breaking down softifacts

To meaningfully address redundancy in software artifacts, we must understand the purpose, content, and audience of each artifact—not just observe repetition. Building on the foundation provided by our case studies, this section examines the structure and role of each major softifact¹ in our process. By breaking down these artifacts, we aim to reveal both their commonalities and their differences, setting the stage for identifying patterns of knowledge organization and redundancy that inform the design of Drasil.

As noted earlier, for our approach to work we must understand exactly what each

¹For brevity, we focus our analysis on the SRS, Module Guide, and Source Code, as these artifacts are both representative and sufficiently distinct to illustrate the patterns of redundancy and knowledge organization central to our argument.

of our softifacts are trying to say and to whom.² By selecting our case studies from those developed using common artifact templates, we have given ourselves a head start on that process. The following subsections present a brief sampling of our process of breaking down softifacts, using the GlassBR case study as a running, motivating example to ground the analysis, acknowledging that a comprehensive overview would be excessively lengthy.

3.3.1 SRS

To start, we look at the Software Requirements Specification (SRS). The SRS (or some incarnation of it) is one of the most important artifacts for any software project as it specifies what problem the software is trying to solve. There are many ways to state this problem, and the template from Smith et al. has given us a strong starting point. Figure 3.1 shows the table of contents for an SRS using the Smith et al. template.

With the structure of the document in mind, let us look at several of our case studies' SRS documents to get a deeper understanding of what each section truly represents. Figure 3.2 shows the reference section of the SRS for GlassBR. Each of the case studies' SRS contains a similar section so for brevity we will omit the others here, but they can be found in the GitHub repository. We will try to ignore any superficial differences (spelling, grammar, phrasing, etc.) in each of the case studies while we look for commonality. We are also trying to determine how the non-superficial differences relate to the document template, general problem domain, and specific system information.

²Refer to Section 2.1 for a general summary of softifacts.

1. Reference Material
 - 1.1. Table of Units
 - 1.2. Table of Symbols
 - 1.3. Abbreviations and Acronyms
2. Introduction
 - 2.1. Purpose of Document
 - 2.2. Scope of Requirements
 - 2.3. Characteristics of Intended Reader
 - 2.4. Organization of Document
3. Stakeholders
 - 3.1. The Customer
 - 3.2. The Client
4. General System Description
 - 4.1. System Context
 - 4.2. User Characteristics
 - 4.3. System Constraints
5. Specific System Description
 - 5.1. Problem Description
 - 5.1.1. Physical System Description
 - 5.1.2. Goal Statements
 - 5.2. Solution Characteristics Specification
 - 5.2.1. Assumptions
 - 5.2.2. Theoretical Models
 - 5.2.3. General Definitions
 - 5.2.4. Data Definitions
 - 5.2.5. Instance Models
 - 5.2.6. Data Constraints
 - 5.2.7. Properties of a Correct Solution
6. Requirements
 - 6.1. Functional Requirements
 - 6.2. Non-Functional Requirements
7. Likely Changes
8. Unlikely Changes
9. Traceability Matrices and Graphs
10. Values of Auxiliary Constants
11. References
12. Appendix

Figure 3.1: The Table of Contents from the expanded Smith et al. template

1.1 Table of Units

The unit system used throughout is SI (Système International d’Unités). In addition to the basic units, several derived units are also used. For each unit, the **Table of Units** lists the symbol, a description and the SI name.

Symbol	Description	SI Name
kg	mass	kilogram
m	length	metre
N	force	newton
Pa	pressure	pascal
s	time	second

(a) Table of Units Section

[The figure with subfigures might be a hassle to keep co-located, might split it up into 3 separate figures and just refer to them collectively —DS]

Looking at the (truncated for space) Table of Symbols, Table of Units, and Table of Abbreviations and Acronyms sections (Figure 3.2) we can see that, barring the table values themselves, they are almost identical. The Table of Symbols is simply a table of values, akin to a glossary, specific to the symbols that appear throughout the rest of the document. For each of those symbols, we see the symbol itself, a brief description of what that symbol represents, and the units it is measured in, if applicable. Similarly, the Table of Units lists the Système International d’Unités (SI) Units used throughout the document, their descriptions, and the SI name. Finally, the table of Abbreviations and Acronyms lists the abbreviations and their full forms, which are essentially the symbols and their descriptions for each of the abbreviations.

While the reference material section should be fairly self-explanatory as to what it contains, other sections and subsections may not be so clear from their name alone. For example, it may not be clear offhand of what constitutes a theoretical model compared to a data definition or an instance model. One may argue that the author

1.2 Table of Symbols

The symbols used in this document are summarized in the [Table of Symbols](#) along with their units. The symbols are listed in alphabetical order.

Symbol	Description	Units
a	Plate length (long dimension)	m
AR	Aspect ratio	—
$AR_{\max}$	Maximum aspect ratio	—
B	Risk of failure	—
b	Plate width (short dimension)	m
$capacity$	Capacity or load resistance	Pa
$d_{\max}$	Maximum value for one of the dimensions of the glass plate	m
$d_{\min}$	Minimum value for one of the dimensions of the glass plate	m
E	Modulus of elasticity of glass	Pa

(b) Table of Symbols (truncated) Section

1.3 Abbreviations and Acronyms

Abbreviation	Full Form
A	Assumption
AN	Annealed
AR	Aspect Ratio
DD	Data Definition
FT	Fully Tempered
GS	Goal Statement
GTF	Glass Type Factor
HS	Heat Strengthened
IG	Insulating Glass
IM	Instance Model
LC	Likely Change
LDF	Load Duration Factor
LG	Laminated Glass

(c) Table of Abbreviations and Acronyms (truncated) Section

Figure 3.2: The reference sections of GlassBR

of the SRS, particularly if they chose to use the Smith et al. template, would need to understand that difference. However, it is not clear whether the intended audience would also have such an understanding. Who is that audience? Refer to Section 2.1, for more details. A brief summary is available in Table 2.2.

Returning to our exercise of breaking down each section of the SRS to determine the subtleties of *what* is contained therein³ it should be unsurprising that each section maps to the definition provided in the Smith et al. template. However, the kinds of information contained in those sections differ substantially. Some material is generic boilerplate, such as the introductory text describing the purpose and organization of each section of the SRS. Some is clearly system-specific, such as GlassBR’s goal statements, its instance models, and its data definitions for quantities like demand, stress distribution factor, and probability of breakage. Between those extremes is material that belongs to the problem domain more than to the individual software system. In GlassBR, for example, the theoretical background concerning glass mechanics, load, and resistance is not unique to that one program; rather, it is domain knowledge that the SRS uses to derive the system-specific models, constraints, and calculations presented later in the document.

Observing the contents of an SRS template adhere to said template may seem mundane, but it is a necessary step before we can move on to other softifacts. Without understanding what the SRS template intends to convey it is hard to assess whether or not the case study SRS conveys that information. With that in mind, we can move on to the MG and source code.

³The breakdown details are omitted for brevity and due to their monotonous nature, although the overall process is very much akin to the breakdown of the Reference Material section.

3.3.2 Module Guide

The module guide (MG) is a softifact that details the architecture of a given software system. It holds a number of design decisions around sensibly grouping functionalities within the system into modules to fulfill the requirements laid out in the SRS. For example, one might have an input/output module for handling user input and giving the user feedback through the display (i.e. via print commands or some other output), or a calculations module that contains all of the calculation functions being performed in the normal operation of the given software system. The Smith et al. MG template also includes a traceability matrix for ease of verifying which requirements are fulfilled by which modules. Finally, the MG includes considerations for anticipated or unlikely changes that the system may undergo during its lifecycle.

Figure 3.3 shows the table of contents for the GlassBR case study’s MG. Again, for the sake of brevity we will omit the other case studies here, even though we engaged in the same breakdown process for each of them. Just as with the SRS we are looking for commonality and understanding of what the document is trying to portray to the reader. As such we will ignore superficial differences between the MG sections. As the MG is a fairly short document we will look at each of the most relevant sections as part of this exercise.

Breaking down the MG by section, we can see that the introduction is itself completely generic boilerplate explaining the purpose of the MG, the audience, and some references to other works that explain why we would make certain choices over other (reasonable) ones given the opportunity. There is nothing system-specific, nor specific to the given problem domain of the case study.

Following through the table of contents into the “Anticipated and Unlikely Changes”

Module Guide for GlassBR

Spencer Smith and Thulasi Jegatheesan

July 25, 2018

Contents

1	Introduction	2
2	Anticipated and Unlikely Changes	3
2.1	Anticipated Changes	3
2.2	Unlikely Changes	4
3	Module Hierarchy	4
4	Connection Between Requirements and Design	5
5	Module Decomposition	5
5.1	Hardware Hiding Modules (M1)	5
5.2	Behaviour-Hiding Module	6
5.2.1	Input (M2)	6
5.2.2	LoadASTM (M3)	7
5.2.3	Output Module (M4)	7
5.2.4	Calc Module (M5)	7
5.2.5	Control Module (M6)	7
5.2.6	Constants Module (M7)	7
5.2.7	GlassTypeADT Module (M8)	8
5.2.8	ThicknessADT Module (M9)	8
5.3	Software Decision Module	8
5.3.1	FunctADT Module (M10)	8
5.3.2	ContoursADT Module (M11)	8
5.3.3	SeqServices Module (M12)	9
6	Traceability Matrix	9
7	Use Hierarchy Between Modules	10
8	Bibliography	10

Figure 3.3: Table of Contents for GlassBR Module Guide

5.2.4 Calc Module (M5)

Secrets: The equations for predicting the probability of glass breakage, capacity, and demand, using the input parameters.

Services: Defines the equations for solving for the probability of glass breakage, demand, and capacity using the parameters in the input parameters module.

Implemented By: GlassBR

Figure 3.4: Calc Module from the GlassBR Module Guide

section, we see that again the introductions to this section and its subsections are generic boilerplate, however the details of each section are not. Both subsections are written in the same way: as a list of labeled changes (AC# for anticipated change, UC# for unlikely change). This is the first place we see both problem-domain and system-specific information. Interestingly, the Module Hierarchy section follows the same general style: it is a list of modules which represent the leaves of the module hierarchy tree and each one is labeled (M#).

Skipping ahead to the module decomposition, we find a section heading for each Level 1 module in the hierarchy, followed by subsections describing the Level 2 modules. The former are almost entirely generic boilerplate (for example common Level 1 modules include: Hardware-Hiding, Behaviour-Hiding, and Software Decision modules), but the latter are problem-domain or system specific. An example of a system-specific module is shown in Figure 3.4.

Each module is described by its secrets, services, and what it will be implemented by. For example, a given module could be implemented by the operating system (OS), the system being described (ex. GlassBR), or a third party system/library that will inter-operate with the given system.

Finally we have a traceability matrix and use hierarchy diagram. Both are visual representations of how the different modules implement the requirements and use each other respectively. The traceability matrix provides a direct and obvious link between the SRS and MG, where other connections between the two softifacts have been implicit until this point. Generally, the next softifact would be the MIS, however as it is structured so similarly to the MG (one section per module, each section organized in a very similar way, a repeated use hierarchy, etc) we will skip it for brevity. The MIS includes novel system-specific, implementation-level information denoting the interfaces between modules, but for our current exercise does not provide any revelations beyond that of the MG.

The MG gives us a very clear picture of *decisions* made by the system designers, as opposed to the knowledge of the system domain, problem being solved, and requirements of an acceptable solution provided in the SRS. The MG provides platform and implementation-specific decisions, which will eventually be translated into implementation details in the source code. With that in mind, let us move on to dissecting the source code.

3.3.3 Source Code

The source code is arguably the most important softifact in any given software system since it serves as the set of instructions that a computer executes in order to solve the given problem. With only the other softifacts and without the source code, we would have a very well defined problem and acceptance criteria for a possible solution, but would never actually solve the problem.

As the source code is the executable set of instructions, one would expect it to

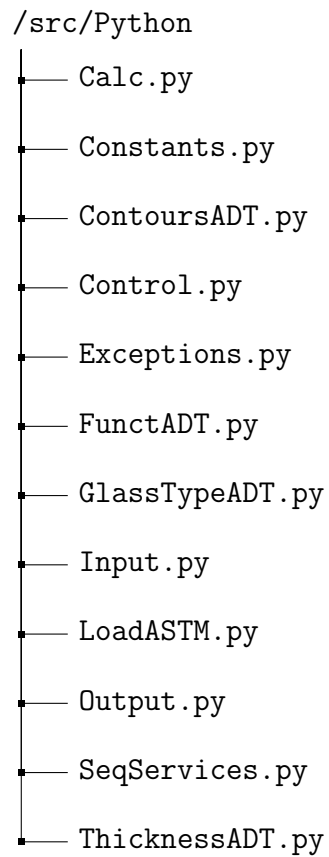


Figure 3.5: Python source code directory structure for GlassBR

be almost entirely system and problem-domain specific with very little boilerplate. Looking into the source of our case studies, we find this to be mostly true barring the most generic of library use (ex. `stdio` in C).

Returning to our example of the MG from GlassBR (Figure 3.3) and comparing it to the python source code structure shown in Figure 3.5 we can see that the source code follows almost identically in structure to the module decomposition. The only difference being the existence of an exceptions module defining the different types of exceptions that may be thrown by the other modules. While this can be considered a fairly trivial difference, likely made for ease of maintenance, readability, and extensibility, it highlights that **the two softifacts are out of sync**. We speculate this difference was caused by a change made during the implementation phase, wherein the MG was not updated to reflect the addition of an exceptions module.

Let us look deeper into the code for one specific module, for example the Calc module introduced in the MG (Figure 3.4). The source code for said module can be found in Figure 3.6. In the source code we see a number of calculation functions, including those that calculate the probability of glass breakage, demand (also known as *load* or *q*), and capacity (also known as *load resistance* or *LR*) as outlined in the *secrets* section of the Calc module definition in the MG. We also see a number of intermediary calculation functions required to calculate these values (for example `calc_NFL` and its dependencies).

The source code provides clear instructions to the machine on how to calculate each of these values and their intermediaries; it provides the actionable steps to solve the given problem. When we compare the code with relevant sections of the SRS, specifically the Data Definitions (DDs) for each term, we can see a very obvious

```

5 from Input import *
6 from ContoursADT import *
7 from math import log, exp
8
9 ## @brief Calculates the Dimensionless load
10 # @return the unitless load
11 def calc_q_hat( q, params ):
12     upper = q * (params.a * params.b) ** 2
13     lower = params.E * (params.h ** 4) * params.GTF
14     return ( upper / lower )
15 ## @brief Calculates the Stress distribution factor based on Pbtol
16 # @return the unitless stress distribution factor
17 def calc_J_tol( params ):
18     upper1 = 1
19     lower1 = 1 - params.Pbtol
20
21     upper2 = (params.a * params.b) ** (params.m - 1)
22     lower2 = params.k * ((params.E * (params.h ** 2)) ** (params.m)) * params.LDF
23
24     return (log( (log(upper1 / lower1)) * (upper2 / lower2) ))
25 ## @brief Calculates the Probability of glass breakage
26 # @return unitless probability of breakage
27 def calc_Pb( B ):
28     output = 1 - (exp(-B))
29     if not (0 < output < 1):
30         raise InvalidOutput("Invalid output!")
31     return (output)
32 ## @brief Calculates the Risk of failure
33 # @return unitless risk of failure
34 def calc_B( J, params ):
35     upper = params.k * ((params.E * (params.h) ** 2) ** params.m) * params.LDF * exp(
36         J)
37     lower = ((params.a * params.b) ** (params.m - 1))
38     return ( upper / lower )
39 ## @brief Calculates the Non-factored load
40 # @return unitless non-factored load
41 def calc_NFL( q_tol, params ):
42     upper = q_tol * params.E * (params.h ** 4)
43     lower = (params.a * params.b) ** 2
44     return ( upper / lower )
45 ## @brief Calculates the Load resistance
46 # @return unitless load resistance
47 def calc_LR( NFL, params ):
48     return ( NFL * params.GTF * params.LSF )
49 ## @brief Calculates Safetey constraint 1
50 # @return true if the calculated probability is less than the tolerable probability
51 def calc_is_safePb( Pb, params ):
52     if (Pb < params.Pbtol):
53         return True
54     else:
55         return False
56 ## @brief Calculates Safetey constraint 2
57 # @return true if the load resistance is greater than the load
58 def calc_is_safeLR( LR, q ):
59     if (LR > q):
60         return True
61     else:
62         return False

```

Figure 3.6: Source code of the Calc.py module for GlassBR

transformation from one form to the other; the symbol used by the DD is the (partial) name of the function in the source code and the equation from the DD is calculated within the source code. This is one of many patterns we see across our softifacts within each case study.

3.3.4 Wrapping up

A close examination of each major software artifact reveals that, while their purposes and audiences differ, they are all fundamentally built upon the same core knowledge. Redundancy is introduced both within and across these artifacts, often as a result of tailoring the same information for different contexts or audiences. Structural similarities emerge across artifacts, highlighting opportunities for systematic knowledge capture and reuse. At the same time, inconsistencies and omissions can easily arise when artifacts are maintained separately, underscoring the risks inherent in manual, disconnected processes. Understanding the intent, content, and audience of each artifact is therefore essential for identifying patterns of redundancy and for informing the design of tools that aim to improve consistency and reduce unnecessary repetition.

By systematically dissecting each artifact type, we not only reveal the sources and forms of redundancy, but also clarify the requirements for any framework (Drasil) that aims to automate or improve the generation of these artifacts. This analysis directly informs the patterns and organizational strategies we discuss in the following sections.

3.4 Identifying Repetitive Redundancy

From the examples in Sections 3.1 and 3.3, we can see a number of simple patterns emerging with respect to organization and information repetition within a case study. Upon applying our process to all of the case studies and adopting a broader perspective, numerous instances emerge where patterns transcend individual case studies and remain universally applicable. Several of these patterns should be unsurprising, as they relate to the template of a particular softifact. It is interesting, however, that patterns of information organization crop up within a given softifact in multiple places, containing distinct information.

Returning to our example from Section 3.3.1, looking only at the reference section of our SRS template, we have already found three subsections that contain the majority of their information in the same organizational structure: a table defining terms with respect to their symbolic representation and general information relevant to those terms. Additionally, we can see that the Table of Units and Table of Symbols have an introductory blurb preceding the tables themselves, whereas the Table of Abbreviations and Acronyms does not. Inspecting across case studies, we observe that the introduction to the Table of Units is nothing more than boilerplate text dropped into each case study verbatim; it is completely generic and applicable to *any* software system using SI units. The introduction to the Table of Symbols also appears to be boilerplate across several examples, however, it does have minor variations. For an easy visual comparison, the earlier GlassBR example in Figure 3.2b can be compared directly with Figure 3.7 (GamePhysics). These variations reveal the obvious: the variability between systems is greater than simply a difference in choice of symbols, and so there is some system-specific knowledge being encoded. While we can intuitively

1.2 Table of Symbols

The table that follows summarizes the symbols used in this document along with their units. More specific instances of these symbols will be described in their respective sections. Throughout the document, symbols in **bold** will represent vectors, and scalars otherwise. The symbols are listed in alphabetical order.

symbol	unit	description
a	m s^{-2}	Acceleration
α	rad s^{-2}	Angular acceleration
C_R	unitless	Coefficient of restitution
F	N	Force
g	m s^{-2}	Gravitational acceleration (9.81 m s^{-2})
G	$\text{m}^3 \text{ kg}^{-1} \text{ s}^{-2}$	Gravitational constant ($6.673 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$)
I	kg m^2	Moment of inertia

Figure 3.7: Table of Symbols (truncated) Section from GamePhysics

infer this conclusion based solely on each system addressing a different problem, our observation of the (structural) patterns within this SRS section confirms it.

The reference section of the SRS provides a lot of knowledge in a very straightforward and organized manner. The basic units provided in the table of units give a prime example of fundamental, global knowledge shared across domains. Nearly any system involving physical quantities will use one or more of these units. On the other hand, the table of symbols provides system/problem-domain specific knowledge that will not be useful across unrelated domains. For example, the stress distribution factor J from GlassBR may appear in several related problems, but would be unlikely to be seen in something like SWHS, NoPCM, or Projectile. Finally, acronyms are very context-dependent. They are often specific to a given domain and, without a coinciding definition, it can be very difficult for even the target audience to understand what they refer to. Within one domain, there may be several acronyms that

Number	DD7
Label	Dimensionless Load ($\hat{q}$)
Equation	$\hat{q} = \frac{q(ab)^2}{Eh^4GTF}$
Description	q is the 3 second equivalent pressure, as given in IM3 a, b are dimensions of the plate, where ($a > b$) E is the modulus of elasticity h is the true thickness, which is based on the nominal thickness as shown in DD2 GTF is the Glass Type Factor, as given by DD6
Source	[2], [8, Eq. 7]
Ref. By	DD4

Figure 3.8: Data Definition for Dimensionless Load ($\hat{q}$) from GlassBR SRS

look identical, but mean different things, for example: PM can refer to a Product Manager, Project Manager, Program Manager, Portfolio Manager, etc.

By continuing to breakdown the SRS and other softifacts, we are able to find many more patterns of knowledge repetition. For example, we see the same concept being introduced in multiple areas within a single artifact and across artifacts in a project. Figure 3.8 shows the data definition for $\hat{q}$ in GlassBR. The same term was previously defined with fewer details in the table of symbols, as well as showing up implicitly or in passing in the MG (Figure 3.4 and the *loadASTM* module respectively), and implemented in the Source Code (Figure 3.6 lines 11-14). A non-exhaustive list of places where $\hat{q}$ appears helps make the repetition more explicit:

- In the SRS table of symbols, where $\hat{q}$ is introduced in abbreviated form.
- In the data definition shown in Figure 3.8, where it is defined more fully.

- In other SRS data definitions, such as the Stress Distribution Factor (J) and Non-Factored Load (NFL), which refer to $\hat{q}$ directly.
- In the SRS calculation of Capacity (LR), where $\hat{q}$ appears indirectly through intermediary quantities such as NFL .
- In the MG, where the *loadASTM* and *Calc* modules mention or rely on the quantity.
- In the source code, where the corresponding calculation is implemented explicitly (Figure 3.6, lines 11-14).

Although the full definition of $\hat{q}$ is initially provided for a human audience only once, it is necessary to reference it in different ways for different audiences. Each audience is expected to grasp the symbol's meaning within their given context or consult other softifacts for more comprehensive understanding. When reading the SRS, the data definitions and other reference materials play a crucial role in swiftly comprehending the complete definition of $\hat{q}$ in relation to the system's inputs, outputs, functional requirements, and acceptance criteria.

The MG, on the other hand, briefly mentions $\hat{q}$ when defining the responsibilities of both the *loadASTM* and *Calc* modules (the former being responsible for loading values from a file, and the latter utilizing those values for calculations), whereas the source code provides a highly detailed definition to ensure accurate execution of the relevant calculation(s).

The varying level of detail across the softifacts should not come as a surprise since each softifact targets a different audience and their specific needs at various stages of the software development process. Although the level of verbosity may

differ, the core information remains consistent: the authors are consistently referring to the definition of $\hat{q}$ via its symbolic representation, regardless of the level of detail incorporated. The goal is to convey relevant aspects of knowledge of a given term, while eliding that which is deemed superfluous, based on the context and the specific requirements of our audience. In other words, the authors only *project* some portion of their knowledge of given terms at a given time, depending on their needs (precision, brevity, clarity, etc.), the expectations of the audience, and contextual relevance.⁴ The audience, on the other hand, engages in *knowledge transformation*, whereby they consume the representation (projected knowledge) and transform it into their own internal representation, based on their personal knowledge-base.

Relying on common representations, eliding definitions, projecting and transforming knowledge are *fundamental to the way humans communicate*. They are readily observable in all forms of communications, whether written or oral, as we assign meaning to given sounds and symbols (words) according to the agreed upon grammar of a given language and use those words (knowledge projections) to simplify communication to a given audience. A context-specific glossary, or more generally a dictionary, is a prime example of a knowledge-base that we use for communication via knowledge projections and transformations. By maintaining a shared vocabulary, we can communicate using the symbolic representations (words) instead of requiring terms to be decomposed (defined) to their most basic form. However, communication of this sort is still imperfect, due to gaps in shared knowledge between participants or misunderstanding of overloaded terms. Interpersonal communications can involve nuance and context-dependent interpretations, yet they still boil down to knowledge

⁴We have only referred to the term as $\hat{q}$ in this section to emphasize our argument and make a meta-argument that the definition is irrelevant to our audience in this example. What matters is the symbolic reference, which we share a common understanding of.

Number	T1
Label	Conservation of thermal energy
Equation	$-\nabla \cdot \mathbf{q} + g = \rho C \frac{\partial T}{\partial t}$
Description	The above equation gives the conservation of energy for transient heat transfer in a material of specific heat capacity C ($\text{J kg}^{-1} \text{°C}^{-1}$) and density ρ (kg m^{-3}), where $\mathbf{q}$ is the thermal flux vector (W m^{-2}), g is the volumetric heat generation (W m^{-3}), T is the temperature (°C), t is time (s), and ∇ is the gradient operator. For this equation to apply, other forms of energy, such as mechanical energy, are assumed to be negligible in the system (A1). In general, the material properties (ρ and C) depend on temperature.
Source	http://www.efunda.com/formulae/heat_transfer/conduction/overview_cond.cfm
Ref. By	GD2

Figure 3.9: Theoretical Model of conservation of thermal energy found in both the SWHS and NoPCM SRS

projection on the part of the communicator and knowledge transformation on the part of the communicatee.⁵ The latter can infer context, or be provided with explicit context, which affirms their use of the appropriate knowledge transformations.

Returning to the context of software systems, if we broaden our view from a single system, to a software family, we can also find patterns of commonality and repeated knowledge across the various softifacts of the family members (For example our SWHS and NoPCM case studies) as they have been developed to solve similar, or in our case nearly identical, problems. Software family members are good examples to help determine what types of information or knowledge provided in the softifacts belong to the system-domain, problem-domain, or are simply general (boilerplate).

Looking at SWHS and NoPCM, we can easily find identical theoretical models (TMs) as the underlying theory for each system is based on the problem domain

⁵All this to say, words are a shorthand for their definitions.

(see example in Figure 3.9). However, when we follow the derivations from the TMs to the Instance Models (IMs), we find the resulting equations have changed due to the context of the system; the lack of PCM has changed the relevant equations for calculating the energy balance on water in the tank as shown in Figure 3.10.

While these examples are fairly small and specific, they are indicative of a larger, more generalizable, set of patterns of knowledge organization and repetition. These patterns are at their core: the use of common knowledge that has been projected through some means, and patterns of organization of those knowledge projections within softifacts. Common knowledge, in this case, refers to one of three categories of knowledge: system-specific, domain-specific, or common to softifacts as a whole. It should also be noted that knowledge projections may include the identity projection (i.e. the full, unabridged definition) as they are dependent on the relevance to the audience of the given softifact. Regardless, the captured knowledge fundamentally underlies these patterns of repetition, and is where we need to focus if we intend to reduce unnecessary redundancy and improve maintainability.

These patterns directly inform the requirements and architecture of Drasil. By understanding them, we reveal challenges and opportunities for automation and consistency in software artifact generation. Recognizing that redundancy arises from projecting the same core knowledge into different forms for different audiences, we see the need for a framework that can capture, organize, and project knowledge in a principled, reusable way. This insight is the conceptual bridge to Drasil: a framework explicitly designed to operationalize these patterns, reduce unnecessary duplication, and ensure consistency across artifacts. The following sections build on this foundation.

Number	IM1
Label	Energy balance on water to find T_W
Input	$m_W, C_W, h_C, A_C, h_P, A_P, t_{\text{final}}, T_C, T_{\text{init}}, T_P(t)$ from IM2 The input is constrained so that $T_{\text{init}} \leq T_C$ (A11)
Output	$T_W(t), 0 \leq t \leq t_{\text{final}}$, such that $\frac{dT_W}{dt} = \frac{1}{\tau_W}[(T_C - T_W(t)) + \eta(T_P(t) - T_W(t))]$, $T_W(0) = T_P(0) = T_{\text{init}}$ (A12) and $T_P(t)$ from IM2
Description	T_W is the water temperature ($^{\circ}\text{C}$). T_P is the PCM temperature ($^{\circ}\text{C}$). T_C is the coil temperature ($^{\circ}\text{C}$). $\tau_W = \frac{m_W C_W}{h_C A_C}$ is a constant (s). $\eta = \frac{h_P A_P}{h_C A_C}$ is a constant (dimensionless). The above equation applies as long as the water is in liquid form, $0 < T_W < 100^{\circ}\text{C}$, where 0°C and 100°C are the melting and boiling points of water, respectively (A14, A19).
Sources	[4]
Ref. By	IM2

(a) SWHS Instance Model for Energy Balance on Water

Number	IM1
Label	Energy balance on water to find T_W
Input	$m_W, C_W, h_C, A_C, t_{\text{final}}, T_C, T_{\text{init}}$ The input is constrained so that $T_{\text{init}} \leq T_C$ (A9)
Output	$T_W(t), 0 \leq t \leq t_{\text{final}}$, such that $\frac{dT_W}{dt} = \frac{1}{\tau_W}(T_C - T_W(t))$, $T_W(0) = T_{\text{init}}$
Description	T_W is the water temperature ($^{\circ}\text{C}$). T_C is the coil temperature ($^{\circ}\text{C}$). $\tau_W = \frac{m_W C_W}{h_C A_C}$ is a constant (s). The above equation applies as long as the water is in liquid form, $0 < T_W < 100^{\circ}\text{C}$, where 0°C and 100°C are the melting and boiling points of water, respectively (A10).
Sources	Original SRS with PCM removed
Ref. By	

(b) NoPCM Instance Model for Energy Balance on Water

Figure 3.10: Instance Model difference between SWHS and NoPCM

3.5 Organizing knowledge - a fluid approach

Recognizing patterns of redundancy and knowledge projection, we are left with a crucial question: how should knowledge itself be captured and organized to support consistent, flexible reuse across artifacts? While our analysis has revealed the importance of knowledge projections and the risks of redundancy, it remains unclear what form the underlying knowledge base should take, or how it should be structured to enable effective projection. In the following section, we turn to this open problem, exploring conceptual approaches to knowledge capture and organization that can lay the groundwork for Drasil’s design.

To this point we have been referring to an intuitive, albeit nebulous concept of *knowledge* without explicitly defining it. Before we can discuss how to organize knowledge, let us first clarify what it is [this is my definition so make that clear! —DS]: knowledge is information that has been structured and contextualized to be meaningful and useful. It encompasses facts, concepts, definitions, theories, and relationships that are relevant to a particular domain or problem. Knowledge can be represented in various forms, such as mathematical equations, natural language descriptions, code snippets, diagrams, and more. The core of what a piece of knowledge is remains consistent, regardless of its representation. This should become more clear in Chapter 4, specifically Section 4.2.1.

For clarity in the following examples, we will use set notation to describe relationships between knowledge and projections. This is not intended as a formalization, but rather as a familiar mathematical juxtaposition to help illustrate the concepts. Here, we use K to denote a set, or in our terminology a *chunk*, of one or more related pieces of information. Each K may represent a single fact, a definition, or a tightly

connected set of information relevant to a particular concept or artifact.

Given the patterns across softifacts we observed in the previous section, we can generalize knowledge projections to their projection functions. Identity projection functions directly repeat knowledge verbatim i.e. $p(K) \equiv K$ for some set of knowledge K . It should be noted in our case that as long as the projection used contains the full definition and context of a piece of knowledge, that projection is considered an identity projection **regardless of changes to notation** (ex. “ $x = y$ ” vs “ $y = x$ ”) or language (ex. “ $x = y$ ” vs “ x is equal to y ” vs “ x est égal à y ”). An identity projection is providing the knowledge chunk in full. Non-identity projections require using representations that elide some details of the knowledge at hand to make it more palatable for the audience of the given softifact i.e. $p(K) \subset K$. Similarly, we can have multivariant projection functions (whether identity or not) which project knowledge from several places into one form, i.e. $p(K, L, M) \subseteq K \cup L \cup M$.

In all cases, we have some fully defined knowledge and then we apply a projection function to retrieve the necessary information for our softifact. We can postulate that organizing and grouping knowledge into some type of structure, with an assortment of projection functions (ex. To look up the full definition in different representations, pull out relevant portions for a given audience, or provide useful abstractions) will allow us to reduce the need for manual duplication and remove unnecessary redundancies, as anything we need to include in our softifacts can be retrieved from a given source with a given projection function.

Keeping in mind that our core knowledge is used across all softifacts via projections, we may naively choose to consolidate all knowledge into a single database. This naive approach works well enough for a limited set of examples, but it quickly

becomes apparent (refer to the PM example from Section 3.4) that context is highly important and the sheer scope of knowledge to be organized may become unwieldy. After breaking down multiple case studies, we believe collecting knowledge chunks into categories based on their domain(s) is a more easily navigable and maintainable approach. This also allows us to keep some context information at a meta-level (ex. Physics knowledge would be categorized into a Physics knowledge-base). Then for any given system, we would likely only need to reference across a handful of contexts (knowledge-bases) relevant to the domain.

There is also knowledge fundamental to all softifacts—it is contextual to the domain of softifact writing itself. This kind of meta-knowledge would be useful to have readily available in its own knowledge-base. The same could be said for things like SI Unit definitions, while they only apply to measuring physical properties, we see some domains built off of physics that operate at a higher level of assumed understanding (ex. Chemistry abstracts some of the physics details, while being directly reliant on them).

Some of the knowledge used in our softifacts is derived from other, more fundamental, knowledge chunks. For example, when using SI units, we may choose to use a derived unit (newton, joule, radian, etc.) that is a better fit for the application domain of the system being documented. While we want to avoid unnecessary redundancy, we can argue that derived units are good candidates for acceptable redundancy. For example, if anywhere we use *Joules* we replace that with the definition ($J = \frac{kg \cdot m^2}{s^2}$) we then run into a problem of context and complexity. Generally, the audience for a given softifact will have an internal representation of context-specific knowledge, so even something as straightforward as changing the units from J to $\frac{kg \cdot m^2}{s^2}$ will put

unnecessary load on said audience and force them to engage in more intensive knowledge transformations, while also potentially making the softifacts harder to parse for experts in the domain. In these cases, we want to use the derived knowledge in place of the core knowledge.

Being able to specify our level of abstraction through the progressive application of projection functions eludes to another necessary piece of knowledge organization: the projection functions themselves. As we project out core knowledge that we know of as otherwise commonly derived concepts (like the Joule example above), we should also like to store them in some form. For example, derived units may end up in the same context as the SI Units, defined by specific projection functions applied to SI Units. Continuing the Joule example, we would be applying a projection function across core knowledge related to energy and specific SI Units, then calling that projection *Joule* and giving it a symbolic representation J that we can refer to later.

We want to take a fluid and practical approach to organizing knowledge, such that we can keep domain-related knowledge chunks together with useful derivations. We want to separate knowledge in unrelated domains, such that it is straightforward to look up whatever we need with relative ease. The specific implementation for organization will be detailed later (See Chapter 4.2).

In summary, this approach to organizing knowledge (grouping related information into domain-specific knowledge-bases and supporting flexible projection) lays the conceptual foundation for Drasil’s framework. By making explicit how knowledge is structured and accessed, we address the root causes of redundancy and inconsistency identified earlier in this chapter. This organizational strategy not only clarifies the requirements for knowledge capture and reuse, but also directly informs the design

principles and implementation choices discussed later in the thesis.

3.6 Summary - The seeds of Drasil

This final section synthesizes the chapter’s findings, highlighting the key insights that inform the design of Drasil. By summarizing our process and rationale, we prepare the reader for the transition from analysis to implementation in the following chapter.

Through this chapter, as part of our effort to reduce unnecessary redundancy across the software development process, we have taken an approach to breaking down softifacts to the core knowledge they present and looked for commonalities in that knowledge between them. We use several case study systems that fit our scope (input $\rightarrow$ process $\rightarrow$ output) as examples to give a concrete, applied base to the work.

Generally, we see softifacts for a given system have a lot in common, namely they require the same core knowledge tailored to a specific audience for each softifact. This knowledge is organized in a meaningful way, and portions relevant to the context of the softifact are presented to the audience.

We delved into the idea of knowledge chunks and projection functions for producing context-relevant pieces of knowledge that are consumable by a given audience. We have also explored strategies for organizing that knowledge in a practical manner.

We have determined the three main components necessary for any useful softifact: knowledge, context (i.e. audience), and organizational structure. From here we can operationalize each component in a reusable and (relatively) redundancy-free manner. This operationalization informs the initial design for our framework Drasil which will be covered in depth in Chapter 4.

Effectively, we want to automate the generation of softifacts through applying projections to knowledge and presenting it in a given structure. The structure of softifacts is relatively straightforward to deal with, we can use templates, blueprints, or deterministic generation which rely on relatively common technologies. The knowledge-capture and projection is much more interesting as it relies on some yet-to-be-determined knowledge-capture mechanism that can provide us with chunks of knowledge that can then be fed to projection functions in some context-aware manner.

Chapter 4

Drasil

Drasil is a framework designed to address the challenges of generating consistent, traceable, and maintainable scientific software artifacts from a single source of domain knowledge. The name “Drasil” is derived from Norse Mythology’s world tree Yggdrasil whose branches spread across the many realms and reflects how our framework spans the many domains and contexts relevant to software generation. In this chapter, we provide a guided tour of Drasil’s architecture and implementation, focusing on how knowledge is captured, organized, and operationalized to produce both documentation and executable code.

We begin by motivating the need for Drasil and outlining its core principles. Next, we delve into the mechanisms for capturing and classifying knowledge, introducing the concept of “chunks” and their role in building a reusable knowledge base. We then describe how this knowledge is structured into recipes that specify the content and organization of generated artifacts. The chapter continues with a detailed look at Drasil’s generation and rendering pipeline, illustrating how abstract representations are transformed into concrete outputs. Finally, we discuss the iterative refinement

process that underpins Drasil’s development and summarize the key takeaways for our approach to software generation.

Throughout, our aim is to show not just what Drasil does, but why each component is necessary and how the pieces fit together to support reliable, maintainable software.

4.1 Drasil in a nutshell

This section provides a high-level overview of Drasil’s motivation and approach, setting the stage for the technical details that follow. To understand why Drasil is needed, consider the challenges of manually writing and maintaining a full range of softifacts: the process is redundant, tedious, and often leads to divergent softifacts. Drasil is a purpose-built framework created to tackle these problems. It is designed to generate a wide range of softifacts from a single, unified source of domain knowledge. By capturing the essential concepts, relationships, and requirements of a software system as reusable, well-typed “chunks,” Drasil enables the automatic production of consistent, traceable, and maintainable artifacts tailored to different stakeholders. The framework emphasizes reducing manual duplication, ensuring traceability, and supporting long-lived, well-understood domains where frequent maintenance or evolution is expected.

Unlike documentation generators like Doxygen, Javadoc, and Sphinx that take a code-centric view of the problem and rely on manual redundancy – i.e. natural-language explanations written as specially delimited comments which can then be weaved into API documentation alongside code – Drasil takes a knowledge-centric, redundancy-limiting, fully traceable, single-source approach to generating softifacts.

The framework builds up knowledge *as needed* to solve concrete problems rather than attempting to encode all possible information from the start, allowing it to be flexible and adaptable to different domains and contexts early in its development.

However, Drasil is not, nor is it intended to be, a panacea for all the woes of software development. Even the seemingly well-defined problems of unnecessary redundancy and manual duplication turn out to be large, many-headed beasts which exist across a multitude of software domains; each with their own benefits, drawbacks, and challenges.

To reiterate: Drasil has not been designed as a silver-bullet. It is a specialized tool meant to be used in well-understood domains for software that will undergo frequent maintenance and/or changes. In deciding whether Drasil would be useful for developing software to tackle a given problem, we recommend identifying those projects that are long-lived (10+ years) with softifacts relevant to multiple stakeholders. For our purposes, as mentioned earlier, we have focused on SC software that follows the input $\rightarrow$ process $\rightarrow$ output pattern. SC software has the benefit of being relatively slow to change, so models used today may not be updated or invalidated for some time, if ever. Should that happen, the models will likely still be applicable given a set of assumptions or assuming certain acceptable margins for error.

With Drasil being built around this specific class of problems, we remain aware that there are likely many in-built assumptions in its current state that could affect its applicability to other domains. As such, we consider expanding Drasil’s reach as an avenue for future work.

The Drasil framework relies on a knowledge-centric approach to software specification and development. We attempt to codify the foundational theory behind the

problems we are attempting to solve and operationalize it through the use of generative technologies. By doing so, we can reuse common knowledge across projects and maintain a single source of truth in our knowledge database.

Given how important knowledge is to Drasil, one might think we are building ontologies or ontology generators. We must make it clear this is *not* the case. We are not attempting to create a source for all knowledge and relationships inside a given field. We are merely using the information we have available to build up knowledge as needed to solve problems. Over time, this may take on the appearance of an ontology, but Drasil does not currently enforce any strict rules on how knowledge should be captured, outside of its type system and some best practice recommendations. We will explore knowledge capture in more depth in Section 4.2.

4.2 Our Ingredients: Organizing and Capturing Knowledge

Before we can generate useful software artifacts, we must first capture and organize the underlying domain knowledge in a form that is both reusable and operational. To illustrate what this entails, consider a concrete example from the domain of scientific computing: Newton’s Second Law of motion [?]. To use this law in both documentation and code, we need to capture several distinct pieces of knowledge:

- The name (or label) for the law (“Newton’s Second Law”).
- A natural language definition (“the net force on a body at any instant of time is equal to the body’s acceleration multiplied by its mass”).

- The mathematical relationship ($F = ma$).
- The symbolic representation (in this case is identical to the equation).
- The definitions and symbols for each variable (F , m , a).
- The units associated with each quantity (newtons, kilograms, metres per second squared).

Capturing all of this information in a way that is reusable, extensible, and traceable is non-trivial. We need a knowledge capture method robust enough to represent not only the specific information needed for the software systems we are building, but also common knowledge relevant across the entire target domain of SC software.

To address this, Drasil introduces a set of mechanisms for representing and structuring knowledge, centered around the concept of “chunks” and the progressive refinement of knowledge structures. By understanding these building blocks, the reader will be equipped to follow how Drasil supports traceability and reuse throughout the software generation process.

Before we can design a knowledge capture mechanism, we must first define what exactly we believe we need to capture. It is nearly impossible to consider every case of knowledge that could be used in the domain of SC software, not to mention an extremely large undertaking to begin with “all the things”. As such we have decided to take an iterative, progressive refinement approach to our knowledge capture mechanisms. This should come as no surprise and follows from our general process for developing the Drasil framework.

```

33 -- === DATA TYPES/INSTANCES === --
34 -- | Used for anything worth naming. Note that a 'NamedChunk' does not have an
    acronym/abbreviation
35 -- as that's a 'CommonIdea', which has its own representation. Contains
36 -- a 'UID' and a term that we can capitalize or pluralize ('NP').
37 --
38 -- Ex. Anything worth naming must start out somewhere. Before we can assign equations
39 -- and values and symbols to something like the arm of a pendulum, we must first give
    it a name.
40 data NamedChunk = NC {
41   _uu :: UID,
42   _np :: NP
43 }

```

Figure 4.1: NamedChunk Definition

4.2.1 Capturing Knowledge via Chunks

Having identified some of the kinds of information we need to capture—names/labels, definitions, symbols, units, and equations (as illustrated by our Newton’s Second Law example)—we now turn to the mechanism Drasil uses to represent this knowledge in a reusable and extensible way. To address these requirements, we borrow and re-purpose the *chunk* term from Literate Programming (LP), creating a simplified, extensible, and ever-expanding hierarchy of chunks based on the needs for capturing individual pieces of knowledge.

Our base chunk, **NamedChunk** is defined in Figure 4.1 and can be thought of as any uniquely identifiable term. It is composed of a UID, or unique identifier, and a NP, or noun phrase, representing the term.

A single node does not a hierarchy make, but with the root **NamedChunk** defined, we can begin to progressively extend it to cover any new chunk types we may need for our knowledge capture requirements. For example, if we want to capture a simple term with its definition, we need more than just a **NamedChunk**. We need to extend **NamedChunk** to include a definition, and we refer to this particular variant chunk as a **ConceptChunk**. For ease of creation, we define a number of semantic, so-called

```

76 probability = dcc "probability" (cnIES "probability") "The likelihood of an event to
    occur"
77 rate = dcc "rate" (cn' "rate") "Ratio that compares two quantities having different
    units of measure"
78 rightHand = dcc "rightHand" (cn' "right-handed coordinate system") "A coordinate
    system where the positive z-axis comes out of the screen."
79 shape = dcc "shape" (cn' "shape") "The outline of an area or figure"
80 surface = dcc "surface" (cn' "surface") "The outer or topmost boundary of an object"
81 unit_ = dcc "unit" (cn' "unit") "Identity element"
82 vector = dcc "vector" (cn' "vector") "Object with magnitude and direction"

```

Figure 4.2: Some example instances of **ConceptChunk** using the **dcc** smart constructor

smart constructors for each chunk type which allow us to more simply define our chunks and their intermediary datatypes (ie: UID and NP from the **NamedChunk** example). An example of such smart constructors being used to create simple instances of **ConceptChunk** can be seen in Figure 4.2. Note there are smart constructors for each datatype when multiple variants exist and could be used in a given place. Looking at the **ConceptChunk** example, we can see two different smart constructors (**cnIES** and **cn'**) being used to create the NP instance. Both smart constructors create an instance of NP, but in this case the smart constructor used defines given properties (i.e. pluralization rules of the noun phrase used as a term) for the instance to simplify construction.

These simple chunks are fine as a starting point, however, knowing the SC domain we can already foresee the need for a variety of other chunks. For brevity, we will only expand upon the chunks needed for the examples used in this paper. More information, including detailed definitions of all the types and current state of Drasil can be found in the wiki (<https://github.com/JacquesCarette/Drasil/wiki/>) or the Haddock documentation which can be generated from the Drasil source code or found online at <https://jacquescarette.github.io/Drasil/docs/full/>

4.2.2 Chunk Combinatorics

The flexibility and extensibility of Drasil’s knowledge capture comes not just from the variety of chunk types, but from the ways in which chunks can be constructed, referenced, and composed. In this section, we explore how extending our chunk hierarchy multiplies the possible combinations and relationships between chunks, and how this enables us to build richer knowledge structures. This approach is essential for reducing duplication and supporting traceability, as it allows definitions and relationships to be constructed from reusable components. To facilitate this, we introduce a Domain Specific Language (DSL) for constructing and combining English-language sentences (**Sentence**) which lets us define terms and their relationships in a flexible, compositional way.

Even with extremely simple chunks we have seen the need for multiple chunk types and a number of ways to construct instances of those types. The more we extend our chunk hierarchy, the larger the number of possible combinations we will need to account for. This is not only relevant to chunks themselves, but also to the information they encode. Take, for instance, a term definition that relies on other terminology that has been captured. In our effort to reduce duplication and maintain a single source of information, we intend to be able to create chunks with references to other chunks such that the definition can use the known terminology.

To define a term with references to other terms, we need to ensure we are creating our definitions by projecting relevant knowledge into the definition, but also ensuring we have the flexibility to extend that knowledge. With that in mind, we created a Domain Specific Language (DSL) for creating and combining (English language) sentences aptly called **Sentence**. In its most basic form, **Sentence** will simply wrap

```

acceleration = dccWDS "acceleration" (cn' "acceleration")
  (S "the rate of change of a body's" +:+ phrase velocity)
position = dcc "position" (cn' "position")
  "an object's location relative to a reference point"
velocity = dccWDS "velocity" (cnIES "velocity")
  (S "the rate of change of a body's" +:+ phrase position)

```

Figure 4.3: Projecting knowledge into a chunk's definition

a string. However, by defining a number of helper functions and other useful utilities, our **Sentence DSL** can be used to combine, change case, pluralize, and more. A simple example of a series of chunks that utilize the **Sentence DSL** to derive their definitions can be seen in Figure 4.3. We use the **phrase** helper function to pull the appropriate sentence out of the **velocity** chunk and the combinator (**++**) to concatenate the sentences. Note that **position** uses a different smart constructor (for non-derived definitions) and so doesn't require the sentence constructor (**S**) around its definition.

We also make opportunistic use of composable accessors (commonly known in the Haskell ecosystem as “lenses”) to help manage complexity when combining and projecting chunks. Conceptually, these are lightweight, composable getters and setters that let us focus on the logical structure of a chunk (its name, symbol, sentence, expression, units, etc.) without repeatedly writing boilerplate navigation code for nested fields¹. In practice they let us build derived chunks by projecting the subparts we need, update specific components (for instance a symbol or an attached unit) in a principled, immutable way, and compose transformations cleanly when constructing larger chunks from smaller ones.

As seen in our earlier case studies (see, for example, Figure 3.8), the types of knowledge we need to capture often go well beyond simple names and definitions.

¹Concrete lens usage and examples are available in the code found in the project repository for readers who want the implementation details.

```

91 force = dcc "force" (cn' "force")
92   "an interaction that tends to produce change in the motion of an object"

```

Figure 4.4: The force **ConceptChunk**

```

acceleration = uc CP.acceleration (vec lA) accelU
force = uc CP.force (vec cF) newton

```

Figure 4.5: The force and acceleration **UnitalChunks**

Those examples highlighted the need for symbolic representations, equations, units, and explicit relationships between concepts. While our motivating example of Newton’s Second Law ($F = ma$) covers some of these aspects, it also reveals gaps in our current chunk definitions—particularly when it comes to capturing all relevant properties in a reusable, traceable way.

For a better sense of what, exactly, we are missing, let us build on our Newton’s Second Law example and determine how we achieve an operational definition of *Force* as a chunk in Drasil. To fully represent force, we need not only a name and definition, but also a symbolic representation (F), a mathematical equation ($F = ma$), and associated units (N). Likewise, mass and acceleration require their own symbols (m , a), definitions, and units (kg , m/s^2). This illustrates the necessity for chunk types that can encode all these facets in a reusable and traceable manner.

We start by building a **ConceptChunk** for the Force concept as seen in Figure 4.4. It is built from a common noun (“force”) and a definition. Next we need to capture the units of measurement to the force concept by defining what we have named a **UnitalChunk**. This definition for force can be seen in Figure 4.5. Note that we are not re-defining force in this instance. We are instead using a smart constructor to build atop the existing force chunk (`CP.force`) and adding a symbolic representation

```

21 -- * Fundamental SI Units
22
23 metre, kilogram, second, kelvin, mole, ampere, candela :: UnitDefn
24 metre = fund "metre" "length" "m"
25 kilogram = fund "kilogram" "mass" "kg"
26 second = fund "second" "time" "s"
27 kelvin = fund "kelvin" "temperature" "K"
28 mole = fund "mole" "amount of substance" "mol"
29 ampere = fund "ampere" "electric current" "A"
30 candela = fund "candela" "luminous intensity" "cd"

```

Figure 4.6: The fundamental SI Units as Captured in Drasil

```

11 accelU = newUnit "acceleration" $ metre /: s_2
12 angVelU = newUnit "angular velocity" $ radian /: second
13 angAccelU = newUnit "angular acceleration" $ radian /: s_2
14 forcePerMeterU = newUnit "force per meter" $ newton /: metre
15 impulseU = newUnit "impulse" $ newton *: second
16 momtInertU = newUnit "moment of inertia" $ kilogram *: m_2
17 momentOffForceU = newUnit "moment of force" $ newton *: metre
18 springConstU = newUnit "spring constant" $ newton /: metre
19 torqueU = newUnit "torque" $ newton *: metre
20 velU = newUnit "velocity" $ metre /: second

```

Figure 4.7: Examples of chunks for units derived from other SI Unit chunks

(`vec cF`, which is shorthand for the selected vector representation² of the capital letter F). We also must capture the units used to measure force (`newton`) which are defined in `Data.Drasil.SI_Units` and can be seen in Figure 4.6. The SI Units are captured using another type of chunk known as a `UnitDefn` which are built atop a `ConceptChunk` and `UnitSymbol`. The `UnitSymbol` is also a chunk that can be one of several other chunk types.³. Returning to the force `UnitDefn`, the smart constructor `uc` we are using will assume we are working in the real number space. There are other smart constructors for other spaces, which are also captured using other chunks elsewhere in Drasil (see: `Language.Drasil.Space`).

Now we approach the terms mass and acceleration in a similar manner. To clarify,

²More on this in Section 4.3

³For brevity we are glossing over many chunk definitions and the differences between them as there are a large variety in use for even simple examples. They are covered in more depth in the full technical documentation found on our github repository.

for each physical quantity we wish to represent in Drasil (such as mass or acceleration), we follow a two-step process. First, we define a **ConceptChunk** for the term, which captures its name and a natural language definition. For example, the concept of acceleration is defined in relation to velocity and position, as shown earlier in Figure 4.3. Second, we create a **UnitalChunk** for each quantity, which extends the concept by associating it with a specific symbol (such as a for acceleration or m for mass) and its units of measurement (for example, metres per second squared for acceleration, or kilograms for mass).

This approach ensures that each quantity is not only described in words but is also fully specified for use in both documentation and code generation. For instance, acceleration’s units are derived from other SI units, and this relationship is captured explicitly in Drasil (see Figure 4.7). By systematically defining both the conceptual meaning and the quantitative properties of each term, Drasil enables consistent reuse and traceability throughout the generated artifacts.

Now we have almost everything we need to finish our definition of Newton’s second law. Everything thus far has been defined in natural language using our **Sentence DSL**, but it is not well-suited to the one piece we are currently missing: a way to define expressions relating chunks in a language agnostic (mathematics) representation.

4.2.3 Relating Chunks via Expressions

Continuing our Newton’s Second Law example from Section 4.2.2, we need a means to capture how force is calculated. We know it is calculated relative to mass and acceleration, so we need a way to encode that, preferably in an operational manner. We also know acceleration is itself derived from time and velocity, which is derived

```

194  -- | Multiply two expressions (Real numbers).
195  mulRe 1 (Lit (Dbl 1))      = 1
196  mulRe (Lit (Dbl 1)) r      = r
197  mulRe 1 (Lit (ExactDbl 1)) = 1
198  mulRe (Lit (ExactDbl 1)) r = r
199  mulRe (AssocA MulRe 1) (AssocA MulRe r) = AssocA MulRe (1 ++ r)
200  mulRe (AssocA MulRe 1) r = AssocA MulRe (1 ++ [r])
201  mulRe 1 (AssocA MulRe r) = AssocA MulRe (1 : r)
202  mulRe 1 r = AssocA MulRe [1, r]

```

Figure 4.8: Defining real number multiplication in `Expr`

from time and position.

When thinking about the types of information we would like to encode with expressions, we have some obvious candidates. We should be able to encode common arithmetic operations (addition, subtraction, multiplication, division, exponentiation, etc.), boolean operations (and, or, not, etc.), comparisons (equal, not equal, less than, greater than), trigonometric functions (sin, tan, cos, arctan, etc.), calculus (derivation, integration, etc.), and vector/matrix operations (dot product, cross product, etc.). For the sake of our example we’ll focus on only those we need to define Newton’s Second Law of motion: multiplication and derivatives.

As an aside, we will elide details on how we arrived at the current implementation of the expression DSL known as `Expr`, but suffice to say it was driven by a practical, “lowest common denominator” approach to developing the operations that could be encoded and factoring out commonalities much the same way we did with the rest of the case studies. With that in mind, we have encoded multiplication (for real numbers) as seen in Figure 4.8. `AssocA` refers to an associative arithmetic operator which can be applied across a list of expressions (in this case, `mulRe`, though addition and other associative operations look very similar). Derivation is much simpler to encode, as it is simply a relationship between two existing chunks, i.e. `deriv a b`

```

newtonSLEqn          = sy QPP.mass 'mulRe' sy QP.acceleration

velocityEqn, accelerationEqn :: ModelExpr
velocityEqn          = deriv (sy QP.position) QP.time
accelerationEqn      = deriv (sy QP.velocity) QP.time

```

Figure 4.9: Encoding Newton’s second law of motion

```

newtonSLQD :: ModelQDef
newtonSLQD = fromEqn' "force" (nounPhraseSP "Newton's second law of motion")
  newtonSLDesc (eqSymb QP.force) Real newtonSLEqn

newtonSLDesc :: Sentence
newtonSLDesc = foldlSent [S "The net", getTandS QP.force, S "on a",
  phrase body 'S.is' S "proportional to", getTandS QP.acceleration 'S.the_ofThe'
  phrase body 'sC' S "where", ch QPP.mass, S "denotes", phrase QPP.mass 'S.the_ofThe'
  phrase body, S "as the", phrase constant 'S.of_' S "proportionality"]

```

Figure 4.10: Newton’s second law of motion as a *Chunk*

represents taking the derivative of **a** with respect to **b**.

Continuing our example, we see (Figure 4.9) it is fairly trivial to encode the expression for Newton’s Second Law using the **Expr** DSL. We are still not done building our Newton’s Second Law chunk however, as this specific law should be referable by its own identifier. It is not simply “force”, nor the relationship between mass and acceleration, but a combination of both with its own natural language semantics that allow us to refer to it specifically. The full definition for this particular chunk can then be found in Figure 4.10. The chunk is defined relative to force and the expression captured in Figure 4.9, alongside a new identifier and description. Notice we are building off of other chunks throughout each piece of the chunk’s definition, thus giving us perfect traceability from beginning to end, regardless of whether we are looking at a defining expression or natural language description.

The **Expr** DSL grants us flexibility in defining relationships without imposing any particular structure on a chunk other than “contains an **Expr**”, or more specifically

“can be modeled using an **Expr.**” Our example also shows us that a few simple chunk types can be combined repeatedly to build much more robust, complex, and information-dense chunks. On the other hand, encoding even a relatively simple theory like Newton’s Second Law of motion requires us to define chunks down to the fundamentals. Luckily, that is another area where chunks shine as they are infinitely reusable. Once a chunk has been defined and added to our knowledge-base, we never need to redefine it. We simply use it wherever it is needed.

On the topic of using chunks, we now need a means for taking our chunks and structuring them such that we can actually do something useful with them. For example, how would we go from chunks to softifacts? Look forward to that in Section 4.3.

4.2.4 Chunk Classification

In the previous sections, we used the idea of “chunks” of knowledge as our building blocks. They range in complexity from simple single terms, to complex relationships between expressions and overarching conceptual frameworks. Each chunk is defined by a collection of structural properties, or fields, such as a unique identifier, name, symbol, unit, or space. This “atomic” information is generally encoded in simpler, more fundamental chunks. Complex chunks are constructed by combining these simpler chunks into more elaborate structures, which we can think of in terms of “molecules” compared to the “atoms” of basic chunks. With the increasing complexity and variety of chunk types, it becomes important to classify chunks based on their specific contents and the kinds of knowledge they encode and can project. This classification system not only helps avoid repetition and ensures each chunk is uniquely suited to its purpose, but also clarifies which chunks are best suited to

capture particular types of domain knowledge.

We have developed a classification system for chunks grouping them based on the properties we can project. For example, all chunks encoding things that are measured with units would be instances of the `Unitary` type⁴. Chunks that capture quantities, whether `Unitary` or not, would be instances of the `Quantity` type. As we simplify, we end up determining what qualifies something as a chunk in the first place. That is, what is the root property that *all* chunks must have to be considered a chunk? That is the `HasUID` classification, which essentially states that a given chunk of knowledge can be referred to by some unique identifier. For a more intuitive example, the `NamedIdea` classification is used for chunks that encode knowledge represented by given terms such as “force”, “computer”, “priority”, and so on.

Figure 4.11 and Figure 4.12 show a subset of the different ways we classify chunks and some example chunk types and how they would be classified, respectively. Note there is a one to many relationship between a given chunk and its classifications. As you can see, the `UnitalChunk` and `DefinedQuantityDict` are very closely related in that they both project their own `NamedIdea`, related `Space`, `Symbol`, and `Quantity`, however, the `UnitalChunk` is also classified as `Unitary` as it *must* have a unit associated with it.

We now have a compact, well-typed vocabulary for representing domain knowledge: chunks and classifications give us named concepts, quantities, units, and relationships that are easy to reason about. That vocabulary is only useful if we can turn it into artifacts people consume. To convert that vocabulary into artifacts people actually use, we need a way to operationalize those chunks. In Drasil that means

⁴ It is important to note that these classifications are based on the ability to project certain information. In the codebase we often use lenses to access or manipulate these fields, as they may be projected from component chunks rather than being directly stored in the chunk itself.

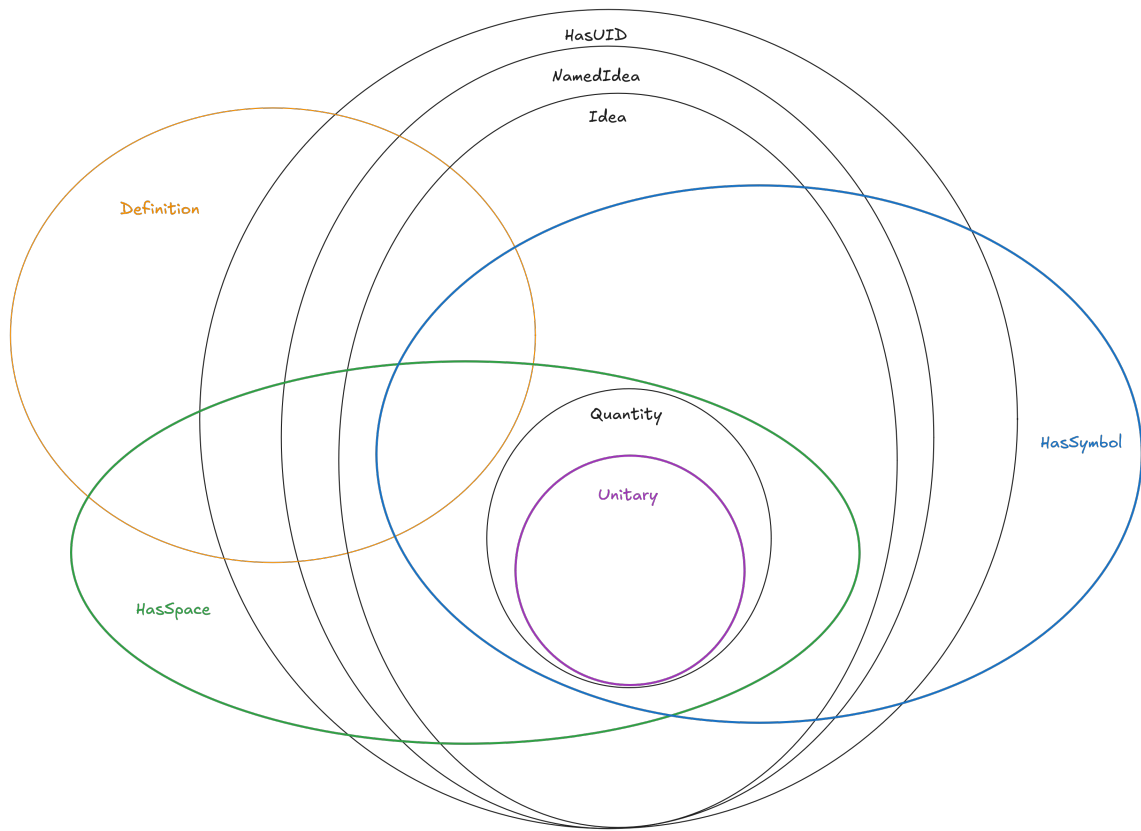


Figure 4.11: A subset of Drasil's Chunk Classification system showing some of the encoded property relationships

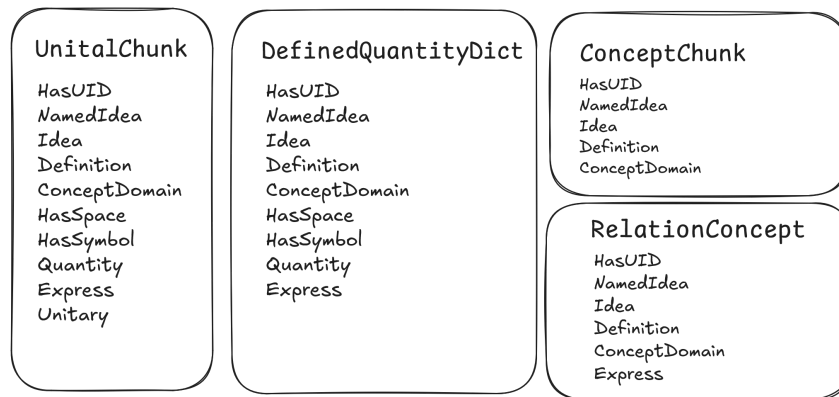


Figure 4.12: A subset of Drasil’s chunks showing how chunks may belong to multiple classifications based on the underlying knowledge they project.

using recipes.

4.3 Recipes: Codifying Structure

Given our chunk classes and a compositional Expr/Sentence model in place, we have a knowledge core that can be selectively projected for any audience. The next step is thus operational: we must describe how those chunks are composed into artifact-specific recipes and how the recipes are rendered into concrete softifacts (documents, code, and supporting files).

Our knowledge base is essentially just a collection of chunks, which are themselves a collection of definitions, encoded in our Expr and Sentence DSLs. To generate softifacts we need a means to define the overall structure of the softifact as well as the knowledge projections we require to expose the appropriate knowledge from our chunks in a meaningful way for the target audience of that particular softifact. This is where our *Recipe* DSL comes into play.

A recipe is, in its simplest form, a specification for one or more softifacts. It is an intermediate representation of a given softifact and can be thought of as a “little program”. With it, we select which knowledge is relevant and how it should be organized before passing the recipe to the generator/rendering engine to be consumed and produce the final softifact. A recipe not only allows us to structure where knowledge should be presented, but also gives us tools for automatically generating non-trivial sections of our softifacts. For example, a traceability matrix can be automatically generated by traversing the tree of relationships between chunks within a recipe, although we’ll discuss that in more depth in Section 4.4.

As we saw in Section 3.4, there are patterns of repetition for the way knowledge is projected and organized within and between our softifacts. The recipe DSL has been designed to take advantage of our knowledge capture mechanisms to reduce unnecessary repetition and duplication. We consider the softifacts as different views of the same knowledge, and project only that which is relevant to the audience of the specific softifact our recipe applies to. For example, a human readable document may use symbolic mathematics notation for representing formulae whereas the source code would represent the same formulae using syntax specific to the programming language in use (or, taking an extreme example in the case of punch cards, punched cards). Regardless, we would define the underlying knowledge once in our chunks, and use the recipe DSL to select the representation.

As our case studies are following templates for our softifacts, they give us a solid foundation with which to write our recipes. We already know how the knowledge should be organized, greatly simplifying how we can begin structuring our recipe DSL. The most straightforward, practical approach would be to start by defining a

```

25 -- | A Software Requirements Specification Declaration is made up of all necessary
    sections ('DocSection's').
26 type SRSDecl = [DocSection]
27
28 -- | Contains all the different sections needed for a full SRS ('SRSDecl').
29 data DocSection = TableOfContents      -- ^ Table of Contents
30 | RefSec DL.RefSec                    -- ^ Reference
31 | IntroSec DL.IntroSec                -- ^ Introduction
32 | StkhldrSec DL.StkhldrSec            -- ^ Stakeholders
33 | GSDSec DL.GSDSec                    -- ^ General System Description
34 | SSDSec SSDSec                      -- ^ Specific System Description
35 | ReqrmntSec ReqrmntSec               -- ^ Requirements
36 | LCsSec                             -- ^ Likely Changes
37 | UCsSec                             -- ^ Unlikely Changes
38 | TraceabilitySec DL.TraceabilitySec -- ^ Traceability
39 | AuxConstntSec DL.AuxConstntSec     -- ^ Auxiliary Constants
40 | Bibliography                       -- ^ Bibliography
41 | AppndxSec DL.AppndxSec              -- ^ Appendix
42 | OffShelfSolnsSec DL.OffShelfSolnsSec -- ^ Off the Shelf Solutions

```

Figure 4.13: Recipe Language for SRS

recipe for a given softifact type (SRS, MG, Code, etc.). Then flesh that out section by section, following the template as an organizational guide, using the patterns we discovered earlier (Section 3.4) as a means of avoiding unnecessary duplication. This is the approach we have chosen to follow, and it has lead to some interesting results.

As an example, let us take a look at the recipe language we have defined for an SRS using the Smith et al. template. Looking at the `DocSection` definition from Figure 4.13, we can see a striking resemblance to (a subset of) the template’s Table of Contents (Figure 3.1). Each section is defined by a datatype, which is, through a similar mechanism, decomposed into multiple subsections (for example, Figure 4.14 shows the decomposition of the requirements section) which then defines the types of organizational structures we use for specifying the contents of a given subsection. When populated, the recipe specifies both the structure of our expected softifact and the knowledge (chunks) contained therein.

As the given recipe so far has provided nothing but structure, you may be wondering how we populate it with the appropriate knowledge chunks to produce a softifact?

```

88 -- | Requirements section (wraps 'ReqsSub' subsections).
89 newtype ReqrmtSec = ReqsProg [ReqsSub]
90
91 -- | Requirements subsections.
92 data ReqsSub where
93   -- | Functional requirements. 'LabelledContent' for tables (includes input values).
94   FReqsSub      :: Sentence -> [LabelledContent] -> ReqsSub
95   -- | Functional requirements. 'LabelledContent' for tables (no input values).
96   FReqsSub'     :: [LabelledContent] -> ReqsSub
97   -- | Non-Functional requirements.
98   NonFReqsSub   :: ReqsSub

```

Figure 4.14: ReqrmtSec Decomposition and Definitions

For that, we should look at a specific example from one of our case studies (GlassBR).

4.3.1 Example: SRS recipe for GlassBR

The recipe for the GlassBR SRS can be seen in Figure 4.15. This should look familiar, as it follows a similar structure to the Smith et al. template table of contents (Figure 3.1). We use a number of helper functions and data types to condense and collect the chunks we require and avoid writing any generic boilerplate text. The structure of the document can be rearranged to an extent, in that we can add, remove, or reorder sections at will by simply moving them around within the `SRSDec1`. We can also make choices regarding the display of certain types of information in the final, generated softifact. For example, we alluded to vector representations earlier (Section 4.2.2) which we currently can display using either bold typeface or italics. This choice must be made using the `TypogConvention` datatype which lists the choices made (ex. `TypogConvention[Vector Bold]`) and is used by a relevant portion of the SRS recipe if vectors are included in the softifact. Our GlassBR example does not make use of this convention, however the GamePhysics case study does.

```

54 srs :: Document
55 srs = mkDoc mkSRS (S.forGen titleize phrase) si

:

85 mkSRS = [TableOfContents,
86   RefSec $ RefProg intro [TUnits, tsymb [TSPurpose, SymbOrder], TAandA],
87   IntroSec $
88     IntroProg (startIntro software blstRskInvWGlassSlab glassBR)
89     (short glassBR)
90     [IPurpose $ purpDoc glassBR Verbose,
91      IScope scope,
92      IChar [] (undIR ++ appStanddIR) [],
93      IOrgSec orgOfDocIntro Doc.dataDefn (SRS.inModel [] []) orgOfDocIntroEnd],
94   StkhldrSec $
95     StkhldrProg
96     [Client glassBR $ phraseNP (a_ company)
97      +:+. S "named Entuitive" +: S "It is developed by Dr." +: S (name
98      mCampidelli),
99      Cstmr glassBR],
100   GSDSec $ GSDProg [SysCntxt [sysCtxIntro, L1C sysCtxFig, sysCtxDesc, sysCtxList],
101     UserChars [userCharacteristicsIntro], SystCons [] [] ],
102   SSDProg
103     [SSDProblem $ PDProg prob [termsAndDesc]
104      [PhySysDesc glassBR physSystParts physSystFig []
105       , Goals goalInputs],
106      SSDSolChSpec $ SCSProg
107      [Assumptions
108       , TMs [] (Label : stdFields)
109       , GDs [] [] HideDerivation -- No Gen Defs for GlassBR
110       , DDs [] ([Label, Symbol, Units] ++ stdFields) ShowDerivation
111       , IMs [instModIntro] ([Label, Input, Output, InConstraints, OutConstraints]
112        ++ stdFields) HideDerivation
113       , Constraints auxSpecSent inputDataConstraints
114       , CorrSolnPties [probBr, stressDistFac] []
115      ],
116   ReqrmtSec $ ReqsProg [
117     FReqsSub inReqDesc funcReqsTables,
118     NonFReqsSub
119   ],
120   LCsSec,
121   UCsSec,
122   TraceabilitySec $ TraceabilityProg $ traceMatStandard si,
123   AuxConstntSec $ AuxConsProg glassBR auxiliaryConstants,
124   Bibliography,
125   AppndxSec $ AppndxProg [appndxIntro, L1C demandVsSDFig, L1C dimlessloadVsARFig]]

```

Figure 4.15: SRS Recipe for GlassBR

```

63 si :: SystemInformation
64 si = SI {
65   _sys      = glassBR,
66   _kind     = Doc.srs,
67   _authors  = [nikitha, spencerSmith],
68   _purpose  = purpDoc glassBR Verbose,
69   _quants   = symbolsForTable,
70   _concepts = [] :: [DefinedQuantityDict],
71   _instModels = iMods,
72   _datadefs = GB.dataDefs,
73   _configFiles = configFp,
74   _inputs    = inputs,
75   _outputs   = outputs,
76   _constraints = constrained,
77   _constants = constants,
78   _sysinfodb = symbMap,
79   _usedinfodb = usedDB,
80   refdb      = refDB
81 }

```

Figure 4.16: System Information for GlassBR

As for the chunks necessary for generating the softifact, each section includes references to them through supplied lists in something we call the **System Information** object (Figure 4.16).

The system information object uses helper functions to keep track of the lists of chunks necessary for filling in the sections of our softifact, the SRS. Each piece of the object consists of one or more chunks to be used in filling in the relevant sections. A quick description of each property can be found in Table 4.1, but it should be noted that some of the defined properties are only there for convenience (and/or remain only until the next refinement pass) as they are derived from other chunks with the use of helper functions and could be inferred.

As should be obvious, the system information object is referencing other chunks which have been defined elsewhere. Most definitions can be found in the knowledge-base, with system-specific aggregations (i.e. lists of chunks) being defined within the scope of the specific example project. For example, `iMods` referenced as the

Table 4.1: System information object breakdown - every property is represented by chunks encoding given information

Property	Chunk(s) Referenced
_sys	System name (ex. "GlassBR").
_kind	Softifact type we are defining (ex. SRS).
_authors	List of authors of the softifact.
_purpose	Purpose of the softifact.
_quants	List of required quantities.
_concepts	List of required concepts not otherwise encoded.
_instModels	List of required instance models.
_datadefs	List of required data definitions.
_configFiles	List of required configuration files.
_inputs	List of required inputs to the system.
_outputs	List of required outputs from the system.
_constraints	List of constraints for the system (physical and software specific).
_constants	List of required constants.
_sysinfodb	Database of all required chunks.
_usedinfodb	Database of all used acronyms and symbols.
refdb	Database of all relevant external references/citations.

```

18 iMods :: [InstanceModel]
19 iMods = [pbIsSafe, lrIsSafe]

:

28 pbIsSafe :: InstanceModel
29 pbIsSafe = imNoDeriv (equationalModelN (nounPhraseSP "Safety Req-Pb") pbIsSafeQD)
30   [qwC probBr $ UpFrom (Exc, exactDbl 0), qwC pbTol $ UpFrom (Exc, exactDbl 0)]
31   (qw isSafePb) []
32   [dRef astm2009] "isSafePb"
33   [pbIsSafeDesc, probBRRef, pbTolUstr]

:

41 lrIsSafe :: InstanceModel
42 lrIsSafe = imNoDeriv (equationalModelN (nounPhraseSP "Safety Req-LR") lrIsSafeQD)
43   [qwC lRe $ UpFrom (Exc, exactDbl 0), qwC demand $ UpFrom (Exc, exactDbl 0)]
44   (qw isSafeLR) []
45   [dRef astm2009] "isSafeLR"
46   [lrIsSafeDesc, capRef, qRef]

```

Figure 4.17: The `iMods` definition

`_instModels` property is defined as shown in Figure 4.17, alongside the two system-specific instance models listed within it. The two specific instance models can be seen to be derived from other existing chunks, through the use of a series of helper functions, thus ensuring their traceability to the knowledge-base.

Given that the system information object refers to all of the requisite chunks for the knowledge contained in our softifacts and the SRS recipe defines how that knowledge should be structured and organized within the softifact, it follows that those are all we should need to generate our SRS. As an aside, the complete recipe for the SRS specification of GlassBR⁵ including the system information object is contained in a single file in less than 370 lines of code⁶. Teasing our results: the generated SRS in PDF format is 44 pages in length. This is partially due to our ability to strip away all of the common boilerplate text that we need not worry about

⁵Not including chunks referenced from within our knowledge bases as they can be shared, reused, and are not necessarily specific to only this system.

⁶Including many comments, whitespace, and import statements.

right now, as that will be a problem for the generator (Section 4.4). As it stands we have created the recipe language to represent what we truly want out of a recipe: a simplified, unnecessary-duplication-avoiding, fully traceable representation of our target softifact.

With the combination of system information and the SRS recipe we can soon move on to creating our generator for the finished SRS document. As the other softifact recipes were developed in tandem with the SRS, they follow a similar structure and in the interest of brevity we will skip the breakdown of each recipe. It is left to the reader to investigate them by diving into our examples via the Drasil github. However, there is one softifact that is significantly different, and as such, we will go into more depth in discussing in the following section: the executable code recipe. Said recipe allows us to not only specify knowledge and the way we wish it structured, but also implementation choices that we would like to make in the final generated source code.

4.3.2 Example: Executable Code recipe for GlassBR

As discussed in the previous section, the executable code recipe is fundamentally different from the documentation-heavy SRS recipe. While the SRS recipe focuses on organizing and projecting knowledge for human consumption, the executable code recipe must specify the structure and content required for generating working source code. This includes not only the relevant knowledge chunks, but also the implementation choices (modules, functions, and other details) necessary for code generation.

The executable-code recipe for GlassBR somewhat mirrors the document recipe

in structure but emphasizes implementation choices: module layout, data representation, target languages, optional features, and constraint-handling strategies. Practically, we supply the modules and computational relationships (the mathematical execution order), target languages (ex. Python, C++, Java), modularity (single file or separated modules), data bundling (structs/classes), and logging/documentation options.

The recipe for GlassBR’s executable code constructed using the `CodeSpec` DSL, is shown in Figure 4.18. Here, the `codeSpec` function takes three arguments:

- **fullSI**: the system information object, which aggregates all the knowledge chunks relevant to GlassBR (as described in the SRS example).
- **choices**: a record specifying code generation options, such as target languages, architecture, data handling, and auxiliary files. The example **choices** definition for GlassBR can be seen in Figure 4.18 and the object itself will be explained in more detail later in this section.
- **allMods**: a list of modules to be generated, each defined in terms of the knowledge chunks and functions they encapsulate.

The `CodeSpec` DSL itself is defined in Figure 4.19, and a subsection of its key fields are summarized in Table 4.2. As mentioned above, we use a helper function `codeSpec` to extract the appropriate fields from the system information, choices, and module list retaining full traceability and avoiding unnecessary manual duplication.

As with the SRS recipe, the code recipe references knowledge chunks defined elsewhere. For example, the list of input variables is defined in `Unitals.hs` just as they were for the SRS. Similarly, the modules to be generated are defined in

Body.hs

```

57 fullSI :: SystemInformation
58 fullSI = fillcdbSRS mkSRS si

```

Choices.hs

```

13 code :: CodeSpec
14 code = codeSpec fullSI choices allMods
15
16 choices :: Choices
17 choices = defaultChoices {
18   lang = [Python, Cpp, CSharp, Java, Swift],
19   architecture = makeArchit (Modular Separated) Program,
20   dataInfo = makeData Bundled Inline Const,
21   optFeats = makeOptFeats
22     (makeDocConfig [CommentFunc, CommentClass, CommentMod] Quiet Hide)
23     (makeLogConfig [LogVar, LogFunc] "log.txt")
24     [SampleInput "../datafiles/glassbr/sampleInput.txt", ReadME],
25   srsConstraints = makeConstraints Exception Exception
26 }

```

Figure 4.18: The Recipe for GlassBR’s Executable Code

```

41 -- | Code specifications. Holds information needed to generate code.
42 data CodeSpec where
43   CodeSpec :: (HasName a) => {
44     -- | Program name.
45     pName :: Name,
46     -- | Authors.
47     authors :: [a],
48     -- | All inputs.
49     inputs :: [Input],
50     -- | Explicit inputs (values to be supplied by a file).
51     extInputs :: [Input],
52     -- | Derived inputs (each calculated from explicit inputs in a single step).
53     derivedInputs :: [Derived],
54     -- | All outputs.
55     outputs :: [Output],
56     -- | List of files that must be in same directory for running the executable.
57     configFiles :: [FilePath],
58     -- | Mathematical definitions, ordered so that they form a path from inputs to
59     -- outputs.
60     execOrder :: [Def],
61     -- | Map from 'UID's to constraints for all constrained chunks used in the problem.
62     cMap :: ConstraintCEMap,
63     -- | List of all constants used in the problem.
64     constants :: [Const],
65     -- | Map containing all constants used in the problem.
66     constMap :: ConstantMap,
67     -- | Additional modules required in the generated code, which Drasil cannot yet
68     -- automatically define.
69     mods :: [Mod], -- medium hack
70     -- | The database of all chunks used in the problem.
71     sysinfodb :: ChunkDB
72   } -> CodeSpec

```

Figure 4.19: The Recipe Language for Executable Code (CodeSpec)

Table 4.2: CodeSpec object breakdown

Property	Chunk(s) Referenced
pName	Program name
authors	List of authors
inputs	All input variables
outputs	All output variables
configFiles	Required configuration files
execOrder	Mathematical definitions ordered such that they form a path from inputs to outputs
cMap	Constraints on variables
constants	Constants used in the program
mods	List of additional code modules to generate
sysinfodb	Database of all knowledge chunks for the given problem

```

30 readTableMod :: Mod
31 readTableMod = packmod "ReadTable"
32   "Provides a function for reading glass ASTM data" [] [readTable]
33
34 readTable :: Func
35 readTable = funcData "read_table"
36   "Reads glass ASTM data from a file with the given file name"
37   [ singleLine (repeated [quantvar zVector]) ', ',
38     multiline (repeated (map quantvar [xMatrix, yMatrix])) ', '
39   ]

```

Figure 4.20: The readTableMod module definition

`ModuleDefs.hs` and make reference to other chunks (both system specific and generic) from our knowledge-base. Looking into the modules, we can see the `readTableMod` module, for example, is defined as shown in Figure 4.20. This module provides a function for reading ASTM glass data from a file, encapsulating both the knowledge of the data format and the implementation logic required for code generation.

While the encapsulation of knowledge in chunks is interesting, we have seen it before in the SRS recipe example. What is novel to the code recipe is the **Choices** object. As mentioned above, it contains the outcome of very important implementation-level choices that must be made to generate the executable code. The definition for the

```

26 data Choices = Choices {
27   -- | Target languages.
28   -- Choosing multiple means program will be generated in multiple languages.
29   lang :: [Lang],
30   -- | Architecture of the program, include modularity and implementation type
31   architecture :: Architecture,
32   -- | Data structure and represent
33   dataInfo :: DataInfo,
34   -- | Maps for 'Drasil concepts' to 'code concepts' or 'Space' to a 'CodeType
35   maps :: Maps,
36   -- | Setting for Softifacts that can be added to the program or left it out
37   optFeats :: OptionalFeatures,
38   -- | Constraint violation behaviour. Exception or Warning.
39   srsConstraints :: Constraints,
40   -- | List of external libraries what to utilize
41   extLibs :: [ExtLib]
42 }

```

Figure 4.21: The `Choices` object definition

`Choices` object can be seen in Figure 4.21 and a breakdown of each property can be found in Table 4.3.

With our understanding of the `Choices` object, we can now look back at the GlassBR example (Figure 4.18) and understand what specific implementation choices have been made. First off, the `lang` property has been set so the generated code will be output in five different programming languages: Python, C++, C#, Java, and Swift. This is a good demonstration of Drasil’s ability to target multiple platforms from a single knowledge base. The `architecture` is configured as modular, with input-related functions separated into their own modules (`Modular Separated`), and the implementation type is set to `Program`, meaning the generated code will be a complete, runnable application rather than just a library.

For data representation, the `dataInfo` field specifies that inputs should be bundled together (for example, as a class or struct), constants should be defined inline within the code, and the language’s constant mechanism should be used where possible. The `optFeats` field enables comprehensive documentation by including function,

Table 4.3: Choices object breakdown

Property	
lang	List of target languages
architecture	Architecture of the generated code. Includes modularity (whether to split into modules or generate a single flat file) and implementation type (library to be consumed or standalone program)
dataInfo	Data structure and representation choices. (Ex. bundle inputs together into a class/struct, define constants inline, use the languages constant mechanism, etc.)
maps	Mapping of Drasil concepts and mathematical spaces to code concepts and types in the target language(s) (ex. One could map the concept of π with the language's built-in π and the $\mathbb{R}$ space to single/double precision floating point numbers)
optFeats	Choices for optional features including documentation (comments and verbosity), logging, and auxiliary files.
srsConstraints	Constraint violation behaviour. Can be used to specify whether to throw a warning or exception for physical/software constraints independently.
extLibs	List of external libraries to use (ex. for solving specific classes of mathematics problems). These libraries are external to Drasil and give the option for users to link to optimized, well-established libraries rather than reimplementing them from scratch in Drasil.

class, and module-level comments, but sets the documentation verbosity to quiet and hides date fields in the generated comments. Logging is also enabled for both variable assignments and function calls, with all logs directed to a file named `log.txt`. Additionally, the auxiliary files generated include a sample input file (pointing to a real data file) and a `README`, supporting both usability and reproducibility.

Finally, the `srsConstraints` field is set so that any violation of software or physical constraints will result in an exception, ensuring that errors are caught and handled strictly during execution. This configuration ensures that the generated GlassBR code is well-documented, maintainable, and ready for use in a variety of environments.

By structuring the executable code recipe in this way, Drasil ensures that all generated code is fully traceable to the underlying knowledge base and any external libraries. Any change in the knowledge chunks or their relationships is automatically reflected in the generated code, just as it is in the documentation. This approach minimizes duplication, maximizes consistency, and enables reliable code generation for scientific computing applications.

4.3.3 Operational Summary

Having established how Drasil captures, organizes, and operationalizes knowledge through both documentation and executable code recipes, we are now poised to explore the final step in Drasil’s operation: turning these structured specifications into tangible softifacts. The next piece of the story will show how Drasil takes these recipes and, through a unified process, produces the diverse softifacts required by different stakeholders. In the following section, we delve into the mechanisms and philosophy behind Drasil’s generation and rendering, showing how the framework brings together

all the ingredients we’ve discussed to deliver consistent, traceable, and maintainable softifacts.

4.4 Cooking it all up: Generation/Rendering

The previous sections have detailed how Drasil captures domain knowledge as chunks, organizes it via a robust hierarchy, and structures it into recipes that specify the desired artifacts. In this section, we focus on the final stages of the generation pipeline: *rendering* and *assembly*. These stages are responsible for transforming the intermediate representations (recipes and their referenced chunks) into tangible, concrete, consumable softifacts, such as documents and executable code.

4.4.1 Rendering: From Abstract Structure to Concrete Formats

At its core, Drasil treats recipes as “little programs”. Rendering in Drasil refers to the process by which these abstract, language-agnostic structures defined in recipes are transformed into specific output formats suitable for end-users and stakeholders. This stage bridges the gap between the high-level, reusable knowledge representations and the diverse forms required for practical use (ex. $\text{\LaTeX}$, HTML, PDF, or source code in various programming languages). Drasil achieves this through a set of dedicated *printers* and, for code, through the use of the Generic Object-Oriented Language (GOOL)[? ? ?]. Each printer is designed to interpret the structured data produced by the recipe and knowledge projection stages and emit output in the desired format,

handling both content and layout⁷.

Document Rendering: Printers and Output Engines

For documentation artifacts (such as SRS or Module Guide), Drasil provides specialized printers for each supported format:

- `genTeX` in `Language.Drasil.TeX.Print` — renders the document as $\text{\LaTeX}$ source, suitable for PDF generation.
- `genHTML` in `Language.Drasil.HTML.Print` — produces HTML for web-based consumption.
- `genJSON` in `Language.Drasil.JSON.Print` — outputs a machine-readable JSON representation, supporting further processing or integration.

Each printer traverses the recipe structure, visiting each section and sub-section, and calls formatting routines appropriate to the target format. For example, a Data Definition chunk will be rendered as a $\text{\LaTeX}$ table row, an HTML table entry, or a JSON object, depending on the printer invoked.

Example: SRS to $\text{\LaTeX}$

As a concrete illustration, consider the printing of the GlassBR SRS to $\text{\LaTeX}$. The process begins with the invocation of the `genTeX` function, which recursively walks the document tree produced by the recipe, emitting $\text{\LaTeX}$ code for sections, equations, and tables. Figure 4.22 shows the entry point to this process.

⁷To some respect, depending on the desired output format. For instance, $\text{\LaTeX}$ handles the majority of the finished softifact's layout where Drasil generates the content

```

41 -- | Generates a LaTeX document.
42 genTeX :: L.Document -> PrintingInformation -> TP.Doc
43 genTeX doc@(L.Document _ _ toC _) sm =
44   runPrint (buildStd sm toC $ I.makeDocument sm $ L.checkToC doc) Text

```

Figure 4.22: `genTeX` function for LaTeX rendering

Code Rendering: GOOL and Target Language Emission

For executable code artifacts, Drasil uses GOOL [? ? ?] as an intermediate layer. GOOL provides a collection of type-safe combinators and abstractions for representing object-oriented and procedural constructs in a language-agnostic manner. The code recipe (ex. a `CodeSpec`) is first translated into a GOOL abstract syntax tree (AST), which is then emitted as source code in one or more target languages. For brevity we will not get into the implementation details regarding Drasil’s use of GOOL, as those have been covered in other works [? ? ?].

In brief, the code emission process follows a straightforward translation process: The code recipe and its referenced chunks are traversed to construct a GOOL AST, representing classes, functions, variables, and control flow; The GOOL backend is invoked for each specified target language, translating the AST into concrete source code (ex. Python, C++, Java, C#); Auxiliary files (such as README, Doxygen config, and sample input files) are generated via dedicated routines.

Example: Emitting GlassBR Code in Python

For GlassBR, suppose the code recipe specifies Python as a target. The generation process calls `generateCode` in `Language.Drasil.Code.Imperative.Generator`, which constructs the GOOL AST and then emits Python code by invoking the Python backend. The resulting code includes all functions, data structures, and comments

```

98 -- | Generates a package with the given 'DrasilState'. The passed
99 -- un-representation functions determine which target language the package will
100 -- be generated in.
101 generateCode :: (OOProg progRepr, PackageSym packRepr) => Lang ->
102   (progRepr (Program progRepr) -> ProgData) -> (packRepr (Package packRepr) ->
103   PackData) -> DrasilState -> IO ()
104 generateCode l unReprProg unReprPack g = do
105   workingDir <- getCurrentDirectory
106   createDirectoryIfMissing False (getDir l)
107   setCurrentDirectory (getDir l)
108   let (pckg, ds) = runState (genPackage unReprProg) g
109   code = makeCode (progMods $ packProg $ unReprPack pckg)
110         ([ad "designLog.txt" (ds ^. designLog) | not $ isEmpty $
111          ds ^. designLog] ++ packAux (unReprPack pckg))
112   createCodeFiles code
113   setCurrentDirectory workingDir

```

Figure 4.23: generateCode and genPackage for code rendering

specified in the recipe and choices object, as well as any automatically generated traceability links (ex. comments referencing requirements or data definitions). Figure 4.23 shows the relevant code path.

4.4.2 Assembly: Producing the Final Softifact Set

Once rendering is complete, the generated document or code sections must be assembled into coherent artifacts ready for distribution and use. Assembly includes writing the primary artifacts to disk (ex. *.tex, *.html, *.py, *.cpp files), generating auxiliary files (ex. Makefiles, CSS, README, example input files), linking cross-references, and ensuring the integrity of tables, figures, and code modules. The remainder of this section illustrates these steps in practice.

Example: SRS Assembly

After rendering the GlassBR SRS to L^AT_EX via the `genTeX` function seen in Figure 4.22, the `prntDoc` function, Figure 4.24, writes the source file, while `prntMake`,

```

56 -- document information and printing information. Then generates the document file.
57 prntDoc' :: String -> String -> Format -> Document -> PrintingInformation -> IO ()
58 prntDoc' dt' fn format body' sm = do
59   createDirectoryIfMissing True dt'
60   outh <- openFile (dt' ++ "/" ++ fn ++ getExt format) WriteMode

```

Figure 4.24: `prntDoc'` function for document output

```

68 -- | Helper for writing the Makefile(s).
69 prntMake :: DocSpec -> IO ()
70 prntMake ds@(DocSpec (DC dt _) _) =
71   do outh <- openFile (show dt ++ "/PDF/Makefile") WriteMode
72       hPutStrLn outh $ render $ genMake [ds]
73       hClose outh

```

Figure 4.25: `prntMake` function for document output

Figure 4.25, generates a Makefile for automated PDF compilation. The resulting directory contains a fully cross-referenced, typeset-ready SRS and all supporting files.

Example: Code Assembly

Once the code has been rendered through the GOOL pipeline into target-language text, the next stage is to assemble the resulting files into a coherent project structure. In this case, `createCodeFiles` (Figure 4.26) is called to write each generated source file and all auxiliary documentation to disk using consistent directory and naming conventions. For multi-language builds, this process is repeated for each target language, ensuring consistency across all output.

In essence, this stage performs for code what the SRS assembly stage does for documentation: it gathers the individually generated fragments and organizes them into a complete, ready-to-use product. The result comprises all files necessary for the generated system source code, build instructions, and supporting assets. They are generated and ready for compilation or integration into a development environment.

```

28 -- | Creates the requested 'Code' by producing files.
29 createCodeFiles :: Code -> IO () -- [(FilePath, Doc)] -> IO ()
30 createCodeFiles (Code cs) = mapM_ createCodeFile cs
31
32 -- | Helper that uses pairs of 'Code' to create a file written with the given
33   document at the given 'FilePath'.
34 createCodeFile :: (FilePath, Doc) -> IO ()
35 createCodeFile (path, code) = do
36   createDirectoryIfMissing True (takeDirectory path)
37   h <- openFile path WriteMode
38   hPutStrLn h (render code)
39   hClose h

```

Figure 4.26: createCodeFiles for code output

Because these outputs are directly derived from the same knowledge base that informed the SRS, Drasil ensures that the implementation remains consistent with the documentation.

4.4.3 A GlassBR Example: From Chunk Definition to Final SRS Rendering

To illustrate the rendering and assembly process, we follow a single chunk through each step of generation in the GlassBR SRS. Earlier, we described how the SRS recipe for GlassBR is constructed by specifying the document structure and referencing the necessary knowledge chunks through the system information object (see Section 4.3.1). Now we observe the journey of the dimensionless load chunk ($\hat{q}$), from its definition in the knowledge base to its appearance as typeset-ready $\text{\LaTeX}$ formatted text in the generated SRS.

Step 1: Knowledge Capture

The definition of $\hat{q}$ is captured in the Drasil knowledge base as a chunk (Figure 4.27). This chunk contains all necessary information: a unique identifier, symbol, natural

```

130 --DD7--
131
132 dimLLEq :: Expr
133 dimLLEq = mulRe (sy demand) (square (mulRe (sy plateLen) (sy plateWidth)))
134   $/ mulRe (mulRe (sy modElas) (sy minThick $~ exactDbl 4)) (sy gTF)
135
136 dimLLQD :: SimpleQDef
137 dimLLQD = mkQuantDef dimlessLoad dimLLEq
138
139 dimLL :: DataDefinition
140 dimLL = ddE dimLLQD [dRef astm2009, dRefInfo campidelli $ Equation [7]] Nothing "
    dimlessLoad"
141 [qRef, aGrtrThanB, stdVals [modElas], hRef, gtfRef]

```

Figure 4.27: Definition of $\hat{q}$ as a chunk in the GlassBR knowledge base

language description, units, and the defining mathematical expression.

Step 2: Recipe Specification

In this step, the previously defined $\hat{q}$ chunk is referenced and included in the system information and recipe for the GlassBR SRS. Here, $\hat{q}$ is added to the list of Data Definitions (can be seen as `GB.dataDefs` in Figure 4.16), which will later be used to populate the relevant sections (Table of Symbols, Data Definitions, Traceability Matrix, etc.) of the document. This step is about specifying what knowledge is relevant and where it should appear, not about how it is rendered.

Step 3: Generation and Rendering

When the generator is invoked to produce the SRS (ex. as a $\text{\LaTeX}$ document), it traverses the recipe, visiting each section. For the Data Definitions section, it projects the full definition of $\hat{q}$, including its symbol, description, units, and defining equation. For the Table of Symbols, it projects only the symbol, a brief description, and the units. The rendering engine formats these projections according to the conventions of the target output.


```

750 \vspace{\baselineskip}
751 \noindent
752 \begin{minipage}{\textwidth}
753 \begin{tabular}{>{\raggedright}p{0.13\textwidth}>{\raggedright\arraybackslash}p{0.82\textwidth}}
754 \toprule \textbf{Refname} & \textbf{DD:dimlessLoad}
755 \phantomsection
756 \label{DD:dimlessLoad}
757 \\\midrule \\
758 Label & Dimensionless load
759
760 \\\midrule \\
761 Symbol &  $\hat{q}$ 
762
763 \\\midrule \\
764 Units & Unitless
765
766 \\\midrule \\
767 Equation & \begin{displaymath}
768 \hat{q} = \frac{q}{\left(a b\right)^2} E h^4 \mathit{GTF}
769 \end{displaymath}
770 \\\midrule \\
771 Description & \begin{symbDescription}
772 \item{$\hat{q}$ is the dimensionless load (Unitless)}
773 \item{$q$ is the applied load (demand) ( $\text{Pa}$ )}
774 \item{$a$ is the plate length (long dimension) ( $\text{m}$ )}
775 \item{$b$ is the plate width (short dimension) ( $\text{m}$ )}
776 \item{$E$ is the modulus of elasticity of glass ( $\text{Pa}$ )}
777 \item{$h$ is the minimum thickness ( $\text{m}$ )}
778 \item{$\mathit{GTF}$ is the glass type factor (Unitless)}
779 \end{symbDescription}
780 \\\midrule \\
781 Notes &  $q$  is the 3 second duration equivalent pressure, as given in \hyperref[DD:calofDemand]{DD:calofDemand}.
782
783  $a$  and  $b$  are the dimensions of the plate, where ( $a \geq b$ ).
784
785  $E$  comes from \hyperref[assumpSV]{A:standardValues}.
786
787  $h$  is defined in \hyperref[DD:minThick]{DD:minThick} and is based on the nominal thicknesses.
788
789  $\mathit{GTF}$  is defined in \hyperref[DD:gTF]{DD:gTF}.
790
791 \\\midrule \\
792 Source & \cite{astm2009} and \cite[(Eq. 7)]{campidelli}
793
794 \\\midrule \\
795 RefBy & \hyperref[DD:stressDistFac]{DD:stressDistFac}
796
797 \\\bottomrule
798 \end{tabular}
799 \end{minipage}

```

Figure 4.28: Raw generated L^AT_EX of the $\hat{q}$ Data Definition

During rendering, the SRS generator traverses the recipe and, for each chunk referenced, projects and formats it for the target output. For the Data Definitions section, it projects the full definition of $\hat{q}$, including its symbol, description, units, and defining equation (Figure 4.28). For the Table of Symbols, it projects only the symbol, a brief description, and the units. The rendering engine formats these projections according to the conventions of the target output (as a $\text{\LaTeX}$ table or section).

Step 4: Assembly and Output

Finally, the rendered sections are assembled into the complete SRS artifact. Cross-references are automatically generated, so that, for example, references to $\hat{q}$ in Instance Models or Requirements link back to its definition in the Data Definitions and/or Table of Symbols. Auxiliary materials, such as the Table of Units and Table of Abbreviations, if required, are generated by traversing the relevant lists in the system information object, ensuring consistency and completeness. The completed document is output as a `.tex` file with $\hat{q}$ fully integrated and accessible alongside all other relevant auxiliary files (ex. `.bib` file for citations). A makefile is also generated to enable the user to easily create a human-readable PDF file from the $\text{\LaTeX}$ source and auxiliary files.

4.4.4 Multiple Renderings from a Single Source: From Chunk to Code

As a continuation of the previous sections example, we now follow the dimensionless load $\hat{q}$ from its knowledge base definition through to its realization in the generated

source code. This example concretely demonstrates how Drasil's single-source-of-truth approach ensures that mathematical concepts are faithfully and consistently reflected in both documentation and code artifacts.

Referencing $\hat{q}$ in the Code Recipe

Recall that $\hat{q}$ was previously defined as a chunk containing its symbol, natural language description, units, and mathematical formula (see Figure 4.27). In the process of generating executable code, this chunk is referenced in the system information object (`fullSI`) and included in the `CodeSpec` recipe for GlassBR (Figure 4.18). The recipe dictates which variables and formulae are required in the target program, and provides implementation-level choices such as type mapping and function structure.

Rendering via GOOL

During code generation, Drasil translates the recipe and referenced chunks into a GOOL (General Object-Oriented Language) intermediate representation. The $\hat{q}$ chunk, for example, is mapped to a function definition within the GOOL AST, preserving its formula and metadata. This intermediate form enables Drasil to emit code in multiple target languages without duplicating logic. As GOOL is not the subject of this thesis, we will elide the GOOL-specific details and continue down the code generation pipeline.

Generated Source Code

In the final step, the GOOL backend emits concrete code for $\hat{q}$ in each selected language. For example, in Python, $\hat{q}$ appears as a function computing its value from

```
34 ## \brief Calculates dimensionless load
35 # \param inParams structure holding the input values
36 # \param q applied load (demand): 3 second duration equivalent pressure (Pa)
37 # \return dimensionless load
38 def func_q_hat(inParams, q):
39
40     return q * (inParams.a * inParams.b) ** 2.0 / (7.17e10 * inParams.h ** 4.0 *
        inParams.GTF)
```

Figure 4.29: $\hat{q}$ as defined in the generated python code

its dependencies(Figure 4.29) just as we’ve seen before in Figure 3.6. The generated code maintains traceability through naming conventions and comments drawn from the original chunk. The final python output can be seen in Figure 4.29.

Discussion

This example demonstrates how Drasil’s architecture ensures that any update to the definition or formula of $\hat{q}$ is automatically propagated to the generated code, just as it is to documentation artifacts. The use of a knowledge-centric pipeline, combined with language-agnostic intermediate representations, guarantees consistency and traceability throughout the software system. Changes need only be made once, and are reflected everywhere $\hat{q}$ appears, reducing maintenance cost and risk of error.

4.4.5 Summary

In summary, rendering and assembly are the final, crucial steps that operationalize Drasil’s knowledge-centric approach and translate the abstract representations produced by our chunks and recipes into practical, high-quality softifacts. The examples from GlassBR demonstrate the practical effectiveness of this approach for both documentation and code.

The next section will discuss how iteration and refinement were used in the creation of Drasil, and how the framework enables rapid evolution of both knowledge and artifacts.

4.5 Seasoning to Taste: Iteration and Refinement

With the understanding of Drasil’s structure and implementation, we can now discuss continuous improvement. The development of Drasil did not follow anything even remotely close to the waterfall model of software development. If anything its development began with a simple set of ideas that created a basis for refinement. That is to say: it *barely* worked.

We approached the development knowing the scope was far too large to plan out all the details ahead of time (in any reasonable time-frame), and made conscious choices on compromises we were willing to accept to get a minimum viable framework and continuously improve upon it, similar to agile software development practices. As such, and as we have eluded to throughout this and the previous chapter, we took a very practical approach to development by starting with a known-good state (our case studies), generalizing as much as possible (as seen in Chapter 3) to distill what we needed to understand to generate software, and updating the framework as our assumptions were challenged or found wanting. Finding holes is a lot easier when you constantly step in them.

Many of our updates involved modifying implementation details (such as Chunk types, combinators, and adding entirely new concepts) when we started looking at the various case study domains and determining how we’d distill and capture the knowledge necessary to reproduce those case studies.

Throughout this process of iterative refinement, we inevitably found a number of places to improve upon the existing case studies and applied those changes to our known-good state. We found errors and undocumented assumptions in the case studies that caused us to rework and improve them for the future. Crucially, moving the case studies into Drasil made those mistakes trivial to locate and correct. A representative example is the SSP symbol inconsistency (Issue #348 on the GitHub issue tracker). The original SSP SRS used the pair S_i and P_i (mobilized shear stress and resistive shear force respectively) in one place and later switched S_i to τ_i (resistive shear stress). τ_i was also omitted from the Table of Symbols. To reiterate, that inconsistency existed in the original case study and was not a generator bug but an error in the source document we had accepted as our starting point. Once the SSP material was encoded as chunks in Drasil the fix was minimal: ensure we had encoded the correct definitions of S_i , P_i , and τ_i and used them in the appropriate equations. Then the correction automatically propagated to all generated artifacts (Table of Symbols, cross-references, and any code that referenced the symbol).

We also noticed a unit mismatch during the investigation of that issue, which sparked some confusion until we determined there was an unspecified assumption of 1m depth “into the page” for one of the equations. That assumption was later captured and encoded into the specific knowledge for that family of software systems as well. Overall the effort to fix both of these related issues was essentially a focused edit to a minuscule subset of the captured knowledge, not a widespread rewrite.

This pattern repeats across many issues: we would find a mismatch between a case study and our Drasil output, or some as-yet unencoded knowledge (ex. implicit assumptions) that made the generated artifacts inconsistent or unclear, update the

chunk(s) that encoded the offending knowledge, and re-generate. That workflow was the dominant mode of refinement of the original case studies and their Drasil implementations. It kept fixes small, localized, and low cost while producing broad, immediate benefits in every artifact that depended on the corrected knowledge.

As Drasil encodes both semantics (definitions, units, relations) and presentation choices (how a chunk is projected), correcting semantics upstream was also used to eliminate downstream inconsistencies with, often, minimal engineering effort. One example of this is changing our Chunk hierarchy to include derived-unit chunks. After we determined the need for **Unital** chunks which tied units to captured knowledge, it was obvious that the units themselves would need to be able to represent their relationships, particularly for those derived from standard SI units (ex. m/s^2). This involved the addition of new combinators for creating **UnitDefn** chunks (examples of some of these combinators have already been shown in Figure 4.7).

In short, we learned how to improve Drasil by trying to reproduce existing case studies, encoding what we needed as chunks, and then correcting or extending the chunks when mismatches were discovered. Fixes were either small edits to the knowledge base (cheap to apply, and automatically, consistently propagated to all generated outputs) or an extension of the way we chose to capture knowledge (new chunk types or combinators to encode previously un-encodable knowledge or relationships between chunks).

4.6 Summary

In this chapter we have described the architecture and implementation choices that make Drasil practical: a knowledge-centric pipeline built from reusable, typed “chunks”,

lightweight DSLs for expressions and natural language fragments, and a recipe language that codifies how knowledge is projected into concrete artifacts. The design is intentionally pragmatic: we capture what is needed to reproduce real case studies, provide well-typed building blocks for assembling that knowledge, and offer generators (printers and the GOOL back end) that render the same source into multiple, consistent outputs. The result is a single source of truth that spans documentation and executable code.

By separating the concerns of knowledge capture, structuring, rendering, and assembly, Drasil ensures that softifacts remain maintainable, consistent, traceable, and free of unnecessary manual duplication from their high-level specification down to their concrete instances. All of which increases confidence in the resulting software systems.

The chunk hierarchy and its classification system give us the levers we need to balance expressivity and reuse. Simple chunks (`NamedChunk`, `ConceptChunk`) let us represent terminology and definitions; richer chunks (`UnitalChunk`, `UnitDefn`, `DataDefn`, `InstanceModel`) let us attach units, symbols, and executable expressions; and the `Expr / Sentence` DSLs give us language-agnostic ways to express mathematics and narrative. Recipes then select and arrange those chunks for particular stakeholders so that the same underlying knowledge can be rendered as an SRS, a module guide, or multi-language source code with full traceability between pieces.

The iterative approach to Drasil’s development ensured modularized components that improve extensibility and maintainability of the system as a whole, as well as continuous refinement and expansion of the system’s capabilities. A key practical lesson is that encoding case studies into Drasil exposes errors and implicit assumptions

in the original artifacts but makes them trivial to fix. Fixes are usually local edits to the knowledge base and those edits automatically propagate through all generated artifacts. This demonstrates the principal payoff of the single-source, chunk-based approach: small, cheap changes at the source yield broad, consistent improvements everywhere.

Drasil is not a universal solution—it is a tool targeted at domains where the models are well understood, long-lived, and shared across multiple stakeholders. Within that scope, Drasil shifts effort from repetitive editing of many artifacts to careful modeling of domain knowledge. The framework we have presented allows that modeling to pay dividends immediately (consistent documents and code) and continuously (streamlined evolution through iteration and refinement).

Chapter 5

Results

In this chapter we discuss our observations following the reimplementation of our case studies using Drasil. Content here is primarily illustrative: concrete examples, issue identifiers, and sample artifacts that document how the single-source approach affected consistency, traceability, and reimplementation effort. Formal, controlled experiments are future work (see Chapter 7.3.8). For interpretation and implications of these observations, see Chapter 6.

5.1 Value in the mundane

This section highlights routine, practical benefits that emerged from the reimplementations. Each subsection focuses on a specific, often-overlooked advantage — consistency, traceability, and reproducibility — and shows how these “mundane” improvements materially reduce maintenance effort and increase confidence in generated artifacts.

5.1.1 Consistency by Construction

Drasil’s single-source knowledge representation produced artifacts that are consistent by construction. Small, localized edits to domain knowledge or recipes propagated deterministically to all generated outputs (documentation and code). During reimplementations, this property reduced the number of manual synchronization tasks normally required when maintaining separate artifacts. In practice, consistency by construction meant that small corrections such as renaming a quantity to use a single canonical identifier, fixing a typographical error in a definition or table entry, or revising the wording of an assumption were made once in the knowledge base and then appeared immediately and uniformly across the SRS, reference tables, and generated code after regeneration. Rather than checking each artifact individually, regeneration ensured that every occurrence reflected the same correction.

Additionally, all domain knowledge must be explicitly encoded, which prevents dangling references or undefined symbols as observed in a number of early passes over the original case study documentation (for example, issue #348 in the issue tracker which will be discussed in more detail in Section 5.2.1 and Section 6.1.3).

5.1.2 Traceability and Maintainability

Every generated element (formulae, symbols, tables, code identifiers) can be traced back to a named knowledge chunk or recipe. The generation pipeline produces traceability matrices and reference-material tables as part of normal output, enabling explicit provenance for items that are typically undocumented. This traceability materially reduced the cognitive effort required to locate the origin of a definition or relation during maintenance tasks and refactoring: it was trivial to identify the

source to edit rather than hunting through multiple artifacts.

For example, in the GlassBR case study (see issue #349), a unit conversion was embedded implicitly in the original documentation. With Drasil, it was straightforward to use the generated traceability information to identify the exact chunk from which the formula was produced. That made it clear where the conversion logic belonged: instead of leaving the division by 1000 implicit in the rendered equation, we made the conversion explicit in the source chunk itself. After regeneration, the corrected expression appeared consistently in the SRS and in every downstream artifact derived from that chunk, demonstrating more explicitly how traceability supports single-point maintenance.

5.1.3 Reproducibility and Determinism

Artifact generation in Drasil is fully deterministic: repeated generation runs from the same knowledge base and recipe set produce byte-for-byte equivalent outputs for generated softifacts as compared to the then-current stable versions. These outputs were confirmed using a diff tool that runs as part of an automated regression test suite.

Furthermore, repeatability checks executed during the case study reimplementations confirmed that regenerating artifacts after environment-consistent builds did not introduce nondeterministic differences, supporting reproducibility guarantees that are important for scientific software.

5.2 Observations: Case Studies and Reimplementations

We reimplemented each of our case studies using Drasil and recorded measurable outcomes for generated artifacts and the knowledge source that produced them.

For the GlassBR case study, the Drasil implementation generated a multi- page SRS (44 pages in the final PDF) and produced working implementations in multiple target languages. The complete SRS recipe for GlassBR is contained in a single source file of under 370 lines of Haskell (including comments and imports). This line count refers only to the GlassBR-specific recipe and does not include the shared base knowledge (units, symbols, common chunks, and generic recipes) that are stored separately in the global knowledge base and are reused across case studies.

Table 5.1: Summary statistics for Drasil case study SRS reimplementations

Case Study	SRS Pages	Recipe LOC ¹
GlassBR	44	370
SWHS	50	560 ²
NoPCM	32	316
SSP	84	380
Projectile	31	209
GamePhysics	46	424

¹ Including comments and import statements

² This is an outlier due to excessive use of comments. Without comments it would be 386 lines.

Comparable results were observed for other reimplemented case studies: each system produced a multi-page SRS and a modest-sized recipe or set of recipes (tens to a few hundred lines) that captured the system-specific knowledge. See Table 5.1

for a summary. Generated code for each system followed the same pattern: language-agnostic knowledge encoded as chunks and recipes was emitted to multiple target languages with consistent identifiers and traceability annotations.

Generated code and documentation remained consistent after edits to the knowledge base, with regeneration producing updated artifacts across all targets. For example, the GlassBR unit-conversion correction from issue #349 was made once in the source knowledge and then propagated to the SRS and generated code after regeneration, while the NoPCM variant was produced from the SWHS knowledge base by excluding PCM-related chunks rather than rewriting the specification. These examples are discussed further in Section 5.2.1, Section 5.2.2, and in the interpretive discussion in Section 6.1.2.

5.2.1 Quality and Bug Detectability

Because all artifacts are derived from the same knowledge, a defect in the knowledge base manifested consistently across every generated artifact. This property made defects more visible (“pervasive bugs”), which in turn aided discovery and correction. During reimplementations, categories of problems that were revealed included inconsistent symbol naming, implicit numeric constants, and unstated domain assumptions. Fixing those issues at the knowledge level eliminated their occurrences across documentation and code simultaneously.

An instructive, concrete example is issue #348 from the issue tracker. In the original SSP artifacts several closely related symbols (for shear stress and associated forces) were used inconsistently across sections and tables. When the SSP content was encoded as Drasil chunks these inconsistencies became immediately visible: the

same semantic item was projected with conflicting identifiers, and cross-references did not align. Correcting the underlying chunk in the knowledge base removed the inconsistencies from all generated artifacts (SRS and code) after regeneration. During the investigation of the same issue, it was also discovered that one of the terms had an implicit unit vector “into the page” that was not documented in the original artifacts. Encoding the unit vector explicitly in Drasil made that assumption visible, allowing it to be documented properly in the generated SRS and ensuring that the generated code reflected the correct physical interpretation.

A similar example arose in the GlassBR case study (see issue #349), where an implicit division by 1000 for unit conversion was embedded in the original documentation. Drasil’s explicit chunk encoding surfaced this hidden assumption, enabling a single-point correction that propagated to all generated artifacts. Similar issues surfaced throughout the reimplementations, including inconsistent symbol names, missing symbol definitions, implicit unit conversions, undocumented unit vectors, mismatched units across tables and equations, and unstated domain assumptions. Taken together, these recurring issue types reinforced that centralized knowledge capture makes such problems easier to detect, localize, and correct.

5.2.2 Design for Change and Software Families

The single-source approach enabled rapid generation of software-family variants. Where artifacts differed only by parameterization or small domain choices (for example, SWHS with PCM vs NoPCM variants), encoding those choices in the knowledge base allowed whole-family updates by editing a small set of chunks. This capability materially reduced the effort to produce consistent variants of the same artifact

family.

Table 5.2: Key differences between SWHS and NoPCM Drasil implementations

Chunk/Concept	SWHS	NoPCM	Notes
PCM properties	Yes	No	Only in SWHS
Latent heat	Yes	No	Only in SWHS
Sensible heat	Yes	Yes	Present in both
Energy balance	Yes	Yes	PCM terms omitted in NoPCM
PCM input parameters	Yes	No	Only in SWHS
PCM assumptions	Yes	No	Only in SWHS
Tank/water properties	Yes	Yes	Shared
Recipes	Modified	Modified	NoPCM recipe omits PCM-related content

For example, producing the NoPCM variant from SWHS required removing all reference to PCM by excluding the PCM-related chunks. The majority of the knowledge base and recipes were reused without change, demonstrating the efficiency of the single-source approach for software families. Table 5.2 summarizes the key differences between the SWHS and NoPCM Drasil implementations, highlighting which chunks and concepts were unique to each variant and which were shared.

5.3 Usability and Adoption

From the reimplementation work it became clear that Drasil’s expressiveness comes with an onboarding cost. Contributors familiar with the codebase could encode and modify domain knowledge effectively; new users required a nontrivial ramp-up period of up to two weeks of mentoring to understand the chunk combinators and recipe

patterns. Common challenges included understanding how to build new chunks, particularly while referencing other chunks, and how to find existing chunks of the appropriate type in the knowledge base.

Feedback collected during onboarding, along with the observations listed above, indicated that improved search tools and documentation would significantly reduce ramp-up time. Nevertheless, undergraduate contributors and short-term students were able to make meaningful contributions after initial mentoring.

5.4 Summary and Takeaways

The combined results from the case studies show that the primary benefits of the Drasil approach are increased consistency, strong provenance for generated artifacts, and deterministic reproducibility. These benefits manifested in lower ongoing synchronization effort, higher visibility of domain assumptions and defects, and faster family-wide changes when those domain choices were encoded explicitly. The observations collected during the reimplementations supports these claims while also documenting the practical cost of initial knowledge capture and onboarding.

Chapter 6

Discussion

This chapter interprets the results reported in Chapter 5. We synthesise the practical benefits observed — automation, traceability, and rapid adaptation — evaluate costs such as onboarding and modelling effort, and outline limitations and directions for formal validation. We refer back to observations in Chapter 5 and cite illustrative issue examples.

6.1 Interpreting the Results

This section organises the principal outcomes from Chapter 5 into interpretation themes that mirror the observations: consistency, traceability, reproducibility, and the tradeoffs that follow from centralizing knowledge.

6.1.1 Consistency by construction

Drasil projects a single knowledge base into diverse artifacts. The case studies demonstrate that local changes to domain knowledge yield system-wide corrections. Practically, this reduces the maintenance surface area: a single canonical edit replaces repeated, artifact-level fixes. The primary tradeoff is increased upfront modelling effort and the need for disciplined knowledge capture.

Disciplined knowledge capture also forces clarification of implicit assumptions that would otherwise lead to inconsistencies. For example, in the SSP case study (Issue #348, Section 5.2.1), inconsistent symbol naming across the original artifacts led to confusion and errors in equations. These were only caught after careful review during reimplementaion. Encoding the symbols as Drasil chunks enforced a single canonical name and identifier, eliminating the inconsistencies across all generated artifacts.

Similarly, as we are forced to encode all domain knowledge explicitly in Drasil, it is impossible to end up in a situation where we have undefined symbols or equations that reference missing quantities. A concrete example of this was discovered while investigating the same issue mentioned above (issue #348), where during reimplementaion we discovered the symbol τ_i was used in equations in the original SRS, but did not appear in the table of symbols or have a formal definition. When encoding the knowledge in Drasil, we had to define these symbols explicitly as chunks, which ensured that they were properly documented in the generated SRS and correctly referenced in the generated code.

A further benefit of this approach is that several supporting sections, such as the many tables in the Reference Materials section of the SRS as well as the traceability matrices, are generated automatically from the structure of the knowledge in use.

These sections, which are often neglected or inconsistently maintained in traditional workflows (as per the issue mentioned above), are kept up to date “for free” as a consequence of Drasil’s single-source knowledge capture, further reinforcing consistency and reducing manual effort.

6.1.2 Traceability and refactoring

Centralized knowledge makes provenance explicit: formulae, symbols, and identifiers trace back to named chunks or recipes. That provenance eases refactoring because changes localized to chunks produce immediate updates across downstream artifacts. The recipe DSL allows reorganizing output structures (documentation sections, code modules, etc.) without manual edits to each target artifact.

A key advantage of Drasil’s architecture is the separation of concerns between knowledge and presentation. The structure of generated artifacts can be modified (“on the fly”) by changing recipes, without altering the underlying knowledge base. For example, generated code can be adapted to include features like logging by updating code generation recipes alone. This enhances maintainability, traceability, and adaptability of the generated artifacts.

Drasil also supports the systematic evolution of software families. By separating domain knowledge from artifact-specific recipes, Drasil enables controlled introduction of variation points. New family members can be generated by modifying recipes or selectively extending the knowledge base, rather than duplicating and manually editing artifacts. This reduces the risk of divergence and inconsistency among related products.

A concrete example is seen in the development of the SWHS and NoPCM case

studies (see Section 5.2.2 for related results). Both are members of the same software family, sharing core physical concepts and requirements but differing in specific details. Because the fundamental knowledge was already encoded for SWHS, creating NoPCM required only targeted extensions and modifications to existing chunks and recipes. This reuse minimized duplication, ensured consistency across both projects, and allowed rapid adaptation of documentation and code for NoPCM. The shared knowledge base made it straightforward to propagate improvements or corrections from one family member to the other, further demonstrating the advantages of Drasil’s approach for software families.

When adapting a requirements specification for a new variant, contributors can introduce new chunks or adjust parameters in the knowledge base, while reusing existing recipes to maintain structure and traceability. Alternatively, recipes can be specialized to produce tailored documentation or code modules for specific family members, all while referencing the same core knowledge. This design for change streamlines the creation and maintenance of software families, and supports rapid, reliable adaptation to evolving requirements.

6.1.3 Reproducibility and error visibility

Artifact generation was deterministic in our case studies: regenerating from the same knowledge base produced consistent textual outputs. At the same time, defects in the knowledge base appear across all generated artifacts, which increases visibility and simplifies root-cause correction.

Recall Issue #348 from SSP mentioned in Section 5.2.1. To summarize, the issue was that some closely related symbols had their uses mixed up in several places in

the original case study. In more detail, the SSP inconsistencies manifested in three concrete ways:

1. Multiple names and short identifiers for the same physical quantity across equations and tables.
2. Implicit or mismatched units appearing in different tables.
3. References and captions that used variant labels, breaking cross-references.

Encoding the SSP concepts as chunks forced a single canonical name, an explicit unit declaration, and a consistent label for each concept that could not get switched to a different symbol.

Using Drasil, the remediation followed a small, local workflow: identify the semantic chunk representing the quantity, unify the identifier and unit within that chunk, and regenerate. The result was immediate and verifiable: the SRS, tables, and generated code all used the same identifier and unit, table cross-references resolved, and no further manual edits were necessary. This example illustrates how centralized knowledge surfacing both increases error visibility and enables concise, single-point fixes.

6.2 Human Factors and Usability

Experienced contributors adapted quickly to Drasil’s patterns, but new users found the knowledge-capture DSL challenging. Two priorities emerge: improved discovery and search tools for the knowledge base, and higher-level authoring interfaces that present domain concepts in more familiar terms. In short: Drasil desperately needs better tooling for authoring.

6.3 Limitations and Directions for Validation

The current evidence is largely qualitative and anecdotal. Formal, controlled experiments are required to measure productivity gains, maintenance cost reduction, and error rates across larger teams. Until such studies are performed, claims about absolute gains should be framed as preliminary.

Future validation should include controlled studies measuring onboarding metrics (time to first meaningful edit, amount of mentoring required), maintenance effort, and error rates across teams of varying experience. Experimental designs could compare Drasil-based workflows to traditional manual approaches, quantifying differences in initial productivity, long-term maintenance, and defect reduction. These metrics will be essential for substantiating the qualitative benefits observed in this thesis. Feedback from such studies will also guide the development of improved authoring tools and onboarding resources, helping to lower the initial ramp-up cost and make Drasil accessible to a broader range of users.

Scalability concerns also merit attention: as the knowledge base grows, improved tools for discovery, efficient projection, and modularization will be essential to maintain developer productivity.

Drasil’s current implementation is also limited primarily to scientific computing software that follows an input $\rightarrow$ process $\rightarrow$ output (IPO) paradigm. This constraint arises because such domains typically feature well-defined requirements, structured models, and predictable outputs, making it feasible to encode domain knowledge explicitly and project it into multiple artifacts.

In contrast, domains with less structured or more dynamic requirements—such as

interactive systems, event-driven applications, or software with complex stateful behaviors—present significant challenges for knowledge capture and automated artifact generation. The underlying information in these domains may not be as readily formalized or reused, limiting the effectiveness of Drasil’s current approach. Extending Drasil to support these broader classes of software will require advances in flexible modeling, knowledge representation, and recipe design.

6.4 Lessons Learned

1. Many dangerous inconsistencies in legacy artifacts trace back to tacit assumptions and implicit knowledge during requirements capture; making those assumptions explicit in chunks forces clarification and improves both maintainability and traceability.
2. Drasil’s architecture shifts the correctness burden from artifact authors to knowledge-base modelers: this trades repeated artifact-level maintenance for concentrated, knowledge-level review.
3. Rapid case-study reimplementations and ease of propagation suggest the approach can scale across domains given improved onboarding and validation, provided the domain knowledge can be effectively captured.
4. The recipe DSL enables flexible artifact structuring and adaptation without modifying the underlying knowledge, supporting maintainability and evolution.

5. Centralized knowledge capture increases error visibility by making defects pervasive across all generated artifacts, simplifying root-cause analysis and correction.
6. The single-source approach facilitates rapid generation of software-family variants by enabling controlled introduction of variation points in the knowledge base and recipes, reducing duplication and ensuring consistency across related products.
7. Improved authoring tools and onboarding processes are critical to lower the initial ramp-up cost and make Drasil accessible to a broader range of users.
8. Formal, controlled studies are needed to quantify productivity gains, maintenance cost reductions, and error rates to validate the qualitative observations reported here.
9. Deterministic, automated artifact generation supports scientific reproducibility by ensuring that repeated builds from the same knowledge base yield identical outputs, reducing the risk of accidental divergence and increasing confidence in results.
10. Centralized, structured knowledge enables the automatic generation and consistent maintenance of supporting materials such as traceability matrices and reference tables that are often neglected or inconsistently updated in traditional workflows.
11. The development and reporting of quantitative metrics such as onboarding time,

regeneration timings, and maintenance effort are essential for guiding future improvements and validating the practical impact of the approach.

12. Drasil’s extensible architecture allows it to adapt to new requirements and scientific advances by incorporating richer or more abstract domain knowledge, future-proofing the system as knowledge capture mechanisms evolve.

6.5 Summary and Takeaways

Drasil’s single-source approach delivers several practical benefits. Consistency is enforced by projecting a single knowledge base into all artifacts, ensuring that local edits yield system-wide corrections and eliminating inconsistencies common in manual workflows. Traceability is improved, as provenance for formulae, symbols, and requirements is explicit and changes are automatically propagated across documentation and code. Reproducibility is enhanced by automating artifact generation, reducing the risk of human error and ensuring that supporting materials such as traceability matrices and reference tables are always up-to-date.

Automation further reduces manual effort, especially when generating new projects or adapting existing ones. Boilerplate text and artifact structure are handled by recipes, allowing contributors to focus on encoding domain knowledge. This approach also supports the systematic evolution of software families: new variants can be created by extending the knowledge base or modifying recipes, minimizing duplication and maintaining consistency across related products.

The primary tradeoff is increased upfront modeling and onboarding effort. The benefits of Drasil’s approach are most pronounced when a substantial portion of

domain knowledge can be reused, but even in new projects, the reduction in manual duplication and the gains in maintainability are significant. Continued investment in authoring tools and onboarding resources will help lower the initial ramp-up cost.

Drasil is designed for extensibility. As our mechanisms for knowledge capture improve, we will be able to encode more fundamental domain concepts and extend the scope of the knowledge base. This extensibility means Drasil can adapt to new requirements or scientific advances by incorporating richer or more abstract representations, rather than requiring a redesign of the system. Continued improvement of knowledge capture tooling and methodologies will directly increase the breadth and depth of what Drasil can represent and generate, further amplifying its benefits for consistency, traceability, and automation.

The observations in Chapter 5 support these conclusions and point to concrete next steps: measure onboarding overhead, report regeneration timings, and develop discovery and authoring tools to reduce the initial ramp-up. Drasil’s architecture positions it well for future growth as both a research platform and a practical tool for reliable, maintainable scientific software.

Chapter 7

Conclusion

This thesis set out to address the persistent challenges of consistency, traceability, and maintainability in scientific software and documentation. The introduction established the need for a principled, automated approach to knowledge capture and artifact generation, motivating the development and evaluation of Drasil.

The results chapter provided concrete evidence that Drasil’s single-source architecture enforces consistency by construction, improves traceability through explicit provenance, and enhances reproducibility via deterministic artifact generation. Case studies demonstrated that local changes to domain knowledge propagate automatically across all generated outputs, reducing manual effort and the risk of inconsistencies. The results also highlighted practical benefits such as the automatic maintenance of supporting materials and the ease of creating new software family members by reusing and extending the knowledge base.

The discussion chapter interpreted these findings, emphasizing that Drasil’s approach shifts the burden of correctness to the knowledge base, clarifies implicit assumptions, and enables rapid, reliable adaptation to new requirements. It also acknowledged tradeoffs, such as the increased upfront modeling and onboarding effort, and identified the need for improved tooling and formal validation.

The primary contribution of this thesis is the demonstration that a centralized, recipe-driven knowledge base can systematically address many of the recurring problems in scientific software engineering. By explicitly capturing domain knowledge and automating artifact generation, Drasil provides a scalable, extensible foundation for reliable, maintainable, and reproducible scientific software. The lessons learned and directions for future work outlined here position Drasil as both a practical tool and a platform for further research in automated knowledge capture and software generation.

7.1 Summary of Main Contributions

This thesis presents the design and implementation of Drasil, a centralized, recipe-driven knowledge base created as part of this research to systematically address persistent challenges in scientific software engineering. Through a systematic analysis of the commonalities and differences among software artifacts, this work operationalizes our understanding of software engineering principles, enabling their direct application in automated artifact generation. By explicitly capturing domain knowledge and automating the generation of software and documentation artifacts, Drasil enforces consistency by construction, improves traceability through explicit provenance, and

enhances reproducibility via deterministic outputs. Case studies presented in Chapter 5 show that local changes to domain knowledge propagate automatically across all generated outputs, reducing manual effort and minimizing the risk of inconsistencies. The architecture described in Chapter 4, refined through iterative development and feedback, enables the reuse and extension of the knowledge base, facilitating the creation of new software family members and supporting the automatic maintenance of supporting materials. Collectively, these contributions provide a scalable and extensible foundation for reliable, maintainable, and reproducible scientific software.

7.2 Limitations and Open Questions

While Drasil demonstrates significant progress toward automating knowledge capture and artifact generation, several limitations remain. The approach requires substantial upfront modeling effort and domain expertise, which may pose a barrier to adoption for new users or projects. Tooling and usability challenges persist, including the need for improved front-end interfaces and developer experience plugins to streamline onboarding and knowledge capture.

Formal validation and empirical evaluation of Drasil’s effectiveness are ongoing, and further experiments are needed to establish its generalizability across diverse scientific computing domains. At present, Drasil is primarily limited to scientific computing software that adheres to an input $\rightarrow$ process $\rightarrow$ output paradigm, where requirements and models are well-understood and structured. As the knowledge base grows, scalability and efficient knowledge management will become increasingly important challenges, particularly for discovery, projection, and modularization of domain knowledge.

Extending Drasil to more dynamic or less formalized fields remains an open question. These limitations and open questions, discussed in Chapter 6, highlight important directions for future work, which are outlined in the following section.

7.3 Future Work

Building on the foundation established in this thesis, several avenues for future work are apparent. As Drasil continues to evolve, its ongoing development will be shaped by both practical experience and feedback from real-world adoption. The directions outlined below reflect the need to broaden Drasil’s applicability, improve its usability, and deepen its integration into scientific software engineering workflows. The following subsections highlight key areas where further research and development can have the greatest impact.

7.3.1 Expanding the Knowledge Base and Knowledge Capture Mechanisms

As Drasil is adopted for new projects, the encoded chunk database will naturally expand. This gradual growth is essential for increasing the breadth and depth of reusable domain knowledge, which in turn enhances the quality and variety of generated artifacts. Each new project contributes additional chunks, models, and recipes, making it easier to support related domains and reducing the effort required for future knowledge capture. Over time, a richer knowledge base will enable Drasil to generate more comprehensive documentation and software, facilitate cross-project reuse, and improve consistency across scientific computing applications.

Similarly, designing more flexible and extensible knowledge capture mechanisms is essential for broadening Drasil’s applicability. Currently, the types of information encoded as chunks are fairly primitive, often limited to basic definitions, equations, and simple relationships. Future work should focus on advancing the sophistication of chunks to enable Drasil to generate higher-quality artifacts and address more nuanced requirements. Additionally, developing new methods and interfaces for knowledge capture (such as guided authoring tools, semi-automated chunk inference, and support for alternative modeling paradigms) will allow domain experts to contribute knowledge more efficiently and facilitate extensibility into new scientific fields.

7.3.2 Inference Improvements

A key direction for future work is improving Drasil’s ability to infer knowledge chunks and their relationships directly from recipes and other high-level specifications. Currently, much of the chunk structure and dependency information must be defined explicitly, which can be tedious and error-prone as projects scale. Automating chunk inference would reduce the need for manual list definitions and make knowledge capture more efficient and less repetitive. Achieving this will require advances in both recipe design and the underlying inference passes. Progress in this area will streamline the authoring process, lower the barrier to entry for new users, and further enhance Drasil’s scalability and maintainability.

7.3.3 Typed Expression Language

Developing a robust typed expression language within Drasil is a promising direction for future work. Currently, expressions are represented in a way that limits the ability

to perform deep sanity checks or enforce domain-specific constraints at the knowledge level. Introducing a typed expression language would enable more rigorous validation of mathematical models, formulae, and relationships before code or documentation is generated. This would help catch errors early, ensure consistency across artifacts, and support advanced features such as type-driven code generation and automated reasoning about units and dimensions. Ultimately, a typed expression language would increase the reliability and expressiveness of Drasil’s knowledge base, making it easier to support complex scientific domains and reduce the risk of subtle errors in generated artifacts.

7.3.4 Usability Enhancements

Improving usability is critical for lowering barriers to adoption and ensuring that domain experts can efficiently contribute and manage knowledge. Future work should focus on developing intuitive tooling, such as visual front-ends, drag-and-drop interfaces, and plugins for popular development environments. Enhanced authoring tools will streamline the process of defining and refining chunks, while improved onboarding resources and guided workflows will help new users become productive more quickly. Tooling for cross-referencing existing chunks will prevent duplication and facilitate knowledge discovery, supporting both the expansion and sophistication of the knowledge base. Collectively, these usability enhancements will make Drasil more accessible and effective for a wider range of users and projects.

7.3.5 Output Formats and Artifact Types

Expanding the range of output formats and artifact types is essential for making Drasil applicable to a wider array of audiences and use cases. Future work should focus on supporting additional programming languages (by expanding the existing GOOL backend) and document formats, such as docx or other widely used file types.

Introducing new artifact types, (ex. Jupyter notebooks, journal papers), will further extend Drasil’s utility beyond traditional software documentation. These enhancements will enable Drasil to serve the diverse needs of scientific computing practitioners, educators, and researchers, and will help bridge the gap between code, documentation, and knowledge dissemination.

7.3.6 Natural Language Variants

Supporting multiple natural language variants and context-sensitive artifact generation will significantly enhance Drasil’s adaptability and accessibility. This capability is especially important for international and interdisciplinary teams, where documentation and software artifacts may need to be produced in different languages tailored to specific audiences. Future work should focus on developing mechanisms for encoding and selecting language variants within the knowledge base, as well as strategies for generating artifacts that reflect cultural and contextual differences. These improvements will broaden Drasil’s reach and facilitate collaboration across diverse scientific communities.

7.3.7 Test Case Generation

Developing more robust automated test case generation is essential for ensuring the reliability and correctness of artifacts produced by Drasil. Future work should focus on enhancing the mechanisms for generating comprehensive and meaningful test cases that reflect both domain knowledge and software requirements. Improvements in this area will help verify artifact integrity and support the adoption of Drasil in safety-critical and high-assurance scientific computing projects.

7.3.8 Experiments and Validation

Formal experiments and empirical validation are needed to rigorously assess Drasil’s effectiveness and generalizability across different domains and project types. Future work should include controlled studies measuring onboarding metrics (such as time to first meaningful edit and mentoring required), maintenance effort, and error rates across teams of varying experience. Comparative experimental designs could evaluate Drasil-based workflows against traditional manual approaches, quantifying differences in initial productivity, long-term maintenance, and defect reduction. Gathering quantitative evidence through these studies will help demonstrate the practical benefits of Drasil’s approach, identify limitations, and guide further improvements. Feedback from such experiments will also inform the development of improved authoring tools and onboarding resources, helping to lower the initial ramp-up cost and make Drasil accessible to a broader range of users. Demonstrating Drasil’s impact through systematic experimentation will be crucial for broader acceptance within the scientific computing and software engineering community.

7.3.9 Extending Beyond Input $\rightarrow$ Process $\rightarrow$ Output

Extending Drasil beyond the input $\rightarrow$ process $\rightarrow$ output paradigm represents a significant challenge and opportunity for future work. Many scientific and engineering applications involve interactive, event-driven, or stateful behaviors that do not fit neatly into the current modeling framework. Addressing these domains will require advances in flexible modeling, chunk representation, and recipe design, enabling the capture and projection of more complex knowledge structures. Progress in this direction will broaden Drasil’s applicability to a wider range of software systems, including those with dynamic requirements and less formalized processes, and will help realize the vision of fully automated, knowledge-centric artifact generation across scientific computing.

As Drasil continues to develop, ongoing research and collaboration will be essential to realizing its full potential and advancing the practice of automated, knowledge-centric scientific software engineering.

7.4 Broader Impact and Personal Reflection

The development of Drasil represents a step toward more principled and automated scientific software engineering. By centralizing domain knowledge and enabling systematic artifact generation, Drasil has the potential to improve reliability, reproducibility, and maintainability across a wide range of scientific computing projects. Its approach may influence future tools and methodologies, encouraging the adoption of knowledge-centric practices in both research and industry.

Drasil’s evolution has been shaped by its iterative development process, where

feedback from case studies and practical use continually informed refinement of both the knowledge base and framework itself. This iterative approach has enabled Drasil to adapt to new requirements, address unforeseen challenges, and incrementally improve its capabilities while maintaining validated, working case study software projects.

On a personal level, this work has highlighted the importance of interdisciplinary thinking and the value of bridging gaps between software engineering theory and practical implementation. The challenges encountered throughout the design and development of Drasil have reinforced my understanding of the need for flexibility, collaboration, and ongoing refinement. The experience has been both intellectually rewarding and formative, shaping my perspective on the future of scientific software.

7.5 Closing Remarks

This thesis has demonstrated the feasibility and benefits of a knowledge-centric approach to scientific software engineering. By operationalizing software engineering principles and automating the generation of software and documentation artifacts, Drasil offers a scalable solution to persistent challenges of consistency, traceability, and maintainability. The work presented here lays the groundwork for further advances in knowledge capture and automated artifact generation, and invites continued research and collaboration to realize the full potential of this approach.

As the scientific computing community evolves, tools like Drasil may play an increasingly important role in supporting reliable, reproducible, and adaptable software systems. The lessons learned from this work highlight the value of centralized knowledge, systematic artifact generation, and the importance of ongoing refinement. Continued investment in these principles will help shape the future of scientific software

engineering, ensuring that software remains trustworthy, maintainable, and responsive to new challenges.

Appendix A

HGHC SRS

HGHC Software Requirements Specification

The following appendix contains the full HGHC SRS document.

SRS for h_g and h_c

Spencer Smith

October 21, 2014

Table of Units

Throughout this document SI (Système International d'Unités) is employed as the unit system. In addition to the basic units, several derived units are employed as described below. For each unit, the symbol is given followed by a description of the unit with the SI name in parentheses.

m	- for length (metre)
kg	- for mass (kilogram)
s	- for time (second)
K	- for temperature (kelvin)
$^{\circ}C$	- for temperature (centigrade)
J	- for energy (joule, $J = \frac{\text{kgm}^2}{\text{s}^2}$)
cal	- for energy (calorie, $\text{cal} \approx 4.2 \frac{\text{kgm}^2}{\text{s}^2}$)
mol	- for amount of substance (mole)
W	- for power (watt, $W = \frac{\text{kgm}^2}{\text{s}^3}$)

Table of Symbols

The table that follows summarizes the symbols used in this document along with their units. The choice of symbols was made with the goal of being consistent with the nuclear physics literature and that used in the FP manual. The SI units are listed in brackets following the definition of the symbol.

h_c	- convective heat transfer coefficient between clad and coolant ($\frac{\text{kW}}{\text{m}^2\text{C}}$)
h_g	- effective heat transfer coefficient between clad and fuel surface ($\frac{\text{kW}}{\text{m}^2\text{C}}$)

1 Data Definitions

Number	DD1
Label	h_g
Units	$ML^0t^{-3}T^{-1}$
SI equivalent	$\frac{\text{kW}}{\text{m}^2\text{°C}}$
Equation	$h_g = \frac{2k_c h_p}{2k_c + \tau_c h_p}$
Description	h_g is the gap conductance τ_c is the clad thickness h_p is initial gap film conductance k_c is the clad conductivity NOTE: Equation taken from the code
Sources	source code

Number	DD2
Label	h_c
Units	$ML^0t^{-3}T^{-1}$
SI equivalent	$\frac{\text{kW}}{\text{m}^2\text{°C}}$
Equation	$h_c = \frac{2k_c h_b}{2k_c + \tau_c h_b}$
Description	h_c is the effective heat transfer coefficient between the clad and the coolant τ_c is the clad thickness h_b is initial coolant film conductance k_c is the clad conductivity NOTE: Equation taken from the code
Sources	source code

Bibliography

- [1] Karen S. Ackroyd, Steve H. Kinder, Geoff R. Mant, Mike C. Miller, Christine A. Ramsdale, and Paul C. Stephenson. 2008. Scientific Software Development at a Research Facility. *IEEE Software* 25, 4 (July/August 2008), 44–51.
- [2] Shereef Abu Al-Maati and Abdul Aziz Boujarwah. 2002. Literate software development. *Journal of Computing Sciences in Colleges* 18, 2 (2002), 278–289.
- [3] I Maria Attarian, Kacper Bak, and Leonardo Passos. 2013. Software Product Line Evolution: the Linux Kernel. *URL: <http://gsd.uwaterloo.ca/sites/default/files/cs846w11report.pdf>*. *Abgerufen am* 20 (2013), 2.
- [4] Stefan Baumgartner. 2016. *Front-End Tooling with Gulp, Bower, and Yeoman*. Simon and Schuster.
- [5] Jan Bosch. 2000. *Design and use of software architectures: adopting and evolving a product-line approach*. Addison-Wesley.
- [6] Jacques Carette, Spencer W. Smith, and Jason Balaci. 2023. Generating Software for Well-Understood Domains. In *Eelco Visser Commemorative Symposium (EVCS 2023) (Open Access Series in Informatics (OASICS))*, Ralf Lämmel, Peter D. Mosses, and Friedrich Steimann (Eds.), Vol. 109. Schloss Dagstuhl –

- Leibniz-Zentrum für Informatik, Dagstuhl, Germany, 7:1–7:12. <https://doi.org/10.4230/OASIcs.EVCS.2023.7>
- [7] Jeffrey C. Carver, Richard P. Kendall, Susan E. Squires, and Douglass E. Post. 2007. Software Development Environments for Scientific and Engineering Software: A Series of Case Studies. In *ICSE '07: Proceedings of the 29th international conference on Software Engineering*. IEEE Computer Society, Washington, DC, USA, 550–559. <https://doi.org/10.1109/ICSE.2007.77>
- [8] Devendra K Chaturvedi. 2017. *Modeling and simulation of systems using MATLAB and Simulink*. CRC press.
- [9] Krzysztof Czarnecki and Ulrich W Eisenecker. 2000. *Generative Programming: Methods, Tools, and Applications*. Addison-Wesley Professional.
- [10] Michael Deck. 1996. Cleanroom and object-oriented software engineering: A unique synergy. In *Proceedings of the Eighth Annual Software Technology Conference, Salt Lake City, USA*.
- [11] Steve M. Easterbrook and Timothy C. Johns. 2009. Engineering the Software for Understanding Climate Change. *Computing in Science & Engineering* 11, 6 (November/December 2009), 65–74. <https://doi.org/10.1109/MCSE.2009.193>
- [12] Matthew Flatt, Eli Barzilay, and Robert Bruce Findler. 2009. Scribble: Closing the book on ad hoc documentation tools. In *ACM Sigplan Notices*, Vol. 44. ACM, 109–120.

- [13] Peter Fritzson, Johan Gunnarsson, and Mats Jirstrand. 2002. MathModelica-An extensible modeling and simulation environment with integrated graphics and literate programming. In *2nd International Modelica Conference, March 18-19, Munich, Germany*.
- [14] Robert Gentleman and Duncan Temple Lang. 2012. Statistical analyses and reproducible research. *Journal of Computational and Graphical Statistics* (2012).
- [15] Les Hatton and Andy Roberts. 1994. How Accurate is Scientific Software? *IEEE Trans. Softw. Eng.* 20, 10 (1994), 785–797. <https://doi.org/10.1109/32.328993>
- [16] Marco S Hyman. 1990. Literate C++. *COMP. LANG.* 7, 7 (1990), 67–82.
- [17] Cezar Ionescu and Patrik Jansson. 2012. Dependently-Typed Programming in Scientific Computing — Examples from Economic Modelling. In *Revised Selected Papers of the 24th International Symposium on Implementation and Application of Functional Languages (Lecture Notes in Computer Science)*, Vol. 8241. Springer International Publishing, 140–156. https://doi.org/10.1007/978-3-642-41582-1_9
- [18] Andrew Johnson and Brad Johnson. 1997. Literate Programming Using Noweb. *Linux Journal* 42 (October 1997), 64–69.
- [19] Stephen C Johnson et al. 1975. *Yacc: Yet another compiler-compiler*. Vol. 32. Bell Laboratories Murray Hill, NJ.
- [20] Project Jupyter. 2024. Project Jupyter. <https://jupyter.org/>. (2024). Accessed: 2026-01-23.

- [21] Rashidah Kasauli, Grischa Liebel, Eric Knauss, Swathi Gopakumar, and Benjamin Kanagwa. 2017. Requirements Engineering Challenges in Large-Scale Agile System Development. 352–361. <https://doi.org/10.1109/RE.2017.60>
- [22] Diane Kelly. 2013. Industrial Scientific Software: A Set of Interviews on Software Development. In *Proceedings of the 2013 Conference of the Center for Advanced Studies on Collaborative Research (CASCON '13)*. IBM Corp., Riverton, NJ, USA, 299–310. <http://dl.acm.org/citation.cfm?id=2555523.2555555>
- [23] Diane Kelly. 2015. Scientific software development viewed as knowledge acquisition: Towards understanding the development of risk-averse scientific software. *Journal of Systems and Software* 109 (2015), 50–61. <https://doi.org/10.1016/j.jss.2015.07.027>
- [24] Diane F. Kelly. 2007. A Software Chasm: Software Engineering and Scientific Computing. *IEEE Softw.* 24, 6 (2007), 120–119. <https://doi.org/10.1109/MS.2007.155>
- [25] D. E. Knuth. 1984. Literate Programming. *Comput. J.* 27, 2 (1984), 97–111. <https://doi.org/10.1093/comjnl/27.2.97>
arXiv:<http://comjnl.oxfordjournals.org/content/27/2/97.full.pdf+html>
- [26] Jeffrey Kotula. 2000. Source code documentation: an engineering deliverable. In *tools*. IEEE, 505.
- [27] Friedrich Leisch. 2002. Sweave: Dynamic generation of statistical reports using literate data analysis. In *Compstat*. Springer, 575–580.

- [28] Russell V Lenth. 2009. StatWeave users' manual. URL <http://www.stat.uiowa.edu/~rlenth/StatWeave> (2009).
- [29] Russell V Lenth, Søren Højsgaard, et al. 2007. SASweave: Literate programming using SAS. *Journal of Statistical Software* 19, 8 (2007), 1–20.
- [30] David Matthews, Greg Wilson, and Steve Easterbrook. 2008. Configuration Management for Large-Scale Scientific Computing at the UK Met Office. *Computing in Science and Engg.* 10, 6 (Nov. 2008), 56–64. <https://doi.org/10.1109/MCSE.2008.144>
- [31] Harlan D Mills, Richard C Linger, and Alan R Hevner. 1986. Principles of information systems analysis and design. (1986).
- [32] Nedialko S. Nedialkov. 2006. *VNODE-LP — A Validated Solver for Initial Value Problems in Ordinary Differential Equations*. Technical Report CAS-06-06-NN. Department of Computing and Software, McMaster University, 1280 Main Street West, Hamilton, Ontario, L8S 4K1.
- [33] James Milne Neighbors. 1980. *Software construction using components*. University of California, Irvine.
- [34] James M Neighbors. 1984. The Draco approach to constructing software from reusable components. *IEEE Transactions on Software Engineering* 5 (1984), 564–574.
- [35] James M Neighbors. 1989. Draco: A method for engineering reusable software systems. *Software reusability* 1 (1989), 295–319.

- [36] Object Management Group. 2000. *The Common Object Request Broker Architecture and Specification*. Technical Report. Object Management Group. <https://www.omg.org/spec/CORBA/2.6/>
- [37] OpenAPI Initiative. 2025. OpenAPI Specification Version 3.2.0. <https://spec.openapis.org/oas/latest.html>. (2025). Accessed: 2026-01-22.
- [38] Oracle. Accessed February 2023. JavaBeans Component Architecture. <https://docs.oracle.com/javase/8/docs/technotes/guides/beans/index.html>. (Accessed February 2023).
- [39] David Lorge Parnas. 2010. Precise Documentation: The Key to Better Software. In *The Future of Software Engineering*. 125–148. https://doi.org/10.1007/978-3-642-15187-3_8
- [40] David L. Parnas and P.C. Clements. February 1986. A Rational Design Process: How and Why to Fake it. *IEEE Transactions on Software Engineering* 12, 2 (February 1986), 251–257.
- [41] David Lorge Parnas, Jan Madey, and Michal Iglewski. 1994. Precise Documentation of Well-Structured Programs. *IEEE Trans. Softw. Eng.* 20, 12 (Dec. 1994), 948–976. <https://doi.org/10.1109/32.368133>
- [42] Terence J. Parr and Russell W. Quong. 1995. ANTLR: A predicated-LL (k) parser generator. *Software: Practice and Experience* 25, 7 (1995), 789–810.
- [43] Shari Lawrence Pfleeger and Joanne M. Atlee. 2010. *Software Engineering: Theory and Practice* (4th ed.). Prentice Hall, New Jersey, United States, Chapter 2.

- [44] Matt Pharr and Greg Humphreys. 2004. *Physically Based Rendering: From Theory to Implementation*. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.
- [45] Vreda Pieterse, Derrick G. Kourie, and Andrew Boake. 2004. A Case for Contemporary Literate Programming. In *Proceedings of the 2004 Annual Research Conference of the South African Institute of Computer Scientists and Information Technologists on IT Research in Developing Countries (SAICSIT '04)*. South African Institute for Computer Scientists and Information Technologists, Republic of South Africa, 2–9. <http://dl.acm.org/citation.cfm?id=1035053>. 1035054
- [46] Klaus Pohl, Günter Böckle, and Frank J Linden. 2005. *Software product line engineering: foundations, principles, and techniques*. Springer.
- [47] Norman Ramsey. 1994. Literate programming simplified. *IEEE software* 11, 5 (1994), 97.
- [48] Patrick J. Roache. 1998. *Verification and Validation in Computational Science and Engineering*. Hermosa Publishers, Albuquerque, New Mexico.
- [49] Rebecca Sanders and Diane Kelly. 2008. Dealing with Risk in Scientific Software Development. *IEEE Software* 4 (July/August 2008), 21–28.
- [50] Eric Schulte, Dan Davison, Thomas Dye, Carsten Dominik, et al. 2012. A multi-language computing environment for literate programming and reproducible research. *Journal of Statistical Software* 46, 3 (2012), 1–24.

- [51] Judith Segal. 2005. When Software Engineers Met Research Scientists: A Case Study. *Empirical Software Engineering* 10, 4 (October 2005), 517–536. <https://doi.org/10.1007/s10664-005-3865-y>
- [52] Stephen Shum and Curtis Cook. 1993. AOPS: an abstraction-oriented programming system for literate programming. *Software Engineering Journal* 8, 3 (1993), 113–120.
- [53] Volker Simonis. 2001. ProgDoc—A Program Documentation System. *Lecture Notes in Computer Science* 2890 (2001), 9–12.
- [54] SmartBear Software. 2025. Swagger: API Documentation and Design Tools for Teams. <https://swagger.io/>. (2025). Accessed: 2026-01-22.
- [55] Spencer Smith and Nirmitha Koothoor. 2016. A Document Driven Method for Certifying Scientific Computing Software Used in Nuclear Safety Analysis. *Nuclear Engineering and Technology* Accepted (2016). 42 pp.
- [56] Spencer Smith, Nirmitha Koothoor, and Ned Nedialkov. Submitted 2014. A Document Driven Method for Facilitating Certification of Scientific Computing Software. *IEEE Transactions on Software Engineering* (Submitted 2014).
- [57] Spencer Smith, Yue Sun, and Jacques Carette. 2015. *Comparing Psychometrics Software Development Between CRAN and Other Communities*. Technical Report CAS-15-01-SS. McMaster University. 43 pp.
- [58] Spencer Smith, Yue Sun, and Jacques Carette. Submitted December 2015. Statistical Software for Psychology: Comparing Development Practices Between

- CRAN and Other Communities. *Software Quality Journal* (Submitted December 2015). 33 pp.
- [59] W Spencer Smith. 2016. A Rational Document Driven Design Process for Scientific Software. In *Software Engineering for Science*. Chapman and Hall/CRC, 63–88.
- [60] W. Spencer Smith, Nirmitha Koothoor, and Nedialko Nedialkov. 2013. Document Driven Certification of Computational Science and Engineering Software. In *Proceedings of the First International Workshop on Software Engineering for High Performance Computing in Computational Science and Engineering (SE-HPCCCE)*. [8 p.].
- [61] W. Spencer Smith and Wen Yu. 2009. A Document Driven Methodology for Improving the Quality of a Parallel Mesh Generation Toolbox. *Advances in Engineering Software* 40, 11 (November 2009), 1155–1167. <https://doi.org/10.1016/j.advengsoft.2009.05.003>
- [62] Walid Taha. 2006. A Gentle Introduction to Generative Programming. *ACM Computing Surveys (CSUR)* 37, 4 (2006), 1–28.
- [63] Harold Thimbleby. 1986. Experiences of ‘Literate Programming’ using cweb (a variant of Knuth’s WEB). *Comput. J.* 29, 3 (1986), 201–211.
- [64] Ye Wu, Dai Pan, and Mei-Hwa Chen. 2001. Techniques for testing component-based software. In *Proceedings Seventh IEEE International Conference on Engineering of Complex Computer Systems*. 222–232. <https://doi.org/10.1109/ICECCS.2001.930181>